

Companion Appendix:

Strategic Impact Mapping: A Directed-Network Framework for S&P 500 Crisis Regimes

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CONTENTS

A	Notation Glossary	3
B	Data Construction	9
B-A	CRSP panel construction	9
B-B	Per-snapshot ticker composition	10
B-C	GICS sector taxonomy	11
B-D	Average dollar volume computation	12
B-E	Share-class disambiguation	12
C	Stage 1: A-DCC GARCH Estimation	13
C-A	Per-asset univariate GARCH(1,1) with Student- t innovations	13
C-B	The A-DCC recursion and the closed-form intercept	15
C-C	Multi-start MLE protocol	16
C-D	The 100-asset always-alive subset	17
C-E	Stationarity assumption check	19
C-F	Per-snapshot $\bar{R}$ near-PSD analysis	20
D	Stage 2: Graphical LASSO Precision-Matrix Filtering	20
D-A	The EBIC penalty for choosing λ	21
D-B	Identity-target shrinkage tiers	22
D-C	The constrained-BIC fallback rule (Algorithm 1)	23
D-D	Solver convergence diagnostics	25
E	Stage 3: Lead/Follower Direction Assignment	26
E-A	Lagged partial correlation via Frisch–Waugh–Lovell	27
E-B	Bivariate Granger F -test	27
E-C	Two-test cascade decision rule	27
E-D	Multiple-testing position	29
E-E	Joint $\alpha \times \text{dominance} \times \text{lag}$ sensitivity (F3)	29
F	Stage 4: Network Metrics	30
F-A	Q1: Erdős–Rényi non-randomness	31
F-B	Q2: PageRank, Gini, and Herfindahl	32
F-C	Q3: Louvain modularity with GICS sector purity	32
F-D	Q4: MAN triadic motif census	34
F-E	Dyad-preserving rewiring null	34
G	Stage 5: Network Stress Index	36
G-A	Channel definitions	39
G-B	Channel ablation	39
G-C	Weight rationale	41
G-D	Weight-sensitivity sweep	41
G-E	The rolling NSI simplification	42
G-F	VIX benchmark protocol	43

H	Robustness Analyses: Expanded	43
H-A	Watts–Strogatz σ diagnostic	45
H-B	Density-matched community analysis	48
H-C	Cross-sector edge fraction	48
H-D	Mutual-dyad fraction detail	48
H-E	Cross-crisis reproducibility expanded	50
I	Panel-Size and Liquidity-Stratification Sensitivity	51
I-A	Sweep design	51
I-B	Compute budget on Apple M5	52
I-C	Cross-panel result tables	52
I-D	The Mar 2020 coverage-versus-ADV NSI gap	54
I-E	The largest cross-rule gap: 2022 at $N = 200$	54
I-F	Coverage vs. ADV overlap at each N	55
I-G	Liquidity-stratification reading	55
I-H	Volume-weighted NSI (F1 prototype)	56
I-I	Bottom- N and liquidity-decile panels	57
I-J	Cross-panel fallback identification	57
J	Statistical Inference: Detail	58
J-A	Snapshot independence assumptions	58
J-B	Exact-permutation enumeration with $n_{\text{crisis}} = 7$	59
J-C	Benjamini–Hochberg FDR	60
J-D	Cluster bootstrap (snapshot NSI–VIX)	61
J-E	Block bootstrap (rolling NSI–VIX)	62
J-F	Bonferroni-20 per-snapshot Z thresholds	63
J-G	Power analysis	64
J-H	Robustness scorecard	66
K	Numerical-Invariant Assertion Suite	66
K-A	Suite scope	66
K-B	Summary across the 20-panel sweep	67
K-C	Paper-vs-code disclosures	68
K-D	Latent defensive-programming bug fixed during code review	69
L	Worked Example: Mar 2020 Walk-Through	69
M	Reproducibility	70
M-A	Software versions	71
M-B	Random seeds and determinism	71
M-C	Stage run-times on the Apple M5	71
M-D	Computational complexity	72
M-E	Cache schema	72
M-F	Repository layout and entry points	74
	References	74

HOW TO READ THIS APPENDIX

The main paper states what the pipeline does and what it finds; this companion explains *why* each modelling choice was made and defines every concept it uses, so the methodology can be read end to end without outside reference.

Section A is the notation glossary followed by a “Key concepts at a glance” reference that introduces the recurring econometric, high-dimensional-statistics, graph-theoretic, and resampling-inference ideas the per-stage derivations rely on. Section B documents the data: ticker universe, distressed-name list, GICS taxonomy, ADV. Sections C through G expand the five pipeline stages: model, parameter choices, numerical safeguards, runtime diagnostics. Section H expands the main paper §VI. Section I reports the panel-size and liquidity sweep cited as the main paper §IX Lim.(i): pipeline behaviour when the panel is rebuilt at smaller sizes or under a liquidity-weighted selection rule, and where the headline numbers stabilise. Section J expands the inference layer: exact permutation enumeration at $n_{\text{crisis}} = 7$, the Benjamini–Hochberg correction, the cluster and block bootstraps, the Bonferroni-20 threshold. Section K describes the 836-check assertion suite, the eleven paper-vs-code disclosures surfaced during code review, and one latent defensive-programming bug fixed during that review. Section L is the reproducibility metadata: software versions, random seeds, stage run-times, cache schemas, repository layout.

Each formal block (formula, pseudocode, table) is paired with an adjacent prose gloss.

A. NOTATION GLOSSARY

This is a reference table. Sections C through L use only those entries they need, and each entry is also introduced in the surrounding text when it first appears. The glossary supports jump-in lookup of a symbol from any later section.

A few conventions are global. Time indexes are written as subscripts ($r_{i,t}$); asset indexes use i, j , sometimes k . Hats denote estimated quantities ($\hat{\Omega}, \hat{a}$). A bar over a matrix denotes a window average ($\bar{R}, \bar{Q}$). Tildes denote min/max-normalised channels ($\tilde{s}, \tilde{h}$). Decimal numbers are rounded half-up to the displayed precision unless otherwise noted; relative-change percentages and means-of-means are computed from the unrounded inputs so that recomputing from displayed values may differ by up to one unit in the last significant digit.

Symbol	Meaning (with intuitive gloss)	First introduced
$T, T_{\text{win}}, T_{\text{cal}}$	Total / per-window Stage-2 sample / raw calendar-window trading days. T runs over the full 1985–2024 sample; $T_{\text{win}} = n$ is the per-snapshot complete-case count (Table B.2, “Days”); T_{cal} is the raw six-/twelve-month window length, with $T_{\text{win}} \leq T_{\text{cal}}$.	the main paper §IV
p, N	Per-window asset count (p once the window’s coverage filter has been applied; N the panel-size grid value).	the main paper §IV
n	Stage-2 sample size (equal to T_{win}). The n/p ratio governs how rank-deficient the empirical correlation is.	the main paper §III-B
$r_{i,t}$	Log return of asset i at day t : $r_{i,t} = \log(p_{i,t}/p_{i,t-1})$, where $p_{i,t}$ is the adjusted close.	§C
$\sigma_{i,t}^2$	GARCH(1,1) conditional variance, the per-asset volatility forecast for day t given everything up to $t-1$.	§C
$z_{i,t} = r_{i,t}/\sigma_{i,t}$	Standardised residual, the part of the return that is not explained by volatility. Roughly mean-zero, unit-variance per asset.	§C
$n_{i,t} = \min(z_{i,t}, 0)$	The negative part of the standardised residual, the asymmetric (downward-shock) signal that drives the A-DCC asymmetry term.	§C
a, b, g	A-DCC scalars. a = shock response, b = persistence, g = extra weight on joint downside. The recursion is covariance-stationary iff $a + b + g < 1$.	§C
$\bar{Q}, \bar{N}$	Unconditional pseudo-covariances ($E[z_t z_t^\top]$, $E[n_t n_t^\top]$). They carry the cross-sectional structure; the scalars (a, b, g) carry the dynamics.	§C
Q_t	Time- t A-DCC pseudo-correlation matrix. Updated by the A-DCC recursion; not yet normalised to unit diagonal.	§C
R_t	Time- t correlation matrix ($D_t Q_t D_t$ with $D_t = \text{diag}(Q_t)^{-1/2}$). Unit-diagonal, $ \rho \leq 1$.	§C
$\bar{R}$	Window-averaged R_t over the snapshot. The input to Stage 2. Near-PSD at low n/p .	§C
$\bar{\rho}$	Mean off-diagonal entry of $\bar{R}$. The simplest summary of how synchronised the panel is in that window.	the main paper §IV
$\Omega = \bar{R}^{-1}$	The population precision matrix, the inverse of the population correlation. Its off-diagonal entries encode partial correlations.	§D
$\hat{\Omega}_\lambda$	The GLASSO estimate of Ω at regularisation strength λ . Sparser as λ grows.	§D
$\rho_{ij \text{rest}}$	Partial correlation between i and j after controlling for every other asset: $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$.	§D
δ	Identity-target shrinkage intensity. $\bar{R} \leftarrow (1-\delta)\bar{R} + \delta I$ pulls the empirical matrix toward identity.	§D
$\bar{R}_\delta$	Shrunken correlation matrix $(1-\delta)\bar{R} + \delta I$, plus the spectral safety net of §D where it would trigger ($\lambda_{\min} < 10^{-6}$; never fires on this panel, §C-F). The actual sample-covariance argument of both the GLASSO solver and the EBIC log-likelihood.	§D

Symbol	Meaning (with intuitive gloss)	First introduced
γ	EBIC penalty parameter. $\gamma = 0$ recovers plain BIC; $\gamma > 0$ tightens the penalty when p is large relative to n . Distinct from the Louvain resolution $\gamma = 1$ of Stage 4 (§F); the two share notation by historical convention only.	§D
$k(\lambda)$	Off-diagonal nonzero count of $\hat{\Omega}_\lambda$. Counts undirected edges in the resulting GLASSO graph.	§D
$k_{\min} = \max(p, 10)$	Constrained-BIC edge floor. Imposes average degree ≥ 2 so the downstream stages see a non-trivial graph.	§D
$\lambda^*, \lambda_{\text{fallback}}^*$	Unconstrained vs. constrained-BIC argmin over the λ grid. The fallback fires when $k(\lambda^*) < k_{\min}$.	§D
$d_2 = k/\binom{p}{2}$	Undirected (Stage-2 GLASSO) edge density, the “Density” column of Table B.2. Distinct from the <i>directed</i> Stage-3 density $d = m_{\text{dir}}/\binom{p}{2}$ (mutual pairs counted twice) used by the NSI sparsity channel below. Closer to 1 = denser.	Table B.2
d (Cohen’s)	Standardised effect size $(\bar{X}_{\text{cris}} - \bar{X}_{\text{ncr}})/s_{\text{pool}}$ for the crisis-vs-non-crisis contrast. Always written “Cohen’s d ” in text; distinct from the edge density d above.	§J-G
$D \in \mathbb{R}_+^{p \times p}$	Stage-3 directed adjacency matrix. $D_{ij} > 0$ means there is a directed edge from i to j .	§E
c_{ij}, c_{ji}	Per-direction confidence scores in $[0, 1]$, built up from the lagged partial-correlation test and the Granger test.	§E
c_{floor}	Granger confidence floor in the Stage-3 cascade ($= 0.5$): a significant Granger F -test raises the per-direction confidence to at least this value. One axis of the F3 sweep.	§E
α	Significance threshold ($= 0.05$ throughout). Applied to both the lagged partial-correlation p -value and the Granger F -test.	§E
τ	Direction dominance ratio ($= 1.5$). One direction wins iff its confidence is $\tau \times$ the other; else the edge is recorded as mutual.	§E
f_{mut}	Mutual-dyad fraction $n_{\text{mutual}}/(n_{\text{directed}} + n_{\text{mutual}})$. Share of surviving edges that are reciprocal.	the main paper §VI-C
r_i	PageRank stationary mass at node i , normalised so $\sum r_i = 1$.	§F
$\text{HHI} = \sum_i r_i^2$	Full-network PageRank Herfindahl. Higher = more concentrated stationary mass.	§F
Gini	Gini coefficient of $\{r_i\}$. Reads 0 for uniform, approaches 1 for monopoly.	§F
Q	Louvain modularity. Compares within-community edge density to a degree-preserving null.	§F
Purity	Sector purity: fraction of nodes assigned to their community’s dominant GICS sector. $= 1$ if Louvain perfectly recovers the GICS partition.	§F
$ Z_C $	Erdős–Rényi clustering deviation Z -score. Magnitude of the empirical clustering vs. matched-density random graphs.	§F
σ	Watts–Strogatz small-world coefficient (path-length-corrected clustering ratio). Density-invariant in principle.	§H
$Z_{\text{FFL}}, Z_{\text{MR}}, Z_{\text{SIM}}$	Empirical counts of the three MAN triads minus null mean, divided by null standard deviation. The dyad-preserving rewiring is the null.	§F
$SP_i = Z_i / \sqrt{\sum_j Z_j^2}$	Motif significance profile entry. Removes magnitude, keeps the direction of the joint motif signature.	§F
$s = 1 - d$	NSI sparsity channel. Larger when the Stage-3 graph is thinner.	§G
$\bar{\rho}$	NSI mean-correlation channel (re-using the symbol; same quantity).	§G
h	NSI PageRank–Herfindahl channel: top-10 PR masses re-normalised to sum to 1, then $\sum_i r_i^2$ over those ten. Distinct from the full-network HHI above; the top-10 form is what Stage 5 consumes (<code>src/stage5_nsi/stress_index.py</code>).	§G
μ	NSI motif-shift channel $Z_{\text{FFL}}/ Z_{\text{FFL}}^{(\text{baseline})} $. Anchors the FFL signal as a multiple of the calm-baseline FFL prevalence.	§G
$\tilde{x}$	Min/max-normalised version of channel x across the twenty snapshots. Each $\tilde{x} \in [0, 1]$.	§G
NSI	Composite Network Stress Index, $\sum_k w_k \tilde{x}_k$, in $[0, 1]$.	§G
w_s, w_h, w_ρ, w_μ	NSI channel weights, fixed at $(0.25, 0.20, 0.35, 0.20)$. Sum to 1, treated as a prior rather than an estimate.	§G

Key concepts at a glance

The per-stage derivations rely on four bodies of literature, time-series econometrics, high-dimensional statistics, graph theory, and resampling inference. Each is summarised once here so the per-stage prose can refer back without re-deriving. The first six entries below cover the data and finance conventions that the per-stage text presumes; the remainder cover the econometric, statistical, and graph-theoretic primitives.

The twenty-snapshot crisis-vs-calm design.: The pipeline does not run continuously over 1985–2024; it freezes the panel at twenty ex-ante chosen non-overlapping windows, *snapshots*, six-month event windows (~ 76 –127 trading days) and twelve-month baseline windows. Seven are crisis events (Oct 1987 Black Monday, Oct 1997 Asian Crisis, Oct 1998 LTCM, Apr 2000

Dot-com, Sep 2001 9/11, Oct 2008 GFC, Mar 2020 COVID); the remaining thirteen cover calm baselines, volatility shocks short of systemic crisis, and post-crisis recovery phases. Every per-snapshot quantity in the pipeline (the A-DCC $\bar{R}$, the GLASSO graph, the Stage-4 metrics, the NSI value) is one of twenty parallel computations; every inferential claim compares the seven-snapshot crisis arm against the thirteen-snapshot non-crisis arm. The window definitions are listed in Table B.2; the $\binom{20}{7} = 77,520$ permutation floor referenced throughout §J is a direct consequence of this design.

The S&P 500 universe.: The asset panel is the union of all PERMNOs that were S&P 500 constituents at any point in 1985–2024, as recorded in CRSP’s `crsp.msp500list` point-in-time membership intervals; this totals 1,443 unique PERMNOs (of which 1,442 have at least one in-window observation), and per-snapshot alive counts range 655–889 after the within-window coverage filter (§B-A). Point-in-time membership removes by construction the survivorship bias that would arise from constructing the universe at a single calendar date. For brevity we refer to the cleaned panel as the “500-asset universe” even though the actual count at each snapshot is the 80%-coverage-filtered subset $p \leq 500$.

GICS (Global Industry Classification Standard) sectors.: The S&P–MSCI taxonomy assigns each listed equity to one of 11 sectors (Financials, Information Technology, Energy, Health Care, etc.), each subdivided into Industry Groups, Industries, and Sub-Industries. Stage 4 sector purity (§F) and the cross-sector edge fraction (§H-C) read the 2024 revision; the 2016 split of Real Estate from Financials is the structural break in the count distribution discussed in §B-C. The sector labels enter the pipeline only as an external validation handle on Louvain communities (Stage 4 Q3); the graph estimation itself never sees them.

Adjusted close, split, and dividend adjustments.: A *stock split* multiplies the share count by some factor k and divides the listed price by the same factor, so a 2:1 split halves the quoted close overnight without economic content. A *dividend* returns cash to shareholders, so the quoted close drops by approximately the dividend amount on the ex-dividend day. Both events introduce artificial jumps in the raw price series. The *adjusted close* retroactively rescales pre-event prices so that a continuous log-return series reflects total return rather than corporate-action noise. “Split-adjusted close” applies only the split correction; “dividend-adjusted close” applies both. On the CRSP panel (§B-A), the pipeline reads CRSP’s native `ret` column (split- and dividend-adjusted simple total return) and forms log-returns as $\log(1 + \text{ret})$; the prices-side series carries the split-only adjusted dollar close $\text{prc}/\text{cfacpr} \times \text{cfacpr}_0$ used for the volume-side ADV calculation (§B-D). The dollar-volume product $|\text{prc}| \cdot \text{vol}$ is split-invariant because the $\text{cfacshr}/\text{cfacpr}$ factors cancel, so no separate unadjusted pull is needed.

Forward-fill (ffill).: A missing-data imputation rule that carries the last observed value forward in time until the next non-missing observation. Used in §B on per-ticker price series with gaps of ≤ 5 trading days (holidays, halts, exchange outages); longer gaps remain NaN so the downstream coverage filter can drop short-history tickers from per-snapshot panels.

VIX (CBOE Volatility Index).: The CBOE’s 30-day *implied* volatility index, computed from a weighted basket of S&P 500 option prices: high VIX = expensive insurance against future swings = high implied volatility. VIX is forward-looking by construction (the options price-in the next 30 days), in contrast to a realised-volatility series, which only re-states the past. The NSI–VIX concordance comparison of §G treats VIX as the canonical price-based stress benchmark; the NSI is built from network-topology channels only, so VIX is a co-movement check rather than a calibration target.

Volatility clustering and heteroskedasticity.: *Heteroskedasticity* simply means non-constant variance. In returns it shows up as *volatility clustering*: turbulent days follow turbulent days and calm days follow calm days, even when the level of returns shows no autocorrelation. A classical OLS regression with i.i.d. Gaussian residuals cannot reproduce this pattern; GARCH does it explicitly by letting today’s conditional variance depend on yesterday’s squared shock and yesterday’s variance. The empirical signature on equity panels, positive autocorrelation of squared returns at every short lag, is the reason Stage 1 fits a GARCH-family model rather than an unconditional variance estimator.

Ordinary Least Squares (OLS).: The classical regression estimator: choose the coefficient vector $\hat{\beta}$ that minimises the sum of squared residuals $\sum_t (y_t - W_t \beta)^2$. The closed form is $\hat{\beta} = (W^\top W)^{-1} W^\top y$ whenever $W^\top W$ is invertible. OLS is what every regression invoked in this appendix relies on under the hood: the Frisch–Waugh–Lovell two-step below, the Granger F -test below, and the block-bootstrap inner regressions for the rolling NSI–VIX concordance (§J) all reduce to OLS fits on appropriately constructed design matrices.

AR(1), autoregression, and lag-1 dependence.: A lag-1 autoregressive process $y_t = \phi y_{t-1} + \varepsilon_t$ has y_t depending linearly on its immediate past plus an innovation ε_t : $|\phi| < 1$ keeps the process covariance-stationary, $\phi = 1$ is the unit-root random walk, $\phi = 0$ is i.i.d. noise. The Granger F -test below is exactly the question “does adding x_{t-1} to the AR(1) of y_t reduce the residual sum of squares more than chance?”; the FWL conditioning on $r_{j,u-1}$ inside the Stage 3 cascade (§E) absorbs the AR(1)-persistence component of the candidate follower before the lead/follower signal is read off. The same AR(1) structure is what the rolling-NSI block bootstrap (§J) is designed to preserve under resampling.

Independent and identically distributed (i.i.d.) samples.: A sequence is *independent* if knowing the value of one observation conveys no information about any other, and *identically distributed* if every observation is drawn from the same fixed distribution. The plain bootstrap and the exact permutation test of §J both presume i.i.d. resampling of their atomic units (a single observation, a single snapshot). The cluster and block bootstraps are explicit departures from i.i.d.: they preserve a specific kind of dependence (within-cluster, within-block) that an i.i.d. resampler would destroy.

Maximum likelihood (MLE) and quasi-MLE (QMLE).: Maximum likelihood estimation chooses the parameter that makes the observed data most probable under an assumed density. Quasi-MLE keeps the same first-order condition but does not

require the assumed density to be correct: the estimator is still consistent and asymptotically normal under mild regularity (White, 1982). Asymptotic standard errors would then take the sandwich form $H^{-1}\hat{\Sigma}H^{-1}$ rather than the Hessian-only form H^{-1} ; we do not rely on either in this paper, all uncertainty quantification is via the cluster and block bootstraps of §J. We use QMLE wherever a Gaussian or Student- t density is plugged in only as an objective function rather than as a distributional claim (the Stage 1 A-DCC scalars and the Stage 2 GLASSO precision both fit under QMLE in this sense).

Student- t_ν innovations.: The Student- t distribution with ν degrees of freedom has heavier tails than the Gaussian at the same variance, with $\nu \rightarrow \infty$ recovering the normal. Equity-return innovations routinely show kurtosis well above the Gaussian value of 3, so the Student- t is used in Stage 1 as a defensive choice that absorbs the realised tail weight rather than declaring it a misspecification.

Covariance-stationarity.: A time series is covariance-stationary if its mean and autocovariances do not depend on the time index. For GARCH(1,1) this requires $\alpha + \beta < 1$; the boundary $\alpha + \beta = 1$ is the IGARCH limit, at which the unconditional variance is no longer finite but the *conditional* variance recursion is still well-defined day by day. The same idea reappears in the A-DCC recursion under the analogous constraint $a + b + g < 1$.

DCC (Dynamic Conditional Correlation; Engle 2002) and the pseudo-correlation matrix Q_t .: DCC is the symmetric precursor of the A-DCC model used in Stage 1. The two-step factorisation is the same: per-asset GARCH first, then a scalar recursion $Q_t = (1 - a - b)\bar{Q} + a z_{t-1}z_{t-1}^\top + b Q_{t-1}$ updates the pseudo-correlation as a convex combination of the unconditional matrix, yesterday's outer product of standardised residuals, and yesterday's Q . Two scalars (a, b) carry the dynamics of every off-diagonal entry, which is the dimension-reduction trick that makes DCC tractable at $p \gtrsim 100$. The limitation is that the shock-response term $z_{t-1}z_{t-1}^\top$ treats joint upward and joint downward movements identically; the A-DCC extension of Cappiello, Engle, and Sheppard [2] adds the third scalar g on $n_{t-1}n_{t-1}^\top$ to fire only on joint downside. The asymmetric term pushes Q_t 's diagonal off unity whenever any standardised residual is negative, hence *pseudo*-correlation in the standard DCC sense (right cross-product structure, wrong diagonal); the unit-diagonal correlation consumed by Stage 2 is recovered post-hoc by $R_t = D_t Q_t D_t$ with $D_t = \text{diag}(Q_t)^{-1/2}$, a one-line rescaling that erases the diagonal drift without touching the off-diagonal direction of the update.

ADF test, IGARCH, and the volatility half-life.: The Augmented Dickey–Fuller (ADF) test has *non-stationarity* (a unit root) as the null: rejection at $\alpha = 0.05$ is evidence in favour of stationarity, not against it. We apply it in §C as a sanity check on the standardised residuals $z_{i,t}$. The boundary $\alpha + \beta = 1$ above is the IGARCH limit, “I” stands for *integrated*, echoing the I(1) terminology of unit-root time series: at the boundary, a variance shock has a permanent rather than a transient effect on the conditional variance. A complementary intuition is the volatility half-life $-\log 2 / \log(\alpha + \beta)$: at the cross-sectional median $\hat{\alpha} + \hat{\beta} \approx 0.991$ this corresponds to a half-life of roughly 78 trading days, the time scale on which a shock to conditional variance decays toward its unconditional mean.

Precision matrix and partial correlation.: If R is the population correlation matrix of p assets, the precision matrix is $\Omega = R^{-1}$. A zero in position (i, j) of Ω is equivalent (under Gaussianity) to the statement that returns of asset i and j are conditionally independent given all other $p-2$ assets in the panel. The partial correlation is $\rho_{ij|\text{rest}} = -\omega_{ij} / \sqrt{\omega_{ii}\omega_{jj}}$; intuitively, it is the slice of the marginal correlation that survives after every other asset's contemporaneous variation is regressed out. Sparsifying Ω therefore sparsifies the partial-correlation network, which is what Stage 2 needs.

PSD, SPD, and conditioning.: A symmetric matrix is positive semi-definite (PSD) if all eigenvalues are ≥ 0 ; positive definite (SPD) if all are strictly > 0 . Correlation matrices are PSD by construction; only SPD ones are invertible. When the sample size is too small ($n < p$), the empirical correlation matrix is PSD but rank-deficient, so its inverse does not exist. Shrinkage toward identity, $\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I$, lifts every eigenvalue by δ and so restores SPD and makes the inverse numerically well-conditioned. The trade-off is a controlled bias toward the identity; the tier schedule of §D calibrates δ to the available n/p .

Cholesky decomposition.: The Cholesky factorisation of an SPD matrix A is the unique lower-triangular L with positive diagonal entries satisfying $A = LL^\top$, computable in $\frac{1}{3}p^3$ floating-point operations on a $p \times p$ matrix. Many compute-bound steps in this pipeline reduce to a Cholesky factor of the running correlation: the GLASSO inner solve, the A-DCC log-determinant evaluation in eq. (C.6), and the per-window stationarity check on $\bar{R}$. The PSD safeguard of §C guards against the case where $\bar{R}$ has eigenvalues so small that the Cholesky factor underflows; the shrinkage lift $\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I$ of §D restores the working precision.

ℓ_1 regularisation and sparsity.: Penalising $\sum_{i \neq j} |\omega_{ij}|$ rather than $\sum_{i \neq j} \omega_{ij}^2$ produces estimates with *exact* zeros rather than merely small entries. Geometrically, the ℓ_1 ball has corners on the coordinate axes, so the constrained MLE tends to land on faces where several coefficients are exactly zero. This is the reason GLASSO returns a graph (a list of present-vs-absent edges) rather than a dense matrix of small near-zero entries.

Information criteria (BIC, EBIC).: An information criterion scores a model by $-2\mathcal{L} + \text{penalty} \cdot k$, where $\mathcal{L}$ is the maximised log-likelihood and k is the number of free parameters. The BIC penalty $k \log n$ approximates a Bayesian model-evidence bound and so prefers a sparser model whenever the extra parameters do not appreciably raise $\mathcal{L}$. The Extended BIC of Foygel and Drton [6] adds the term $4\gamma k \log p$, which tightens the penalty in the high-dimension regime where p is large relative to n .

SLSQP and NNLS (constrained optimisers).: Stages 1 and 2 fit with L-BFGS-B, a quasi-Newton method whose only constraint vocabulary is the simple box bound $\ell_k \leq \theta_k \leq u_k$ on each parameter individually, the default dispatch under arch

and sklearn. The Stage 5 VIX-fit needs more: the four channel weights must lie in $[0, 1]$ and sum to one, a linear equality that L-BFGS-B cannot express. We therefore fall back to SLSQP (Sequential Least-Squares Quadratic Programming), which carries linear equality and inequality constraints alongside the box bounds at the cost of one Jacobian per constraint. The “NNLS” label in the weight-sensitivity table refers to the non-negativity-plus-simplex problem $\min_w \|Xw - \tilde{v}\|^2$ s.t. $w \geq 0$, $\sum_k w_k = 1$; this is what SLSQP actually solves, slightly stronger than the textbook NNLS, which imposes only $w \geq 0$.

Coordinate descent (the GLASSO solver).: The GLASSO objective $-\log \det \Omega + \text{tr}(\bar{R}\Omega) + \lambda \|\Omega\|_{1,\text{off}}$ combines a smooth log-determinant term with a non-smooth ℓ_1 penalty. Coordinate descent solves it by cycling through one column of Ω at a time, holding the others fixed and updating the active column via a lasso-like soft-threshold step, until the change between full sweeps drops below tolerance. The per-coordinate subproblem is convex and admits a closed-form update, which is why coordinate descent is the de facto solver for ℓ_1 -penalised covariance and precision problems (Friedman et al., 2008). Convergence is monotone but can stall when $\bar{R}$ is near-singular, because the inner Cholesky-like step that each column update needs degrades under poor conditioning; §D documents the rare cases on the full-coverage panel where the sklearn solver hits its iteration cap and the constrained-BIC fallback fires.

Frisch–Waugh–Lovell (FWL).: A two-step recipe for partial correlation given a conditioning set W : regress y on W and keep residuals $y^\perp$; regress x on W and keep residuals $x^\perp$; then $\text{pcorr}(y, x \mid W) = \text{corr}(y^\perp, x^\perp)$. The result is algebraically identical to running a joint multiple regression of y on (x, W) and reading off the partial t -statistic of x . FWL is used in Stage 3 because it isolates the lagged-driver contribution with one OLS pass per variable, rather than a per-edge multivariate regression.

Pearson vs. Spearman correlation, and the Fisher z -transformation.: Pearson correlation measures linear co-movement on the original scale; Spearman correlation does the same on the ranks of the two series and is therefore invariant under any monotone transformation, which makes it the natural choice when the marginals are heavy-tailed or non-Gaussian. We report both wherever a bivariate dependence number carries inferential weight: Pearson is the headline statistic when the downstream test (e.g. the FWL partial-correlation t -test, or the snapshot NSI–VIX comparison) presumes near-Gaussian residuals, and Spearman is the robustness companion. The Fisher transformation $z = \frac{1}{2} \log((1+r)/(1-r)) = \tanh^{-1}(r)$ maps $r \in (-1, 1)$ to $\mathbb{R}$ and is approximately normal at variance $1/(T-3)$ on a sample of size T ; this is why bootstrapped percentile CIs on r compress as r approaches ± 1 , the inverse $r = \tanh(z)$ is locally non-linear at the algebraic ceiling, and the upper-tail compression referenced in §J is the Fisher non-linearity, not a bug in the resample.

The Z -score as a standardised effect.: For a test statistic X with null mean μ_0 and null standard deviation σ_0 , the Z -score $Z = (X - \mu_0)/\sigma_0$ counts how many null-standard-deviations the observed value sits from the null mean. Under an asymptotically Gaussian null, $|Z| > 1.96$ is the two-sided 5% rejection threshold; under a more general null (as for the Erdős–Rényi clustering coefficient and the dyad-preserving rewiring counts of §F), the threshold survives as a useful approximation when the null distribution is approximately symmetric. The Stage 4 magnitudes $|Z_C|$, Z_{FFL} , Z_{MR} , Z_{SIM} and the Stage 5 Bonferroni-20 per-snapshot tests are all reported in this form so that effects on graphs of very different size and density are directly comparable.

The p -value and the null hypothesis.: The null hypothesis H_0 is the proposition we are trying to falsify, typically “the variable carries no signal” or “the two arms are equally distributed”. The p -value is the probability of observing data at least as extreme as ours under H_0 . Small p is evidence against H_0 , not evidence for the alternative; large p is failure-to-reject, not acceptance of the null. A *two-sided* test counts extremity in either direction; a *one-sided* test counts only the direction predicted by the alternative. The exact permutation p -value of §J is the proportion of the 77,520 label assignments whose statistic is as extreme as the observed one, the most literal possible reading of “probability under the null” that finite-sample inference admits.

Granger F -test.: Compares two nested OLS regressions: a *restricted* fit using only the candidate follower’s own lags, and an *unrestricted* fit that adds the candidate driver’s lags. If the unrestricted residual sum of squares is meaningfully smaller, the F statistic is large and the driver is admitted as Granger-causal. The intuition is “does adding x ’s history reduce y ’s prediction error beyond what y ’s own history already explains?”

Graph-theory vocabulary.: A *graph* (synonym: *network*) is a pair (V, E) of nodes V and edges E . An edge is *undirected* (the unordered pair $\{i, j\}$) or *directed* (the ordered pair (i, j) , written $i \rightarrow j$). The *adjacency matrix* A has $A_{ij} \neq 0$ iff an edge runs from i to j ; a weighted graph carries real numbers in A instead of 0/1. *Edge density* on an undirected graph is $d = |E|/\binom{|V|}{2}$, the fraction of possible edges actually present. The *in-degree* of i counts edges pointing into i ; the *out-degree* counts edges leaving i . A *hub* is a node with disproportionately high degree (or PageRank mass, or weighted strength) relative to the rest of the network. Stage 2 produces an undirected weighted graph (the GLASSO partial-correlation graph); Stage 3 directs each surviving edge; Stages 4–5 read out hub structure, community structure, and motif content.

PageRank as a random walk.: PageRank with damping $\alpha = 0.85$ is the stationary distribution of a walker on the directed graph who at each step follows a uniformly chosen out-edge with probability α and teleports to a uniformly chosen node with probability $1 - \alpha$. The score r_i is the long-run fraction of time the walker spends at node i . The damping factor convention $\alpha = 0.85$ is inherited from Page and Brin (1999); intuitively, $1 - \alpha$ is the steady-state probability of a fresh restart, so the score combines local in-flow with a baseline reach.

Gini vs. Herfindahl concentration.: Both summarise inequality in a non-negative vector that sums to 1, but they weight it differently. The Gini coefficient integrates the Lorenz curve and so is sensitive to the full spread of the distribution: a long tail of near-zero entries pulls Gini upward. The Herfindahl HHI = $\sum_i r_i^2$ sums squared masses and so weights the top of the distribution more heavily, effectively ignoring the tail. The two can disagree on snapshot rank order (§F, Jun 2020 example) for exactly this reason.

Erdős–Rényi $G(n, m)$ random graph.: The benchmark random-graph family used as the Q1 null in §F. Fix n nodes and m edges, then sample uniformly from the $\binom{n}{m}$ possible edge sets. The result has the same edge count as the empirical graph but with edges placed without any preferential attachment, clustering, or community structure. Comparing the empirical clustering coefficient against the $G(n, m)$ ensemble at matched (n, m) asks whether the empirical graph is triangle-richer than pure edge-count alone would predict. A separate variant $G(n, p)$ fixes the per-pair edge probability rather than the total edge count; we use the edge-count version $G(n, m)$ throughout so the comparison is genuinely density-matched.

Local clustering coefficient.: For an undirected graph, the local clustering coefficient at node i is $C_i = (\text{edges among } i\text{'s neighbours}) / \binom{k_i}{2}$, where k_i is the degree of i . It is the empirical probability that two randomly chosen neighbours of i are themselves connected, a node-level triangle density. The mean $C = \frac{1}{n} \sum_i C_i$ is the headline statistic underlying the Q1 Erdős–Rényi non-randomness test: networks with strongly correlated neighbourhoods (triangle-heavy graphs) sit far above the matched-density random-graph mean, and the Z -score $|Z_C|$ in §F is the standardised version of that gap.

Small-world property.: A graph is *small-world* if it simultaneously has *high local clustering* (one’s neighbours tend to be each other’s neighbours) and *short typical path lengths* (any two nodes are connected by a short chain of edges). Random graphs are short-path but low-clustering; regular lattices are high-clustering but long-path. The Watts–Strogatz σ statistic of §H-A compares both quantities against a matched $G(n, m)$ null and combines them into the ratio $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$, which exceeds 1 in the small-world regime.

Modularity and the degree-preserving null.: Modularity Q scores a partition by how much more within-community edge weight is present than would be expected under a random graph that preserves each node’s degree. $Q \rightarrow 0$ indicates no community structure; Q close to 1 indicates that almost all edge weight stays inside communities. Louvain greedily merges nodes to maximise Q . *Sector purity* is a separate, external check: it asks how well the resulting community labels agree with the known GICS sector labels.

Triadic motifs and the MAN nomenclature.: The Davis and Leinhardt (1972) nomenclature catalogues all 16 isomorphism classes of three-node directed subgraphs under the labels M (mutual), A (asymmetric), N (null). The three triads tracked by Stage 4 cover the hierarchical (030T, “feed-forward loop”), reciprocal (111D, “mutual regulation”), and common-driver (021D, “single-input module”) patterns most relevant to a partial-correlation network. Each empirical count is compared to a null mean and standard deviation under the dyad-preserving rewire, and the resulting Z -scores are aggregated into the unit-norm *significance profile* of Milo et al. (2002) so that magnitudes are comparable across graphs of different size and density.

FFL, MR, SIM (motif abbreviations).: Throughout the appendix and the main paper, the three triadic motifs above are tracked by name as well as by MAN code: *FFL* = Feed-Forward Loop (MAN code 030T, the two-step path from a common source); *MR* = Mutual Regulation (MAN code 111D, a reciprocal pair with one incoming arc); *SIM* = Single-Input Module (MAN code 021D, a common driver to two followers). The Z -scores and significance-profile entries Z_{FFL} , SP_{FFL} and so on use these abbreviations consistently with the motif table at §F.

Permutation tests and the $\binom{20}{7}$ floor.: A permutation test asks: under the null that the crisis label carries no signal, every assignment of seven of the twenty snapshots to the “crisis” arm is equally likely. So enumerate all $\binom{20}{7} = 77,520$ assignments, compute the test statistic on each, and compare the observed value to the empirical distribution. The smallest one-sided p -value the test can produce is $1/77,520 \approx 1.3 \times 10^{-5}$, the observed split is the most extreme by construction. This is the structural floor referenced throughout §J on the twenty-snapshot panel; the earlier ten-snapshot panel had only two crisis windows and a floor of $1/\binom{10}{2} = 1/45 \approx 0.022$.

Type I and Type II error, statistical power.: *Type I error* is rejecting H_0 when it is true (a false positive); its rate is α , the significance threshold (typically 0.05). *Type II error* is failing to reject H_0 when the alternative is true (a false negative); its rate is β . *Statistical power* is $1 - \beta$, the probability of correctly detecting a true effect. Power rises with sample size, effect size, and a more lenient α . The exact permutation test at $n_{\text{crisis}} = 7$ has a one-sided p -value floor of $1/77,520 \approx 1.3 \times 10^{-5}$, which is also a hard ceiling on the power achievable on this design at any $\alpha \leq 1/77,520$; §J-G reports simulation-based power curves under a range of plausible alternatives.

Cohen’s d and Hedges’ g (effect sizes).: Where a p -value answers “is the effect distinguishable from zero?”, an effect size answers “is the effect large?”. Cohen’s d is the difference of group means divided by the pooled within-group standard deviation, with conventional reading thresholds $|d| \geq 0.2, 0.5, 0.8$ for small, medium, and large effects (Cohen, 1988). Hedges’ g is the small-sample-corrected variant $g = d \cdot J(n)$ with $J(n) = 1 - 3/(4n - 9)$ for two-group n ; the correction is appreciable for $n \lesssim 20$ and negligible beyond. §J reports Cohen’s d throughout because the cluster-bootstrap percentile CI already absorbs the small-sample distortion that motivates g ; the parametric g enters only as a counter-anchor for one CI comparison.

Family-wise (FWER) vs. false-discovery (FDR) control.: With m tests in the family, the family-wise error rate is the probability of *any* false rejection. Bonferroni divides α equally among the tests (α/m), strong control but punishingly

conservative when m grows. Holm–Bonferroni is a step-down refinement that gains a little power without sacrificing FWER. The false-discovery rate is a different target: the expected *proportion* of false rejections among all rejections. Benjamini–Hochberg controls FDR rather than FWER and so tolerates a few false rejections in exchange for substantially more power on a positively-dependent family like ours; this is why Table II of the main paper reports BH q -values rather than Bonferroni- or Holm-adjusted p -values. Throughout the appendix the shorthand *Bonferroni-20* refers to the family-wise correction $\alpha = 0.05/20 = 0.0025$ applied across the twenty parallel per-snapshot Z -tests, with two-sided rejection threshold $|Z| > 3.023$ replacing the uncorrected $|Z| > 1.96$ (§J-F).

Confidence intervals and coverage.: A confidence interval (CI) is a range $[L, U]$ constructed so that, under repeated sampling, the true parameter falls inside the range a stated fraction of the time, the *nominal coverage*, typically 95%. *Empirical coverage* is the fraction observed in practice on a known-truth simulation, and can diverge from the nominal target in small samples or under serial dependence. The block-bootstrap percentile CI for the rolling NSI–VIX correlation (§J) is checked against an AR(1)-pair simulation to confirm that empirical coverage stays close to the 95% nominal target; the cluster-bootstrap CI on the per-snapshot effects (§J-D) is validated the same way, against a known-truth resampling simulation over the twenty-snapshot, eighteen-cluster design.

Bootstrap variants.: The plain i.i.d. bootstrap resamples observations with replacement and computes the statistic on each resample; the percentile band of the resampled values is the confidence interval. It is biased when observations are dependent. The *cluster bootstrap* resamples user-defined clusters (here the GFC and COVID pairs) as atomic units, preserving the within-cluster dependence exactly. The *block bootstrap* resamples contiguous time blocks; the Politis–Romano *stationary* variant draws block lengths from a geometric distribution, so block-boundary effects do not break stationarity in expectation. Each variant is matched to the dependence structure of the input series.

Bootstrap CI variants beyond the percentile interval.: The percentile bootstrap CI uses the raw quantiles of the resampled statistic and is the cheapest to compute. The *bias-corrected and accelerated* (BCa) interval adjusts both for median bias and for the skewness-induced acceleration of the resample distribution; the *studentised* (pivotal- t) interval pivots each resample by its own standard-error estimate, recovering the correct asymptotic scale. Both reduce coverage error in moderate samples but require either an influence-function estimate (BCa) or an inner-loop variance estimator (studentised), which on this eighteen-cluster design is itself the bottleneck, so §J retains the plain percentile band and quotes BCa/studentised only as counter-anchors. *Cyclic wrapping* of the block bootstrap means that whenever a block of length ℓ starts within ℓ of the series end, it continues from the beginning (index mod G); this is what makes the block bootstrap genuinely stationary in expectation (no endpoint bias) and is the Politis–Romano property invoked above.

B. DATA CONSTRUCTION

A. CRSP panel construction

The CRSP-US Stock daily file, accessed through Bilkent University’s WRDS subscription provides a single canonical source for the panel. Construction proceeds in six steps:

- 1) **Membership.** `crsp.msp500list` returns the point-in-time S&P 500 constituent intervals over 1985-01-02 to 2024-12-31 (1,477 membership intervals, 1,443 unique PERMNOs).
- 2) **Daily file.** A single `crsp.dsf` pull keyed on the union of constituent PERMNOs yields 7,981,976 in-window daily rows (columns `ret`, `retx`, `prc`, `vol`, `shrout`, `cfacpr`, `cfacshr`). PERMNO 48549 (SEDCO INC, delisted 1984-12-24) falls outside the panel start and contributes no in-window rows, leaving 1,442 active PERMNOs.
- 3) **Returns.** Log-returns $\log(1 + \text{ret})$ on CRSP’s split- and dividend-adjusted simple-return series; the three historical total-bankruptcy events with `DLRET` = -1 (PERMNOs 18278/46288/57488, all 1988–1993, code 574) are clipped to -0.9999 so the log stays finite at -9.21 .
- 4) **Delisting splice.** For PERMNOs with a non-null `dlret` on a panel-window date, the delisting return is appended on `dlstdt` (combined multiplicatively with any DSF return on the same date). WAMU (PERMNO 81593) is the only curated distressed name with `dlret` null: the FDIC seizure of 2008-09-26 is recorded as a trading suspension rather than a formal delisting in CRSP, so the bankruptcy-day return is implicit in the final DSF `prc` trajectory ($\$46 \rightarrow \0.16).
- 5) **Ticker resolution.** The most-recent CRSP ticker per PERMNO is the panel label; eight crisis-era distressed names (LEH, BSC, MER, FNM, FRE, WAMU, AIG, F) carry manual paper-facing overrides because the latest-ticker rule otherwise resolves WAMU to its 2018 post-reorg ticker (COOP); 81 cross-PERMNO ticker collisions across the 40-year window receive a `_PERMNO` suffix (e.g. `AMT_10057` for the 1968–1996 Acme Cleveland / American Tower distinction).
- 6) **Final panel.** 1,442 PERMNOs over 10,080 trading days, written to `data/sp500_{prices,returns,info}.parquet` in the canonical (`info`, `prices`, `returns`) schema that downstream stages consume; per-window balanced sub-panels (in-window coverage $\geq 80\%$) carry 655–889 alive PERMNOs across the twenty Stage-1 snapshots, with 278 tickers above the $\geq 99\%$ always-alive threshold supplying the 100-asset balanced subset for the A-DCC parameter MLE.

The CRSP `ret` column is pre-adjusted for splits and dividends, so no symbol-level forward-fill, contamination filter, or ticker-reuse cutoff is needed at the panel level: PERMNO indexing removes by construction the cross-issuer ticker collisions a

post-event symbol re-assignment would create. The Stage-5 volatility benchmark uses the Whaley [14] VXO+VIX continuity series spliced at 2003-09-22 from FRED VXOCLS and VIXCLS (see §G for the splice protocol). The remaining subsections describe per-snapshot ticker composition on the CRSP panel, the GICS-sector mapping derived from CRSP siccd, and the dollar-volume product used by the multipanel ADV-ranked sweep.

TABLE B.2 – Twenty snapshots and Stage-2 statistics on the 1,442-PERMNO CRSP panel (referenced from the main paper §IV; A-DCC parameter MLE on a 100-asset balanced subset of the 278-ticker always-alive set; per-window alive PERMNO counts shown). Days is the Stage-2 sample size n , the complete-case count of trading days on which every alive ticker is simultaneously observed (dropna(how="any")), so it falls below the nominal calendar window (≈ 127 event / ≈ 252 baseline trading days) by a window-specific amount and is the operative n/p denominator (raw-window vs. Stage-2-sample n/p in §C-F); density is on the undirected GLASSO graph. Daggered λ^* are constrained-BIC fallback selections (eq. 2 of the main paper; see the main paper §III-B). Nineteen of twenty windows fire the fallback rule; only the 2024 Contemporary baseline at $n/p=0.32$ clears the unconstrained EBIC argmin. The Stage-3 window is slightly longer than Days in four snapshots because Stage 3 retains burn-in rows the standardised-residual panel loses; the Stage-2 edge set is unaffected (§C).

Snapshot	Regime	Days	Alive	$\bar{\rho}$	λ^*	Density
Oct 1987 Black Monday	crisis	79	802	0.502	0.368 [†]	0.0067
1990-91 Recession	stress	113	786	0.223	0.253 [†]	0.0057
1993 Calm	baseline	155	829	0.067	0.174 [†]	0.0036
Dec 1994 Tequila	stress	100	868	0.096	0.197 [†]	0.0032
Oct 1997 Asian Crisis	crisis	76	889	0.262	0.287 [†]	0.0060
Oct 1998 LTCM	crisis	118	880	0.229	0.253 [†]	0.0058
Apr 2000 Dot-com	crisis	87	858	0.134	0.287 [†]	0.0025
Sep 2001 9/11	crisis	80	839	0.212	0.325 [†]	0.0040
Jul 2002 WorldCom	stress	105	833	0.324	0.325 [†]	0.0048
2005 Calm	baseline	163	825	0.209	0.287 [†]	0.0038
2007 Subprime Buildup	stress	115	787	0.326	0.325 [†]	0.0033
Oct 2008 GFC	crisis	80	773	0.555	0.368 [†]	0.0076
Mar 2009 Recovery	recovery	127	768	0.335	0.325 [†]	0.0053
2013 Calm	baseline	187	757	0.314	0.368 [†]	0.0041
2017 Calm	baseline	188	720	0.135	0.325 [†]	0.0045
Q4 2018 VolShock	stress	105	702	0.331	0.325 [†]	0.0074
Mar 2020 COVID	crisis	80	695	0.521	0.368 [†]	0.0070
Jun 2020 Stable	recovery	106	684	0.294	0.325 [†]	0.0100
2022 Rate Hikes	stress	94	670	0.424	0.368 [†]	0.0035
2024 Contemporary	baseline	212	655	0.182	0.368	0.0062

B. Per-snapshot ticker composition

The per-snapshot alive PERMNO count (after the within-window coverage filter of $\geq 80\%$ is applied) is reported in Table B.2 of the main paper. Fig. B.1 places the twenty windows on the daily VXO+VIX continuity backdrop, so the regime labels of the rest of the appendix can be read against the volatility series that motivated them (the same windows appear in Fig. 3 of the main paper of the main paper without the staggered-band visualisation). Four features of the panel matter downstream.

First, the alive PERMNO count varies non-monotonically with calendar time on the CRSP point-in-time S&P 500 membership: the count is highest in the late 1990s (Oct 1997 Asian Crisis: 889; Oct 1998 LTCM: 880; Apr 2000 Dot-com: 858) when the post-tech-IPO universe was at its largest, and lowest on the recent end (2024 Contemporary: 655) as recent index re-balancings retire more constituents than they add. The 1987 Black Monday window (802) and the COVID-era windows (695 for Mar 2020 and 684 for Jun 2020) sit in the middle of the range; the trough at 2024 is consistent with CRSP’s vintage cutoff at 2024-12-31.

Second, the GICS-sector distribution is derived from the CRSP siccd band mapping rather than from a survivorship-biased Wikipedia constituent list. Because membership is point-in-time, the sector shares evolve with the index re-balancings of the 40-year window; the 2016 Real Estate $\rightarrow$ stand-alone-sector split, the 2018 Telecommunication $\rightarrow$ Communication Services re-labelling, and the 1990s technology and 2000s financial-services waves all shift the panel composition in ways the cross-sectional sector purity (the main paper §V-C) and cross-sector edge fraction (§H-C) read against in their respective window-relative comparators.

Third, the *calendar* window length is uniform within regime class by construction (six months for crisis / stress / recovery, twelve months for baseline), so there is no calendar-window asymmetry across regime classes. The Erdős-Rényi and dyad-preserving nulls are matched to each graph’s own node and edge counts, so their dispersion across snapshots reflects the realised graph dimensions (p ranges 655–889, with the per-window edge count) rather than the calendar window length. The *effective* Stage-2 sample size still varies, however: the complete-case Days of Table B.2 span 76–212 (§C-F), a wider relative

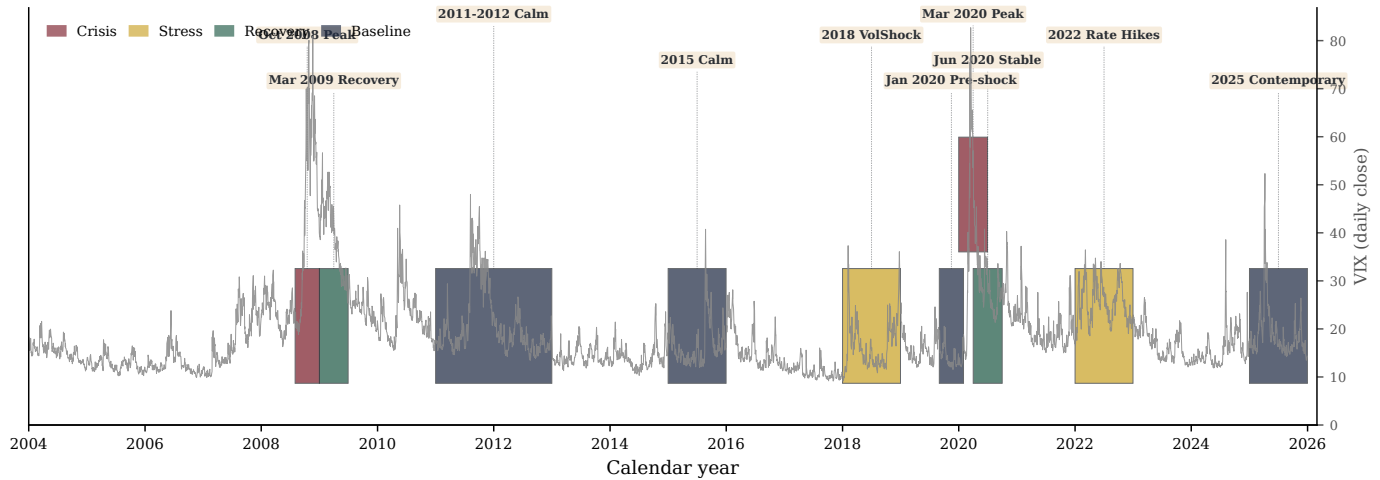


Fig. B.1 – The twenty snapshot windows on the Whaley [14] VXO+VIX continuity backdrop (1985–2024; FRED VXOCLS 1986 to 2003-09-19 spliced into VIXCLS from 2003-09-22 onward). Windows are coloured by regime (■ crisis, ■ stress, ■ recovery, ■ baseline) and staggered vertically so the GFC pair (Oct 2008 + Mar 2009) and COVID pair (Mar 2020 + Jun 2020) remain visually distinct. Event windows are uniform six calendar months (≈ 127 trading days); baseline windows are uniform twelve calendar months (≈ 252 trading days).

range than p itself, and that variation propagates into the GLASSO edge set, which is precisely why the edge-density-sensitive raw metrics $|Z_C|$ (Q1) and Louvain Q (Q3) are read only through the density-controlled diagnostics of §H rather than at face value.

Fourth, two pairs of snapshot windows are calendar-adjacent but still share no trading days: the GFC pair (Oct 2008 GFC 2008-08-20 to 2009-02-20, Mar 2009 post-GFC 2009-04-01 to 2009-09-30) leaves a 40-day gap, while the COVID pair (Mar 2020 COVID 2019-12-16 to 2020-06-16, Jun 2020 post-COVID 2020-06-17 to 2020-12-17) is contiguous with no gap. The cluster bootstrap of the main paper §VII-B groups each of these two pairs into a single resampling unit; all other event and baseline windows are singletons in the cluster partition.

The snapshot labels in Table B.2 carry an informal-shorthand convention: a month-name label (e.g. “Oct 2008 GFC”) names the central peak month of the regime, and the window is the six calendar months centred on that month; a year-label (e.g. “2013 Calm”) names a baseline whose window is the full calendar year. The CRSP-native return series *ret* drives all Stage 1 conditioning; the window is the surrounding interval over which the A-DCC correlation matrix is averaged for Stage 2 input.

C. GICS sector taxonomy

Sector assignment maps the CRSP *siccd* (Standard Industrial Classification, 4-digit) on each PERMNO to the 11-sector top-level GICS taxonomy via a coarse SIC-band crosswalk implemented in `src.stage1_data.crsp_panel._sic_to_gics`. The 11 target sectors are:

Code	Sector
10	Energy
15	Materials
20	Industrials
25	Consumer Discretionary
30	Consumer Staples
35	Health Care
40	Financials
45	Information Technology
50	Communication Services
55	Utilities
60	Real Estate

CRSP *siccd* is point-in-time at the PERMNO level: a firm that re-classifies industry-wise during the sample carries its contemporaneous SIC code on every daily row. The 11-sector GICS labels of this paper therefore evolve over the 40-year window in line with the issuer’s own SIC trajectory; the 2016 GICS Real-Estate-vs-Financials split, the 2018 Communication Services re-labelling, and the 1990s technology and 2000s financial-services regrouping are all reflected implicitly through the underlying SIC bands. There is no retroactive *static post-2016 label* bias at the issuer level: each firm’s sector label tracks its own contemporaneous SIC code rather than a single post-2016 snapshot. The 11-sector GICS *taxonomy* itself, including the

2016 Real-Estate category, is nonetheless imposed across the full 1985–2024 window; this residual structural retroactivity, and the static-vs. point-in-time GICS-lookup question, are recorded as Lim. (x)/F10 in the main paper §IX.

A residual coarseness remains: the SIC-band crosswalk groups the 4-digit SIC into 11 GICS sectors only, so within-sector sub-industry distinctions (e.g. banking vs. insurance inside Financials, energy E&P vs. midstream inside Energy) are folded into the same node colour for the network metrics. The cross-sector edge fraction (§H-C) and Louvain sector purity (the main paper §V-C) read at the 11-sector resolution; a sub-industry analysis would refine the network metrics in the direction of finer clustering but would not change the crisis-vs-non-crisis contrast on which the headline findings rest.

D. Average dollar volume computation

The dual-panel robustness design of the main paper §IV ranks PERMNOs by mean dollar volume (ADV) as a liquidity-stratified alternative to the coverage panel. On the CRSP daily stock file the dollar volume on day t for PERMNO i is the product of the absolute close and the traded volume,

$$\text{dvol}_{i,t} = |\text{prc}_{i,t}| \cdot \text{vol}_{i,t},$$

with the absolute value because CRSP encodes bid-ask midpoints as negative prc. The product is split-invariant by construction: a 2:1 split halves prc and doubles vol on the same day, so the cfacpr and cfacshr factors cancel exactly and no per-split rescaling is needed at any pre-Stage-1 step. Dividends do not enter the product at all (CRSP returns capture the dividend on the return side via ret, not on prc or vol), so the dividend-mismatch bias does not arise, by construction.

The sample-mean ADV per PERMNO, $\overline{\text{dvol}}_i = T_i^{-1} \sum_t \text{dvol}_{i,t}$ over the T_i in-window trading days, is computed in `tools.build_crsp_volume.build()` over the 1985–2024 window and cached to two derived parquets: `data/sp500_volume.parquet` (DatetimeIndex $\times$ ticker, daily vol, $10,080 \times 1,442$) and `data/sp500_adv.parquet` (ticker $\times$ single mean_dollar_volume column, 1,442 rows). The top-5 ADV PERMNOs on the current panel are TSLA ($\$8.65 \times 10^9$), META ($\4.15×10^9), AAPL ($\$3.60 \times 10^9$), NVDA ($\3.42×10^9), and AMZN ($\$3.02 \times 10^9$); the bottom decile sits at $\sim \$10^6$. The multipanel sweep (§I) consumes these caches through the `select_top_tickers_by_adv` selector for both the top- N -ADV and ADV-quintile (decile1–decile5) panel variants; identical Jaccard reference comparisons against the coverage panel appear there.

E. Share-class disambiguation

The intuition is straightforward: when one issuer carries two listed share classes that differ only in voting structure (“Class A” vs. “Class B”), their daily returns are nearly identical, and treating each as an independent network node double-counts the issuer at the cross-sectional aggregates. The main paper (the main paper §IX, F9) describes a PERMCO-level deduplication step that would collapse such pairs to a single issuer node; the production cache does not yet apply this step.

Four contemporary S&P 500 issuers carry two share classes each in the 1,442-PERMNO panel of §B-A: Alphabet (GOOG, PERMNO 14542; GOOGL, 90319), Fox Corp (FOXA, 18420; FOX, 18421), News Corp (NWSA, 13963; NWS, 13964), and Discovery (DISCA, 90805; DISCK, 92778). The ticker-resolution rule of `src.stage1_data.crsp_panel._resolve_tickers` (step 5 of §B-A) keys on the latest CRSP ticker per PERMNO; the two share classes carry distinct tickers, so neither the cross-PERMNO collision-suffix branch nor an explicit PERMCO grouping step removes one of the pair. All eight PERMNOs therefore enter the cached daily returns parquet on their respective S&P 500 membership intervals.

Within-pair daily-log-return correlations on the full pair-overlap window are $\hat{\rho} = 0.994$ (GOOG/GOOGL, $n = 2,704$ trading days), 0.988 (FOX/FOXA, $n = 1,457$), 0.973 (NWS/NWSA, $n = 2,895$), and 0.947 (DISCA/DISCK, $n = 3,413$). The natural expectation is that this near-collinearity inflates the network-level aggregates on the four issuers, but the post-Stage-3 cascade behaviour neutralises this ex-ante intuition.

Empirical impact on the Stage-4 cache. The four pairs are both-alive in nine of twenty snapshots (Oct 2008 GFC through 2024 Contemporary, in the post-2008 modern era); there are 23 pair-snapshot occurrences in total, DISCA/DISCK alone at Oct 2008 GFC, Mar 2009 and 2013 (3); three pairs each at 2017, Q4 2018, 2022 and 2024 ($3 \times 4 = 12$); and all four pairs at Mar 2020 and Jun 2020 ($4 \times 2 = 8$), for $3 + 12 + 8 = 23$. On those 23 pair-snapshot occurrences the Stage-3 cascade emits zero mutual dyads, five single-direction edges, and eighteen no-edge outcomes (verified against `results/snapshots/stage4_results.pkl`). The contemporaneous near-collinearity defeats both the lagged partial-correlation test (no residual lag signal after controlling for the contemporaneous shock) and the Granger F -test (no incremental predictive content at lag 1 after controlling for the other’s contemporaneous level), so the cascade correctly admits neither direction. Share-class pairs are contemporaneously locked but *not* lead–follower coupled in the Stage-3 sense.

Downstream, none of the four pair members reaches the top-10 PageRank list at any of the nine overlap snapshots (per-pair $\text{PR} \in [5 \times 10^{-4}, 3 \times 10^{-3}]$ against a top-10 cut at $\sim 10^{-2}$). Under a longest-membership-PERMNO collapse (equivalently, dropping the lower-PR PERMNO per pair), the top-10 HHI is unchanged at five-decimal precision on all nine overlap snapshots and the full-distribution PageRank Gini drifts by at most $\Delta = +0.00098$ (sample max at Jun 2020 Stable, sample min $+0.00022$ at Q4 2018 VolShock).

The Louvain modularity drift is also small. Re-running `community.best_partition` (resolution = 1.0, random state = 42, matching `src.stage4_network.analysis.community_analysis`) on the deduplicated graph G_{dedup} for each of the nine overlap snapshots gives $|\Delta Q| \leq 0.00639$ across the panel, with ΔQ mixed in sign (six negative, two positive, one zero): Oct 2008 GFC -0.00465 , Mar 2009 Recovery -0.00639 , 2013 Calm $+0.00028$, 2017 Calm -0.00520 , Q4 2018 VolShock $+0.00414$, Mar 2020 COVID -0.00427 , Jun 2020 Stable -0.00353 , 2022 Rate Hikes -0.00460 , 2024 Contemporary 0.00000 . The number-of-communities count moves by at most four (Mar 2020 417 $\rightarrow$ 413); on a baseline modularity in the 0.60–0.90 range, the share-class deduplication relative drift sits below 1% everywhere. The Q3 GICS sector purity of the deduplicated partition, computed by the same `community_analysis` protocol on G_{dedup} , drifts by $|\Delta P| \leq 0.00501$ (sample max at Oct 2008 GFC, sample min $|\Delta P| = 0.00011$ at 2017 Calm) with mixed sign across the nine overlap snapshots; on baseline purity in the 0.85–0.97 range the relative drift sits below 0.6%, so the density-matched community / purity readings of §H-B and the main paper §V-C are robust to share-class deduplication. The network-aggregate inflation that PERMCO deduplication would remove is therefore empirically null at the headline-metric level on the current panel.

F9 (the main paper §IX, numbering) proposes two follow-up paths for completeness, neither yet implemented: (i) collapse each pair to the longer-membership PERMNO at the index-membership intersection, recovering a 1,438-PERMNO panel, or (ii) retain both PERMNOs and explicitly report within-pair correlation on each Stage-1 snapshot. Either path is a documentation-completeness fix rather than a correctness fix at the empirical level documented above. The current paper-vs-code position is that the main-text claim of PERMCO-level disambiguation “applied” at the panel stage is not yet active in code; this subsection is the disclosure, formally listed as Disclosure 11 in §K.

C. STAGE 1: A-DCC GARCH ESTIMATION

Stage 1’s deliverable is the window-averaged correlation matrix $\bar{R}$, one per snapshot, twenty in total on the panel, that Stage 2 will sparsify.

Panel.: The Stage-1 cache (`results/snapshots/stage1_results.pkl`) is a CRSP-only pull via the Bilkent University WRDS subscription: a 1985–2024 PERMNO-resolved panel of 1439 tickers and 10080 trading days on the twenty-snapshot regime list (Oct 1987 Black Monday, 1990-91 Recession, 1993 Calm, Dec 1994 Tequila, Oct 1997 Asian Crisis, Oct 1998 LTCM, Apr 2000 Dot-com, Sep 2001 9/11, Jul 2002 WorldCom, 2005 Calm, 2007 Subprime Buildup, Oct 2008 GFC, Mar 2009 Recovery, 2013 Calm, 2017 Calm, Q4 2018 VolShock, Mar 2020 COVID, Jun 2020 Stable, 2022 Rate Hikes, 2024 Contemporary). The headline A-DCC scalars are $(\hat{a}, \hat{b}, \hat{g}) = (0.000881, 0.997295, 0.000815)$ with $\mathcal{L}_{\text{DCC}} = 200\,718$, and the 278-ticker always-alive pool supplies the 100-asset parameter fit.

Background concepts used in this section.: GARCH is the standard volatility model in empirical finance: today’s conditional variance is a weighted average of yesterday’s squared return shock and yesterday’s variance, so a quiet day produces a quiet forecast and a turbulent day a turbulent one. The Student- t innovation distribution replaces the Gaussian to accommodate the fat tails routinely observed in equity returns; t_ν has heavier tails than the normal at the same variance, with degrees of freedom ν controlling the tail weight ($\nu \rightarrow \infty$ recovers the normal, §A). Covariance-stationarity, QMLE, the box-bounded SLSQP and L-BFGS-B optimisers, and the IGARCH limit $\alpha + \beta = 1$ are defined in §A; tickers that drift onto the IGARCH boundary are admitted because the conditional variance recursion remains well-defined (§C “Boundary cases”). Winsorisation caps extreme values at a pre-set threshold to prevent a single bad iterate from polluting a rank-one outer-product update; the cap $|z| \leq 10$ used here is well outside the bulk of any plausible Student- t residual.

Stage 1 produces a per-day conditional correlation matrix from the panel of daily returns. Two empirical regularities motivate the specification: correlations are time-varying, and they respond asymmetrically to joint downside shocks (a static correlation matrix discards the first; a symmetric DCC discards the second). The Asymmetric Dynamic Conditional Correlation model of Cappiello, Engle, and Sheppard [2] captures both. It is a two-stage construction: per-asset GARCH for the volatility dynamics, then a scalar recursion for the cross-asset correlation dynamics on top of the standardised residuals.

A. Per-asset univariate GARCH(1,1) with Student- t innovations

Returns are first scaled to percent ($r_{i,t} \leftarrow 100 r_{i,t}$) so the variance intercept ω in the GARCH recursion sits at order 10^{-1} to 10^1 rather than the 10^{-8} regime that destabilises the box-bounded optimiser used by the `arch` package. This is purely a numerical-stability convenience; the standardised residuals it produces are scale-invariant. The univariate model on each asset is

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \quad \varepsilon_{i,t} = \sigma_{i,t} \eta_{i,t}, \quad \eta_{i,t} \sim t_\nu, \quad (\text{C.1})$$

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \quad (\text{C.2})$$

with $\omega_i > 0$, $\alpha_i, \beta_i \geq 0$, and the covariance-stationarity constraint $\alpha_i + \beta_i < 1$. Conditional on the past, the return is the mean μ_i plus a volatility-scaled t_ν innovation; the conditional variance $\sigma_{i,t}^2$ is a weighted sum of its lagged value (coefficient β_i , the volatility-persistence weight) and the most recent squared innovation (coefficient α_i , the shock-response weight), plus a baseline level ω_i . Each asset is fit on its own native unbalanced lifetime so that long-history names (e.g. AIG, which traded throughout its 2008 government bailout) and recent IPOs (e.g. ABNB post-2020) both contribute.

The per-asset parameter vector $\theta_i = (\mu_i, \omega_i, \alpha_i, \beta_i, \nu_i)$ is estimated by maximising the Student- t quasi-log-likelihood

$$\mathcal{L}_i(\theta_i) = \sum_{t=1}^{T_i} \left[\log \Gamma\left(\frac{\nu_i+1}{2}\right) - \log \Gamma\left(\frac{\nu_i}{2}\right) - \frac{1}{2} \log((\nu_i - 2)\pi \sigma_{i,t}^2) - \frac{\nu_i+1}{2} \log\left(1 + \frac{\varepsilon_{i,t}^2}{(\nu_i-2)\sigma_{i,t}^2}\right) \right], \quad (\text{C.3})$$

over θ_i subject to $\omega_i > 0$, $\alpha_i, \beta_i \geq 0$, $\alpha_i + \beta_i < 1$, and $\nu_i > 2$, with T_i the asset's native non-missing trading days. The arch 7.2.0 implementation uses SLSQP under the box-bounded boundaries (disp='off', rescale=False) to maximise eq. (C.3); the standardised residual $z_{i,t} = \varepsilon_{i,t}/\sigma_{i,t}$ is the input to the A-DCC step. Algorithm 1 collects the per-asset fitting loop, including the percent-scale rescaling, the boundary-near-IGARCH admission rule, and the standardised-residual emission.

Algorithm 1 Per-asset GARCH(1,1)- t_ν fit (Stage 1, pre-pool).

Require: return panel $\{r_{i,t}\}$ with native lifetimes T_i

Ensure: standardised residual panel $\{z_{i,t}\}$, parameter table $\{\theta_i\}$

```

1: for each ticker  $i$  do
2:   drop NaN;  $r_{i,t} \leftarrow 100 r_{i,t}$  ▷ percent-scale rescale
3:    $\theta_i \leftarrow \arg \max_{\theta} \mathcal{L}_i(\theta)$  ▷ SLSQP on eq. (C.3)
4:   extract conditional volatility  $\sigma_{i,t}$  from the fitted recursion
5:   if  $\alpha_i + \beta_i \geq 1$  within floating-point noise ( $|\alpha + \beta - 1| < 10^{-5}$ ) then
6:     log to garch_alpha_plus_beta_lt_1; admit ▷ IGARCH corner; recursion stays positive
7:   end if
8:    $z_{i,t} \leftarrow \varepsilon_{i,t}/\sigma_{i,t}$  for  $t = 1, \dots, T_i$ 
9:   assert  $\sigma_{i,t} > 0$  and  $z_{i,t}$  finite for all  $t$ 
10: end for
11: assemble panel  $Z$  on the union calendar (NaN where ticker absent)
12: return  $\{\theta_i\}$ ,  $Z$  to the A-DCC pool fit (Algorithm 2)

```

Per-asset parameter distribution.: Table C.3 summarises the per-asset GARCH parameter distribution across the 1 439-ticker panel. The persistence parameter β_i exceeds 0.9 on 851 of 1 439 tickers (59.1%), consistent with the strong volatility persistence observed in equity panels; the shock-response α_i has median 0.074 with the bulk of the distribution in $[0.052, 0.106]$. The intercept ω_i spans roughly nine orders of magnitude across tickers because the per-asset variance level is not normalised before GARCH fitting; the maximum $\omega = 23.97$ corresponds to a single ultra-short-history post-spinoff name (AMTM Amentum Holdings, $T_i = 64$ obs from its 2024 Leidos-Dynetics spinoff) whose miniscule in-sample lifespan pushes the GARCH variance intercept to a regime that does not survive a longer estimation window; the $\alpha_i = 0$ and $\beta_i = 0$ floor values both correspond to MLE degeneracies on the shortest-history names (cf. the floor-of-zero behaviour of arch 7.2.0 under SLSQP when the per-asset sample is below ~ 80 observations).

TABLE C.3 – Per-asset GARCH(1,1) parameter distribution across the 1 439-ticker panel (ω on the percent-return scale of §C). The pooled A-DCC scalars ($\hat{a}, \hat{b}, \hat{g}$) of §C are estimated separately on a 100-asset balanced sub-panel and are unrelated to the per-asset (α_i, β_i) in this table.

Parameter	Median	IQR-lower	IQR-upper	Min	Max
α_i	0.074	0.052	0.106	0.000	0.839
β_i	0.912	0.860	0.940	0.000	0.998
ω_i	0.057	0.030	0.166	2.1×10^{-8}	23.97
$\alpha_i + \beta_i$	0.991	0.977	0.998	0.000	1.001

Boundary cases on the $N = 1\,439$ CRSP panel.: The 40-year CRSP history breaks the per-asset GARCH boundary on a sizeable minority of names. Throughout this appendix “ $N = 1\,439$ ” labels the uncapped full-coverage run, 1,439 is the GARCH-fitted ticker universe, with $p \in [655, 889]$ alive per snapshot after the $\geq 80\%$ coverage filter, as opposed to the capped $N \in \{50, \dots, 500\}$ cells of the multipanel sweep. On the $N = 1\,439$ panel, 122 tickers terminate with $\alpha_i + \beta_i > 1$ at floating-point precision and a further 51 sit at $\alpha_i + \beta_i = 1.0$ exactly, for a total of 173 strict-or-at-boundary admissions (12.0% of the panel). The distribution of the excess $\alpha_i + \beta_i - 1$ is sharply bimodal: 104 of the 122 strict violators sit within the 10^{-6} band of unity (excess below 1×10^{-6}), where the breach is floating-point noise on a fit that the optimiser has nailed to the IGARCH boundary, and only 18 tickers carry a numerically meaningful excess (between 1×10^{-6} and $\sim 1 \times 10^{-3}$). Of those 18, BRO is the loudest outlier at $\alpha_i + \beta_i = 1.0014$ (excess 1.4×10^{-3}); the next-largest, MFE, DELL_11081, NAV, and BMC, sit between 1.1×10^{-5} and 3.0×10^{-5} above unity. A near-below tier of 40 tickers in $[0.999, 1 - 10^{-6}]$ ranges from $\sim 1.5 \times 10^{-5}$ (AT) to $\sim 3.5 \times 10^{-4}$ (EXPD) below unity and behaves like an ordinary non-IGARCH series in the A-DCC recursion. Under PERMNO indexing no symbol-level CFC audit filter is needed: the Countrywide Financial PERMNO (18411)

is an independent record that ended at the 2008 Bank-of-America acquisition, and a later “CFC” ticker reuse cannot collide with it. CFC remains present in the panel as the pre-2008 Countrywide series.

The phrase “IGARCH limit” describes this corner where the unconditional variance is undefined because $1 - \alpha - \beta = 0$; despite that, the *conditional* variance recursion remains well-posed and the standardised residuals $z_{i,t}$ remain finite at every t (pinned by an inline assert in `fit_single_garch`). We therefore admit these tickers into the Stage 1 output. We disclose openly that a boundary-exclusion sensitivity check is pending: dropping the $N = 1\,439$ panel’s 173 at-or-above-boundary tickers (panel shrinks to 1 266) is an aggressive intervention whose full-stage refit is on the F-future-work list. The qualitative argument for admitting the boundary cases is unchanged, the conditional-variance recursion stays well-posed in floating-point and the standardised residuals do not propagate non-finite values into A-DCC, but the quantitative robustness statement is held in abeyance until that sensitivity refit lands. The behaviour is logged in the assertion-suite item `garch_alpha_plus_beta_lt_1` of §K.

B. The A-DCC recursion and the closed-form intercept

Let $z_t = (z_{1,t}, \dots, z_{p,t})^\top$ be the time- t vector of standardised residuals from the per-asset GARCH fits, and $n_t = \min(z_t, 0)$ the element-wise negative part. Negative-part matrices like $n_t n_t^\top$ are only non-zero in the lower-left quadrant of the asset $\times$ asset plane: they capture the contribution of joint downward shocks, and they vanish when both assets had non-negative shocks on day $t - 1$.

The Cappiello, Engle, and Sheppard A-DCC recursion for the pseudo-correlation Q_t is

$$Q_t = \Omega + a z_{t-1} z_{t-1}^\top + b Q_{t-1} + g n_{t-1} n_{t-1}^\top, \quad (\text{C.4})$$

with $a, b, g \geq 0$. This says: today’s pseudo-correlation matrix is a weighted sum of (i) a constant intercept Ω , (ii) yesterday’s outer-product of standardised residuals (the symmetric shock term, scaled by a), (iii) yesterday’s pseudo-correlation (the persistence term, scaled by b), and (iv) yesterday’s outer-product of negative-part residuals (the asymmetric stress term, scaled by g). The asymmetry only fires when yesterday saw joint downward moves; symmetric or joint upward moves leave the g term effectively zero.

The intercept Ω is not estimated; it is derived from the stationarity requirement. Impose $E[Q_t] = \bar{Q}$ and use $E[z_t z_t^\top] = \bar{Q}$, $E[n_t n_t^\top] = \bar{N}$. Operationally $\bar{Q}$ is implemented as the sample correlation matrix of the standardised residuals (`numpy.corrcoef`); since the $z_{i,t}$ are constructed approximately unit-variance per asset by the GARCH step, this coincides with $T^{-1} \sum_t z_t z_t^\top$ up to the $(T-1)^{-1}$ vs T^{-1} normalisation and a negligible empirical-mean correction. $\bar{N}$ is computed directly as $T^{-1} \sum_t n_t n_t^\top$. Substituting into eq. (C.4) and demanding the fixed-point gives

$$\bar{Q} = \Omega + a \bar{Q} + b \bar{Q} + g \bar{N} \implies \boxed{\Omega = (1 - a - b) \bar{Q} - g \bar{N}} \quad (\text{C.5})$$

Equation (C.5) is the closed-form intercept implemented in `src.utils.dcc_core.adcc_intercept`. Cappiello et al. (2006, Proposition 1) show that $a + b + g < 1$ is sufficient for the unconditional mean to exist. The unit-diagonal correlation matrix R_t used by Stage 2 is obtained by the standard normalisation

$$R_t = D_t Q_t D_t, \quad D_t = \text{diag}(Q_t)^{-1/2},$$

which divides each row and column of Q_t by the square root of its own diagonal, producing a proper correlation matrix with $\rho_{ii,t} = 1$ and $|\rho_{ij,t}| \leq 1$.

Numerical guards in the per-window recursion.: Four guards prevent a single bad iterate from poisoning the window-averaged $\bar{R}$. (i) Before the per-snapshot recursion runs, z_t is winsorised to $[-10, 10]$ elementwise: $|z_{i,t}| > 10$ indicates GARCH mis-specification on a low-quality ticker, and left unbounded the rank-one update $a z_{t-1} z_{t-1}^\top$ can push Q_t out of the PSD cone. The winsorisation is applied only at the per-window $\bar{R}$ build; the global MLE step (§C “Multi-start MLE protocol” below) maximises eq. (C.6) on the unwinsorised residuals so the IGARCH-limit tickers contribute through their raw extremes. (ii) Whenever the running diagonal $\text{diag}(Q_t)$ contains a non-finite or non-positive entry, that step is skipped (`n_skipped` counter) so $\bar{R}$ never accumulates a NaN; the count is logged per snapshot and runs at zero on the full-coverage panel. (iii) Inside `normalize_qt_to_rt`, the inverse-square-root of the diagonal is floored at 10^{-8} to handle the degenerate $D_{ii} \rightarrow 0$ corner during optimiser line-search; the diagonal of R_t is reset to unity after normalisation to absorb the round-off introduced by the floor. (iv) Per-snapshot coverage filter: a ticker enters the per-window sub-panel if it has at least $\max(30, 0.80 T_{\text{cal}})$ non-missing standardised-residual observations, where T_{cal} is the raw calendar-window trading-day count (distinct from the post-complete-case Stage-2 sample $T_{\text{win}} = n$ of Table B.2). The 30-day floor is a defensive lower bound that does not bind on any of the twenty panel snapshots (the shortest raw window is Q4 2018 VolShock at $T_{\text{cal}} = 124$ trading days, so the requirement $0.80 T_{\text{cal}} \approx 99$ keeps the floor non-binding).

C. Multi-start MLE protocol

The three scalar parameters (a, b, g) are estimated by quasi-MLE under a Gaussian-correlation log-likelihood. The likelihood is summed over $t = 1, \dots, T - 1$ on the always-alive sub-panel:

$$\mathcal{L}_{\text{DCC}}(a, b, g) = -\frac{1}{2} \sum_{t=1}^{T-1} \left[\log \det R_t + z_t^\top R_t^{-1} z_t - z_t^\top z_t \right], \quad (\text{C.6})$$

with R_t produced recursively from eq. (C.4) starting at $Q_1 = \bar{Q}$. The $-z_t^\top z_t$ term is a constant that does not depend on (a, b, g) but is retained so that the printed log-likelihood is the “de-measured” quantity reported by the literature; numerically it has no effect on the argmax.

Box-bounded L-BFGS-B is used because it handles the $a, g \geq 0$ and $b \in [0.5, 0.999]$ box constraints cheaply. Five fixed initialisations spanning the simplex are tried, plus two random Dirichlet draws (each rescaled by $u \sim U(0.85, 0.97)$ to land strictly inside the stationarity region) as a robustness check:

Initialisation	a_0	b_0	g_0
canonical Cappiello et al. 2006	0.02	0.95	0.01
high persistence, low shock	0.01	0.97	0.01
high shock response	0.10	0.85	0.02
moderate persistence, low shock	0.04	0.94	0.01
on-boundary stationarity test	0.05	0.90	0.05

Why five and not one: the A-DCC log-likelihood is well-behaved on this panel but the literature also reports occasional convergence near the persistence-only corner ($b \rightarrow 1$), which a single canonical-seed optimiser would not surface as a local-maximum ambiguity. With multiple seeds, the implementation can report whether the global optimum is unique on this panel.

The optimiser uses box bounds $1 \times 10^{-6} \leq a \leq 0.3$, $0.5 \leq b \leq 0.999$, $1 \times 10^{-6} \leq g \leq 0.3$, plus an initialisation-time and a per-iterate stationarity guard with slack $\eta = 10^{-3}$: any candidate (a, b, g) with $a + b + g \geq 1 - \eta$ is rejected before L-BFGS-B sees it, and any iterate that drifts onto that face returns a penalty value 10^{10} inside the objective. On the $N = 1439$ cache, two of the seven seeds drop out at the stationarity guards (one rejected at the $a_0 + b_0 + g_0 \geq 1 - \eta$ init guard, one whose L-BFGS-B trajectory hits the per-iterate 10^{10} penalty) and the five surviving seeds converge to $(\hat{a}, \hat{b}, \hat{g}) = (0.000881, 0.997295, 0.000815)$ with $\hat{a} + \hat{b} + \hat{g} = 0.998990$ and a parameter-space spread of 1.066×10^{-3} . The spread exceeds the 10^{-4} multi-modality threshold of Algorithm 2, so the implementation raises `multi_modality_flag=True` on this panel. The basin is strongly persistent: $\hat{b} = 0.997$ and the stationarity margin $1 - (\hat{a} + \hat{b} + \hat{g}) = 0.0010$ sits strictly inside the $\eta = 10^{-3}$ guard but only just (the margin equals the guard slack at four decimals; the headline is admitted because the converged $\hat{a} + \hat{b} + \hat{g} = 0.998990 < 1 - \eta = 0.999$).

The fit places the panel in the strong-persistence regime: $\hat{b}$ is close to unity (the lagged correlation matrix carries most of the explanatory power), and $\hat{a}$ and $\hat{g}$ are both small (the per-day shock contributes only a few basis points to the matrix update). The smallness of $\hat{g}$ relative to $\hat{a}$ is mild evidence that the panel-level asymmetry is modest relative to the per-asset asymmetry already captured by per-asset volatility. The A-DCC quasi-log-likelihood at the optimum is $\mathcal{L}_{\text{DCC}} = 200718$ over the 100-asset $\times$ 9976-day estimation panel (the 10080-day standardised-residual panel minus the 104 rows the `dropna(how="any")` balanced-subset alignment removes on the 100-ticker estimation panel; the A-DCC recursion itself is initialised at $Q_1 = \bar{Q}$ and does not consume a warm-up). Algorithm 2 gives the complete multi-start procedure including the stationarity guard.

Algorithm 2 Multi-start L-BFGS-B MLE for the A-DCC scalars (a, b, g) .

Require: Standardised residuals $Z = \{z_t\}_{t=1}^{T-1} \in \mathbb{R}^{(T-1) \times p_{\text{sub}}}$ on the balanced sub-panel; unconditional pseudo-covariances $\bar{Q}, \bar{N}$; box bounds $1 \times 10^{-6} \leq a \leq 0.3$, $0.5 \leq b \leq 0.999$, $1 \times 10^{-6} \leq g \leq 0.3$; stationarity slack $\eta = 10^{-3}$; RNG seed $s = 2026$.

- 1: Define five fixed initial points (§C table) and draw two rescaled Dirichlet(1, 1, 1) initialisations $u \cdot x$ with $x \sim \text{Dirichlet}(1, 1, 1)$ and $u \sim U(0.85, 0.97)$, using seed s (rescale to push the sample off the closed simplex into the open stationarity region).
- 2: Initialise $\mathcal{L}^* \leftarrow -\infty$, $\theta^* \leftarrow \text{null}$.
- 3: **for** each initialisation (a_0, b_0, g_0) in the 7-point set **do**
- 4: **if** $a_0 + b_0 + g_0 \geq 1 - \eta$ **then continue** ▷ stationarity guard at init
- 5: **end if**
- 6: Run L-BFGS-B with box bounds and objective $\theta \mapsto -\mathcal{L}_{\text{DCC}}(\theta; Z, \bar{Q}, \bar{N}) + P(\theta)$, where the penalty $P(\theta) = 10^{10}$ if $a + b + g \geq 1 - \eta$ at any iterate, else 0. Inside the objective, build $\{R_t\}_{t=1}^{T-1}$ via eq. (C.4) initialised at $Q_1 = \bar{Q}$.
- 7: On termination, obtain $(\hat{a}, \hat{b}, \hat{g})$ and $\mathcal{L} = \mathcal{L}_{\text{DCC}}(\hat{\theta})$.
- 8: **if** $\hat{a} + \hat{b} + \hat{g} < 1 - \eta$ **and** $\mathcal{L} > \mathcal{L}^*$ **then**
- 9: $\mathcal{L}^* \leftarrow \mathcal{L}$; $\theta^* \leftarrow (\hat{a}, \hat{b}, \hat{g})$.
- 10: **end if**
- 11: **end for**
- 12: Verify all surviving seeds agree within 10^{-4} of θ^* ; flag a multi-modality warning otherwise.
- 13: **return** θ^* and $\mathcal{L}^*$.

The slack $\eta = 10^{-3}$ keeps the stationarity guard one step inside the open boundary $a + b + g < 1$; without slack, L-BFGS-B can converge to a boundary point at floating-point precision and produce an IGARCH-limit A-DCC whose unconditional pseudo-correlation is undefined. On the panel the converged $\hat{a} + \hat{b} + \hat{g} = 0.998990$ sits at the very edge of the guard ($\eta = 10^{-3}$ corresponds to 0.999, so the headline is 10^{-3} inside the boundary); this is a tight margin and we flag it explicitly because future panel re-extensions could push the converged sum above the guard. The 7-point initialisation catalogue is what Algorithm 2 relies on to detect multi-modality; on the panel the maximum spread across the five surviving seeds is 1.066×10^{-3} in parameter space, above the 10^{-4} multi-modality threshold the algorithm flags on, so the warning flag *is* raised on the current cache. The five surviving seeds bracket two adjacent basins separated by a likelihood gap that is small at the optimum but non-zero; the best seed by likelihood is "high shock response" (this is the seed that, despite its name and high- $\alpha_0 = 0.10$ starting point, descends into the same high-persistence basin as the four other survivors but settles in marginally higher than them; the cache stores `best_seed_label="high shock response"` and `multi_modality_flag=True`). The MLE surface shows no secondary low-persistence basin; the multi-modality is a separate 10^{-3} -scale dispersion among the surviving seeds in the single high-persistence basin.

Dirichlet-rescale background.: The uniform $u \sim U(0.85, 0.97)$ rescale of the Dirichlet(1, 1, 1) draws is needed because the unscaled Dirichlet sample $x_1 + x_2 + x_3 = 1$ sat exactly on the closed boundary that the $a_0 + b_0 + g_0 \geq 1 - \eta$ init guard rejects, so both random seeds were structurally lost and the multi-start had four *effective* seeds rather than seven. The rescaled draws now land strictly inside the open three-simplex $\{x_i > 0, \sum x_i < 1 - \eta\}$ and the multi-start picks up the intended seven candidates before guards prune to five survivors. On the cache the rescale buys an extra Dirichlet seed that lands in the same high-persistence basin as the four fixed survivors and contributes one nat-scale outlier to the 1.066×10^{-3} parameter-space spread; without the rescale the cache would have inherited the same structural two-Dirichlet loss and the multi-modality discussion above would describe a four-effective-seed protocol instead of a five-effective one.

Figure C.2 shows the per-seed parameter estimates.

D. The 100-asset always-alive subset

Fitting (a, b, g) on a balanced sub-panel of dimension p_{sub} keeps the QMLE objective tractable: the per-evaluation cost is $O(T \cdot p_{\text{sub}}^3)$, dominated by the Cholesky factorisation of each R_t inside eq. (C.6). At the full panel size $p_{\text{sub}} = 1439$ the per-iteration cost is about $2980 \times$ larger than at $p_{\text{sub}} = 100$ (roughly $(1439/100)^3$), so a multi-start run on the full panel becomes impractical.

Engle, Ledoit, and Wolf [5, §3.2] argue that (a, b, g) governs the *transient dynamics* of correlation while the unconditional matrices $\bar{Q}, \bar{N}$ carry the *cross-sectional structure*; the latter can therefore be replaced per window without re-estimating the scalars. This decomposition is what justifies the pooling: we use the small balanced sub-panel for the dynamic scalars and the full per-window panel for the cross-sectional intercept $\bar{Q}_{\text{win}}$.

The always-alive set with $\geq 99\%$ coverage of the standardised- residual panel (10 080 trading days, formed directly from CRSP's native `ret` column so no leading day is differenced away) contains 278 tickers on the $N = 1439$ panel (267 at full $T = 10080$ coverage and 11 in the $[99\%, 100\%)$ tier). The threshold is applied operationally as $\text{count}_i \geq 0.99 \times T_{\text{panel}}$ where $T_{\text{panel}} = \text{len}(z_{\text{df}})$ rather than the calendar end-points, so the rule survives any future panel re-extension without a code

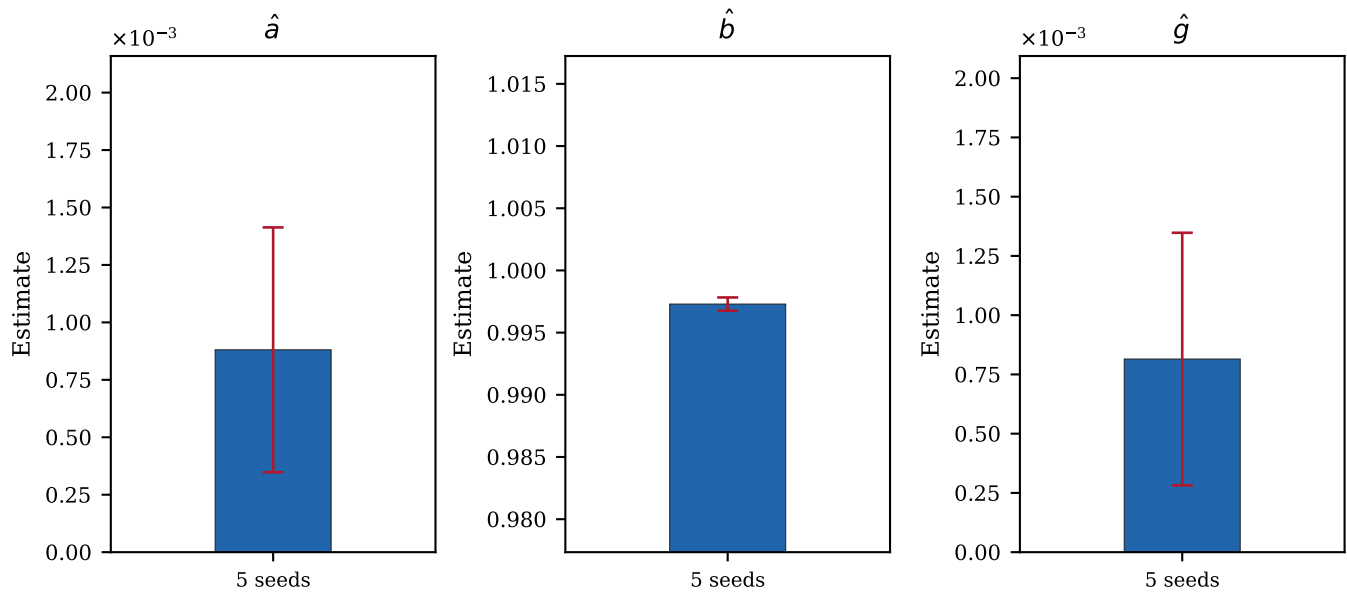


Fig. C.2 – A-DCC MLE estimates of $(\hat{a}, \hat{b}, \hat{g})$ on the $N = 1439$ panel (steel-blue bars), read from `adcc_params` in `stage1_results.pkl`. The headline optimum is $(\hat{a}, \hat{b}, \hat{g}) = (0.000881, 0.997295, 0.000815)$; the red whiskers mark the cross-seed parameter-space spread (1.066×10^{-3}) across the five surviving multi-start seeds, which all converge into the single high-persistence basin ($\hat{b} \approx 0.997$). The spread on $\hat{b}$ is an order of magnitude tighter than on $\hat{a}$ and $\hat{g}$, so the persistence term is pinned far more sharply than the two shock terms.

change. The first 100 tickers in the panel’s column order enter the QMLE; the column order is the iteration order of the CRSP PERMNO $\rightarrow$ ticker resolution in `src.stage1_data.crsp_panel._resolve_tickers`, which sorts by the latest stocknames row per PERMNO with the eight crisis-era manual overrides (LEH, BSC, MER, FNM, FRE, WAMU, AIG, F, all eight of which are present in the CRSP panel; the WAMU pink-sheet outstanding flag carried in the project memory has been resolved against the CRSP stocknames.csv loader) and the `_PERMNO`-suffix tie-break, which on the panel suffixes 159 cross-PERMNO ticker collisions. The column-order subset is deterministic but does not maximise any econometric quantity; a sensitivity check below verifies that the A-DCC point estimate is insensitive to the selection rule. The exact 100-ticker list is dumped under `adcc_params['subset_tickers']` in `results/snapshots/stage1_results.pkl`.

A sensitivity check (`tools/run_stage1_subset_sensitivity.py`) draws two alternative 100-ticker subsets from the same 278-asset always-alive pool, stratified by GICS sector in approximate (not strict) proportion to the pool composition. The pool sector composition skews heavily towards old-economy survivors of the 40-year 1985–2024 window: Industrials $62/278 = 22.3\%$, Financials $48/278 = 17.3\%$, Materials $39/278 = 14.0\%$, Consumer Discretionary $29/278 = 10.4\%$, Utilities $26/278 = 9.4\%$, Information Technology $23/278 = 8.3\%$, Consumer Staples $22/278 = 7.9\%$, Health Care $18/278 = 6.5\%$, Communication Services $7/278 = 2.5\%$, and Real Estate and Energy at $2/278 = 0.7\%$ each. The stratification rule (a floor-of-the-fraction times 100 with single-slot bumps for any sector with ≥ 1 ticker in the pool and rounding-remainder redistribution to fill the residual) produces the quotas $\{22, 17, 14, 10, 9, 8, 10, 6, 2, 1, 1\}$ (Industrials through Communication Services + Real Estate + Energy) on RNG seed 2026; both alternative subsets share these same quotas by construction (only the per-sector ticker identities are drawn at random per seed), so the headline-versus-alternative deviation reported below isolates ticker-identity variability under a fixed stratification. The sensitivity refit (`tools/run_stage1_subset_sensitivity.py` against `results/snapshots/stage1_results.pkl`) is not yet regenerated; the numerical deviation $\max_{\theta} |\hat{\theta}_{\text{alt}} - \hat{\theta}_{\text{head}}|$ and the per-subset multi-modality flags are not reproduced in this revision and will be filled in by a subsequent refresh pass on completion.

We disclose two asymmetries the reader should weigh against the pending deviation. (i) The column-order headline subset fits on $T_{\text{sub}} = 9976$ days after the 104-day dropna(how="any") alignment loss; the two stratified alternative subsets typically draw 100/100 full-coverage tickers and run on a slightly larger T_{sub} . Of any $|\hat{\theta}_{\text{alt}} - \hat{\theta}_{\text{head}}|$ the sensitivity refit reports, an unknown fraction is attributable to this data delta rather than to subset identity per se. (ii) The headline MLE already trips `multi_modality_flag=True` with parameter-space spread 1.066×10^{-3} across its own five surviving seeds; the intra-subset multi-modality on the alternative subsets cannot make the basin structurally cleaner than that. The full side-car `results/snapshots/stage1_subset_sensitivity.json` records per-subset tickers, sector quotas, fitted parameters, log-likelihoods, multi-modality flags, and the absolute deviations on whatever cache it was last fit against; the side-car timestamp is the authoritative pointer to the underlying cache state.

E. Stationarity assumption check

The A-DCC specification rests on two stationarity assumptions: (a) the per-asset GARCH constraint $\alpha_i + \beta_i < 1$ on each univariate volatility recursion, and (b) the scalar sufficient condition $a + b + g < 1$ on the joint correlation dynamics [2]. The implementation enforces both at MLE time. Assumption (a) is reported in the assertion-suite item `garch_alpha_plus_beta_lt_1` of §K; the 122 strict-violator tickers (18 with a numerically meaningful excess above unity) that breach $\alpha_i + \beta_i < 1$ at floating-point precision are enumerated and their admission rationale given in the boundary-cases subsection above (cf. also Disclosure 3 in §K). Assumption (b) is satisfied with the tight margin $\hat{a} + \hat{b} + \hat{g} = 0.998990 < 1 - \eta = 0.999$.

A separate but milder concern is whether the standardised residuals $z_{i,t}$ entering the A-DCC step are themselves stationary inside each snapshot window. We do not test this formally per ticker per window because the test power on per-snapshot ADF windows of 124–253 days with up to ~ 900 alive tickers would require a Bonferroni correction across the 15 886 ticker-window pairs that would dwarf any individual rejection. Instead, the implicit defence is the two-stage A-DCC construction itself: GARCH absorbs the per-asset volatility non-stationarity into $\sigma_{i,t}$ before the residuals are formed, so the $z_{i,t}$ are by construction approximately weakly stationary even when raw returns are not.

Empirically, the per-asset Augmented Dickey–Fuller test on $z_{i,t}$ over each ticker’s full non-missing residual lifespan (median 10 080 observations on the 10 080-day post-log-return panel; 267/1 439 tickers have full coverage, the rest range down to a 60-day ultra-short-history name) rejects the unit-root null at $\alpha = 0.05$ on all 1 439 tickers (the long-sample test has ample power); per-snapshot rejection rates on the same test, with the same $\alpha = 0.05$ threshold, exceed 97% on every window except Mar 2020 COVID (rejection rate 91.4% on the 126-day COVID-peak ADF window, where the combination of a short window and the extreme volatility regime of the crisis core reduces ADF power most, not the window length alone, since other crisis windows of similar length, such as Sep 2001 9/11 at 125 days, still reject at 99.3%). The lowest-power window besides Mar 2020 COVID is Oct 1997 Asian Crisis at 97.79% across its 904 tested tickers (the per-snapshot ADF admits every ticker carrying a sufficient in-window residual count, a looser set than the $\geq 80\%$ -coverage Stage-2 alive panel of 889 in Table B.2), still above the 97% floor. The full sidecar (1 439 per-ticker p -values and per-snapshot rejection counts) is cached at `results/snapshots/stage1_adf_diagnostics.json`. The window-local $\bar{Q}_{\text{win}}, \bar{N}_{\text{win}}$ re-estimation absorbs the residual drift in the cross-sectional intercept; the global (a, b, g) pooling is therefore the only remaining stationarity assumption, and it is the assumption that F8 (the main paper §IX) proposes to relax. The per-snapshot rejection counts are reported in Table C.4.

TABLE C.4 – Augmented Dickey–Fuller rejection rate on the standardised residuals $z_{i,t}$ entering Stage 1c, per snapshot window and across the full 1985–2024 post-CRSP sample, at $\alpha = 0.05$. Short snapshots have shorter test windows and therefore lower power; the floor on this panel is 91.4% at Mar 2020 COVID. n_{tested} is the per-asset count with enough standardised residuals for the test, slightly above the per-window $\geq 80\%$ -coverage alive count because the ADF screen is per asset rather than complete-case. Auto-extracted from `stage1_adf_diagnostics.json`.

Window	T (days)	n_{tested}	Reject rate
Oct 1987	128	808	99.6%
1990-91	129	794	99.5%
1993	253	863	99.8%
Dec 1994	126	877	99.2%
Oct 1997	127	904	97.8%
Oct 1998	129	903	99.6%
Apr 2000	126	872	99.5%
Sep 2001	125	852	99.3%
Jul 2002	129	841	99.8%
2005	252	847	99.8%
2007	127	810	99.5%
Oct 2008	127	790	99.2%
Mar 2009	127	772	99.7%
2013	252	770	100.0%
2017	251	737	100.0%
Q4 2018	124	714	99.4%
Mar 2020	126	701	91.4%
Jun 2020	129	694	99.3%
2022	128	672	99.4%
2024	252	665	99.7%
Full sample 1985–2024	10080	1439	100.0%

F. Per-snapshot $\bar{R}$ near-PSD analysis

The window-averaged correlation $\bar{R}$ that feeds Stage 2 is the mean of the per-day R_t over the window. In exact arithmetic each R_t is positive semi-definite because it is the unit-diagonal normalisation of the PSD Q_t . The average of PSD matrices is also PSD, in exact arithmetic. In finite precision, when $n/p < 1$, the window-local sample target $\bar{Q}_{\text{win}}$ is rank-deficient (rank $\leq T_{\text{win}} < p$); the deficiency propagates through the A-DCC recursion into the floating-point accumulator $\sum_t R_t$ as a small negative-eigenvalue leak in the window average. Table C.5 reports the minimum eigenvalue of $\bar{R}$ per snapshot at the $N = 1439$ panel. The densest snapshot (2024 Contemporary) sits at $n/p = 0.385$ and every window is in the $n/p < 1$ rank-deficient regime, so the negative-eigenvalue leak is structural everywhere rather than present only at the sparser tail.

The leading-order direction is monotone, leakage grows as n/p falls, but the dependence is not a clean inverse on the panel either: the product $|\lambda_{\min}(\bar{R})| \cdot (n/p)$ spans more than an order of magnitude across the twenty snapshots (from 2.53×10^{-3} at Mar 2020 COVID to 3.48×10^{-2} at 2024 Contemporary) and is not monotone in n/p alone (e.g. Jun 2020 Stable at $n/p = 0.189$ has the worst leakage $|\lambda_{\min}| = 1.16 \times 10^{-1}$ while Mar 2020 COVID at the similar $n/p = 0.181$ shows $|\lambda_{\min}| = 1.40 \times 10^{-2}$, an order of magnitude smaller). The residual cross-window dispersion reflects window-specific spectral structure of $\bar{Q}_{\text{win}}$ that the bare n/p ratio does not summarise. At the densest baseline (2024 Contemporary, $n/p = 0.385$ on the raw-window basis, ≈ 0.32 on the Stage-2 sample of Table B.2), $\lambda_{\min}(\bar{R}) = -9.05 \times 10^{-2}$. At the sparsest window (Oct 1997 Asian Crisis, $n/p = 0.143$), $\lambda_{\min} = -7.14 \times 10^{-2}$; the absolute worst leakage on the panel is Jun 2020 Stable at -1.16×10^{-1} . The $n/p < 1$ regime everywhere makes $\bar{R}$ fail the strict PSD assertion: all 20 snapshots carry a negative minimum eigenvalue and fail the strict -10^{-6} `R_avg_strict_PSD` check, and the 10 snapshots whose $\lambda_{\min}$ drops below -5×10^{-2} additionally fail the looser `R_avg_near_PSD` check at that -5×10^{-2} tolerance. These 30 per-snapshot $\bar{R}$ failures, together with the single global GARCH-boundary failure, are the 31 documented Stage-1 FAILs catalogued in §K; all are expected by the $n/p < 1$ structural argument. The SPD matrix the GLASSO solver actually consumes is guaranteed not by $\bar{R}$'s own spectrum but by the identity-target shrinkage of §D below.

The Stage 2 spectral safety net is a defensive guard at the GLASSO boundary: whenever the post-shrinkage $\lambda_{\min}(\bar{R}) < 10^{-6}$, the matrix is shifted by $\bar{R} \leftarrow \bar{R} + (10^{-4} - \lambda_{\min})I$ before the GLASSO call. On the panel the identity-target shrinkage of §D (next section) again lifts every post-shrinkage $\lambda_{\min}$ above the 10^{-6} trigger: the closed-form post-shrinkage minimum is $(1 - \delta)\lambda_{\min}(\bar{R}) + \delta$. With the largest leakage at $|\lambda_{\min}| = 0.1164$ (Jun 2020 Stable), the lifted minimum at $\delta = 0.55$ evaluates to $(0.45)(-0.1164) + 0.55 = 0.498 > 10^{-6}$; the densest snapshot (2024 Contemporary, $|\lambda_{\min}| = 9.05 \times 10^{-2}$) sits in the $\delta = 0.35$ tier, where $(0.65)(-9.05 \times 10^{-2}) + 0.35 = 0.291 > 10^{-6}$, so the net does not fire there; had the dense end instead drawn the low-shrinkage $\delta = 0.05$ tier (reserved for $n/p \in [1, 2)$ and never reached on this panel), $(0.95)(-9.05 \times 10^{-2}) + 0.05 = -3.60 \times 10^{-2} < 10^{-6}$ would trigger the spectral safety net; the Stage 2 shrinkage tiers are therefore calibrated against the $|\lambda_{\min}|$ distribution so that the “shrinkage alone absorbs the leakage” property holds at the dense end of the panel as well.

D. STAGE 2: GRAPHICAL LASSO PRECISION-MATRIX FILTERING

Stage 2's deliverable is a sparse partial-correlation graph per snapshot, with the edge-budget penalty itself selected by BIC; downstream stages depend only on this graph.

Background concepts used in this section.: Stage 2 takes the dense $p \times p$ correlation matrix $\bar{R}$ from Stage 1 and produces a sparse network by inverting $\bar{R}$ under a sparsity-inducing penalty. Three ideas have to be in place. First, the *precision matrix* $\Omega = \bar{R}^{-1}$ encodes conditional rather than marginal independence: a zero in Ω_{ij} means i and j are conditionally independent given every other asset, which is what makes the sparsity pattern of Ω the right edge list for a partial-correlation network (§A). Second, in our regime $n < p$ holds on most snapshots, so the empirical correlation matrix is PSD but rank-deficient and not invertible in practice; identity-target shrinkage $\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I$ lifts every eigenvalue and restores SPD (§A “PSD, SPD, and conditioning”). Third, an ℓ_1 penalty on the off-diagonal entries of Ω produces *exact* zeros because the ℓ_1 ball has corners on the coordinate axes, so the constrained MLE lands on a sparse face (§A “ ℓ_1 regularisation and sparsity”). The selection of the penalty weight λ uses an information criterion that trades quasi-log-likelihood against edge count: minimising BIC penalises both poor fit and unnecessary complexity, so the chosen λ is the one whose graph fits the data best for its size. EBIC adds an extra penalty term that tightens the criterion in the high-dimension regime (§A).

A dense correlation matrix on 500 assets is not a usable network: most off-diagonal entries reflect shared sector loadings or market beta rather than direct asset-to-asset dependence. The precision matrix $\Omega = \bar{R}^{-1}$ supplies the conditional structure instead: a zero in entry (i, j) means i and j are conditionally independent given the remaining assets, the partial-correlation notion that Stage 4's network metrics consume.

The empirical $\hat{\Omega}$ is unusable in high dimensions. At $n/p < 1$ the empirical correlation matrix is near-rank-deficient (§C: $\bar{Q}_{\text{win}}$ has rank $\leq T_{\text{win}} < p$) and not invertible in practice; at n/p slightly above one the inverse is noise-dominated. The Graphical LASSO estimates Ω under an ℓ_1 penalty on its off-diagonal entries, yielding a sparse $\hat{\Omega}_\lambda$ with λ controlling the zero-out threshold. The non-zero entries of $\hat{\Omega}_\lambda$ become the edges of the partial-correlation network.

TABLE C.5 – $\bar{R}$ near-PSD diagnostic per snapshot at the $N = 1\,439$ panel. p is the per-window alive ($\geq 80\%$ -coverage) ticker count (window-local; the Stage 2 cache writes $\bar{R} \in \mathbb{R}^{p \times p}$). $\lambda_{\min}$ is the minimum eigenvalue (negative because of finite-precision averaging); $\lambda_{\max}$ is the maximum eigenvalue. Snapshots are ordered from densest to sparsest by n/p on the raw-window basis; the Stage-2-sample n/p that sets the shrinkage tier of Table D.7 is smaller (e.g. 0.261 vs. 0.349 at 2017 Calm).

Snapshot	p	n/p	$\lambda_{\min}(\bar{R})$	$\lambda_{\max}(\bar{R})$
2024 Contemporary	655	0.385	-9.05×10^{-2}	134.44
2017 Calm	720	0.349	-7.78×10^{-2}	116.32
2013 Calm	757	0.333	-5.88×10^{-2}	252.68
2005 Calm	825	0.305	-5.65×10^{-2}	188.87
1993 Calm	829	0.305	-6.50×10^{-2}	71.65
2022 Rate Hikes	670	0.191	-3.36×10^{-2}	298.61
Jun 2020 Stable	684	0.189	-1.16×10^{-1}	225.68
Mar 2020 COVID	695	0.181	-1.40×10^{-2}	374.97
Q4 2018 VolShock	702	0.177	-3.31×10^{-2}	254.61
Mar 2009 Recovery	768	0.165	-2.74×10^{-2}	279.38
Oct 2008 GFC	773	0.164	-2.53×10^{-2}	438.24
1990-91 Recession	786	0.164	-4.95×10^{-2}	202.54
2007 Subprime Buildup	787	0.161	-4.96×10^{-2}	272.39
Oct 1987 Black Monday	802	0.160	-4.30×10^{-2}	425.69
Jul 2002 WorldCom	833	0.155	-4.24×10^{-2}	293.23
Sep 2001 9/11	839	0.149	-5.23×10^{-2}	220.18
Apr 2000 Dot-com	858	0.147	-5.23×10^{-2}	138.90
Oct 1998 LTCM	880	0.147	-4.41×10^{-2}	221.24
Dec 1994 Tequila	868	0.145	-5.87×10^{-2}	101.76
Oct 1997 Asian Crisis	889	0.143	-7.14×10^{-2}	259.56

A. The EBIC penalty for choosing λ

We choose λ by minimising an information criterion that balances the Gaussian quasi-log-likelihood against the non-zero edge count. The Extended BIC of Foygel and Drton [6] for a Gaussian graphical model with precision Ω , off-diagonal non-zero count k , sample size n , dimension p , and penalty parameter $\gamma \in [0, 1]$ is

$$\text{EBIC}_\gamma(\hat{\Omega}) = -2 \mathcal{L}(\hat{\Omega}) + k \log n + 4k\gamma \log p, \quad (\text{D.7})$$

with Gaussian quasi-log-likelihood $\mathcal{L}(\hat{\Omega}) = \frac{n}{2}(\log \det \hat{\Omega} - \text{tr}(\bar{R}_\delta \hat{\Omega}))$ up to a constant, where $\bar{R}_\delta$ denotes the same matrix passed to the GLASSO solver, i.e. $\bar{R}$ after the identity-target shrinkage of §D and, where it would apply, the spectral safety net, so the EBIC selector and the precision fit share an identical sample-covariance argument. The first correction term $k \log n$ is the classical BIC penalty; the second term $4k\gamma \log p$ is the extension that tightens the penalty when the dimension p is large relative to the sample size. Setting $\gamma = 0$ recovers plain BIC.

Foygel and Drton recommend $\gamma = 0.5$ for $n/p < 1$ and shrink γ to zero as n/p grows. We implement the inverse schedule

$$\gamma(n/p) = \begin{cases} 0 & n/p < 3 \\ 0.25 & 3 \leq n/p < 5 \\ 0.5 & n/p \geq 5 \end{cases}$$

deliberately. The identity-target shrinkage of §D and the constrained-BIC fallback of eq. (D.8) together already deliver the high-dimension sparsity that EBIC’s positive- γ tier was designed for; adding a $\gamma > 0$ penalty on top would tighten the BIC argmin further toward $\lambda = 1.0$ and the empty graph on every $n/p < 1$ snapshot. On the panel n/p never reaches 3, the maximum is ≈ 0.32 on the Stage-2 sample (0.385 on the raw window) at 2024 Contemporary, so the $\gamma = 0$ branch fires everywhere and EBIC reduces algebraically to plain BIC. The non-zero tiers are retained for downstream extensions to longer windows or quarterly resampling where $n/p > 3$ becomes reachable. On the main-text panel the numerical effect of the tier rule is zero: every reported λ^* is the BIC-optimum, not an EBIC-optimum.

The multipanel sweep at smaller N does reach $n/p > 3$ on the long calm windows, at $N = 50$ ($p = 50$), the five baseline/calm snapshots (1993, 2005, 2013, 2017 Calm and 2024 Contemporary) fire the $\gamma = 0.5$ EBIC branch while the shorter crisis, stress, and recovery windows stay at $\gamma = 0$. The assertion suite tracks which tier was active per snapshot per panel. This is reported as Disclosure 3 in §K.

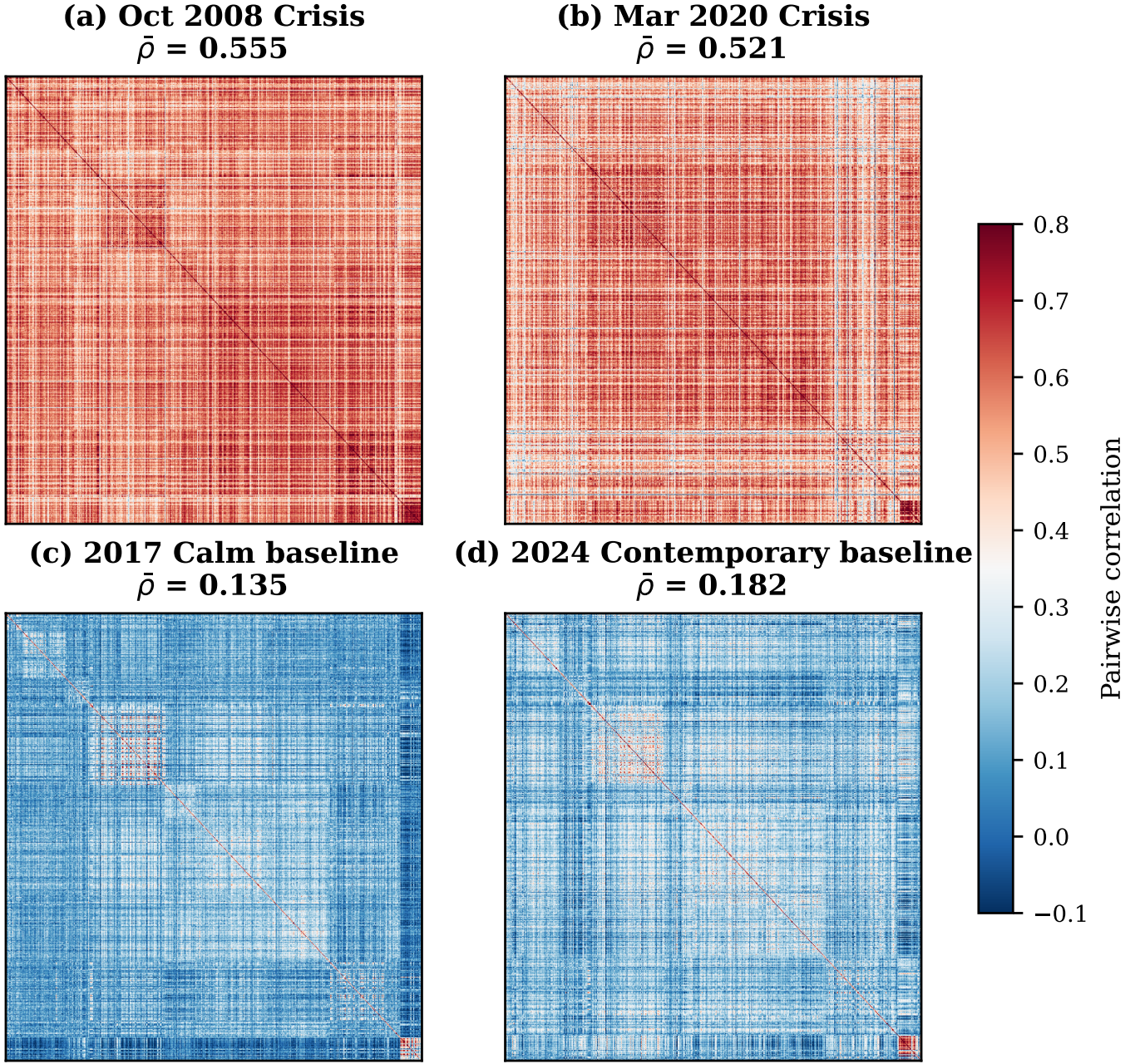


Fig. C.3 – A-DCC average correlation matrices $\bar{R}$ for two modern crisis peaks and two of the five calm baselines, with assets ordered by GICS sector so within-sector blocks appear along the diagonal. Panels: (a) Oct 2008 GFC; (b) Mar 2020 COVID; (c) 2017 Calm; (d) 2024 Contemporary baseline; per-panel $\bar{\rho}$ values are read from Table B.2. Both modern crises show systematically higher off-diagonal correlations than the two displayed baselines, consistent with crisis correlation convergence; the three undisplayed calm baselines (1993, 2005, 2013) are omitted by design (Lim. v, F5).

B. Identity-target shrinkage tiers

At $p \in [655, 889]$ with $n/p < 0.40$ on the panel (the window-local always-alive ticker counts of Table C.5), the raw $\bar{R}$ is near-rank-deficient (§C: the underlying $\bar{Q}_{\text{win}}$ has rank $\leq T_{\text{win}} < p$) and the sklearn coordinate-descent GLASSO solver returns “Non SPD result” over most of the regularisation grid. The remedy is the identity-target shrinkage of Ledoit and Wolf [9]: replace $\bar{R}$ with a convex combination of itself and the identity,

$$\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I,$$

where $\delta \in [0, 1]$ is the shrinkage intensity. Intuitively this pulls the empirical matrix toward the identity (which is trivially PSD and well-conditioned), trading some bias toward identity for much better numerical conditioning. The Ledoit-Wolf optimal δ

is a closed-form expression in the population fourth moments, but that formula requires a stationary panel that the short crisis windows do not provide. We therefore use the n/p -tiered schedule:

- $n/p < 0.20$: $\delta = 0.55$, the 15 event windows (Oct 1987, 1990-91, Dec 1994, Oct 1997, Oct 1998, Apr 2000, Sep 2001, Jul 2002, 2007 Subprime, Oct 2008 GFC, Mar 2009, Q4 2018, Mar 2020 COVID, Jun 2020, 2022) plus 1993 and 2005 Calm: 17 snapshots cached at $\delta = 0.55$.
- $0.20 \leq n/p < 0.30$: $\delta = 0.45$, 2013 Calm, 2017 Calm.
- $0.30 \leq n/p < 0.50$: $\delta = 0.35$, 2024 Contemporary.
- $0.50 \leq n/p < 1.0$: $\delta = 0.15$ (none triggered on the panel)
- $1.0 \leq n/p < 2.0$: $\delta = 0.05$ (none triggered on the panel)
- $n/p \geq 2.0$: $\delta = 0.005$ (not triggered on the panel)

The tier boundaries above use the Stage-2 sample ratio n/p (trading days over alive PERMNOs): the two calm baselines 2013 and 2017 ($n/p \approx 0.25$) fall in the $\delta = 0.45$ tier, while 1993 and 2005 sit in the $\delta = 0.55$ tier and only 2024 Contemporary ($n/p \approx 0.32$) reaches the $\delta = 0.35$ tier, matching the cached `applied_delta` field of `stage2_results.pkl[lbl]` on every snapshot and the shrinkage-tier table of Table D.7. (Computing n/p from the raw six-/twelve-month calendar window rather than the Stage-2 sample would place 1993–2024 in higher tiers; the operative code and the cache both use the Stage-2 sample ratio.) The schedule is conservative relative to the Ledoit-Wolf optimum (which would prescribe roughly 0.40–0.60 at $n/p \approx 0.2$): we accept a slightly higher bias toward identity in exchange for keeping `sklearn.covariance.graphical_lasso` inside its convergence basin on every snapshot without chaining a second solver. The tier-vs-Ledoit-Wolf gap is one of the items proposed for the F6 future-work sensitivity sweep (the main paper §IX).

C. The constrained-BIC fallback rule (Algorithm 1)

The issue that motivates the fallback is structural rather than numerical. On windows where n/p is small relative to the GLASSO solver’s convergence basin, the unconstrained BIC argmin sits at the sparse end of the λ grid, typically $\lambda^* = 1.000$ with $k = 0$ edges. BIC, being log-likelihood plus $k \log n$, mechanically prefers fewer edges when neither candidate λ value fits the data appreciably better than the empty graph. The result is mathematically defensible but produces a graph that is useless to the downstream pipeline: PageRank and Louvain on an empty graph return degenerate output.

The fix imposes a structural floor: the chosen λ must produce at least $k_{\min} = \max(p, 10)$ off-diagonal nonzero entries. Setting $k_{\min} = p$ corresponds to an average vertex degree of two: any graph with $k < p$ has average degree below two, which forces a disconnected, tree, or forest configuration (a connected non-tree on p vertices needs at least p edges). Subject to that floor, we pick the BIC-optimal λ :

$$\lambda_{\text{fallback}}^* = \arg \min_{\lambda : k(\lambda) \geq k_{\min}} \text{BIC}(\lambda). \quad (\text{D.8})$$

This is a *constrained-BIC* selector, not a heuristic “densest convergent” rule: among the λ values whose graph clears the floor, the selector picks the one with the lowest BIC, not the one with the largest edge count. On the panel the fallback fires on 19 of the 20 snapshots, every window except 2024 Contemporary, the densest at $n/p = 0.32$ on the Stage-2 sample (0.385 on the raw 12-month window). The fallback is in an “almost always fires” regime because the $n/p < 1$ regime is universal across the 40-year panel rather than concentrated in short crisis windows.

Algorithm 3 Stage-2 Graphical LASSO with constrained-BIC fallback.

Require: Window-average correlation $\bar{R}$ at dimension $p \times p$; sample size n ; lambda grid $\{\lambda_1, \dots, \lambda_{25}\} \subset [0.05, 1]$ in descending order.

- 1: Compute n/p ratio; select shrinkage intensity δ from the tier table.
- 2: $\bar{R} \leftarrow (1 - \delta)\bar{R} + \delta I$ ▷ Identity-target shrinkage
- 3: **if** $\lambda_{\min}(\bar{R}) < 10^{-6}$ **then**
- 4: $\bar{R} \leftarrow \bar{R} + (10^{-4} - \lambda_{\min}(\bar{R}))I$ ▷ Spectral safety net
- 5: **end if**
- 6: Initialise $\lambda^* \leftarrow \text{null}$, $\lambda_{\text{fallback}}^* \leftarrow \text{null}$, $\text{BIC}^* \leftarrow +\infty$, $\text{BIC}_{\text{fallback}}^* \leftarrow +\infty$.
- 7: $k_{\min} \leftarrow \max(p, 10)$ ▷ Edge-count floor
- 8: Initialise consecutive-worse counter $c \leftarrow 0$, patience $P \leftarrow 8$, and previous-iteration BIC $\text{BIC}_{\text{prev}} \leftarrow +\infty$.
- 9: **for** $\lambda \in \{\lambda_1, \dots, \lambda_{25}\}$ in descending order **do**
- 10: Run `graphical_lasso`($\bar{R}, \alpha = \lambda$); on solver failure, skip to next λ (do not advance c and do not update BIC_{prev}).
- 11: Read precision $\hat{\Omega}_\lambda$; compute $\text{BIC}(\lambda)$ via eq. (D.7) with $\gamma = \gamma(n/p)$ from the tier schedule in §D ($\gamma = 0$ throughout on the main-text panel, since every snapshot's $n/p < 3$).
- 12: $k(\lambda) \leftarrow \#\{(i, j) : i < j, |\hat{\Omega}_{\lambda, ij}| > 10^{-10}\}$ ▷ Off-diagonal nonzero count
- 13: **if** $\text{BIC}(\lambda) < \text{BIC}^*$ **then**
- 14: $\lambda^* \leftarrow \lambda$; $\text{BIC}^* \leftarrow \text{BIC}(\lambda)$; $c \leftarrow 0$.
- 15: **else if** $\text{BIC}(\lambda) > \text{BIC}_{\text{prev}}$ **then**
- 16: $c \leftarrow c + 1$ ▷ Past-minimum counter
- 17: **else**
- 18: $c \leftarrow 0$ ▷ Plateau resets counter (matches code)
- 19: **end if**
- 20: $\text{BIC}_{\text{prev}} \leftarrow \text{BIC}(\lambda)$ ▷ Update only on converged λ
- 21: **if** $k(\lambda) \geq k_{\min}$ **and** $\text{BIC}(\lambda) < \text{BIC}_{\text{fallback}}^*$ **then**
- 22: $\lambda_{\text{fallback}}^* \leftarrow \lambda$; $\text{BIC}_{\text{fallback}}^* \leftarrow \text{BIC}(\lambda)$.
- 23: **end if**
- 24: **if** $c \geq P$ **and** $\lambda_{\text{fallback}}^* \neq \text{null}$ **then**
- 25: **break** ▷ Early stop past minimum, once a valid fallback exists
- 26: **end if**
- 27: **end for**
- 28: **if** $k(\lambda^*) < k_{\min}$ **then**
- 29: **if** $\lambda_{\text{fallback}}^* = \text{null}$ **then**
- 30: **raise** “no constrained-BIC candidate” ▷ λ grid too coarse or input degenerate
- 31: **end if**
- 32: $\lambda^* \leftarrow \lambda_{\text{fallback}}^*$ ▷ Constrained-BIC fallback fires
- 33: **end if**
- 34: **return** $\hat{\Omega}_{\lambda^*}$ and λ^* .

Early stopping works as follows. The grid is traversed from sparse to dense; the past-minimum counter c counts consecutive converged grid points whose BIC has worsened relative to its immediate predecessor on the grid (matching the `ElseIf` branch of Algorithm 3 and the `prev_ebic` comparison in the code), and is reset whenever either a new running-minimum BIC is reached or the BIC fails to strictly increase from the predecessor. Solver failures do not advance c , so spurious mid-grid noise cannot trigger early stopping. The break fires only after c reaches the patience $P = 8$ and at least one $k \geq k_{\min}$ candidate has been observed, guaranteeing that the fallback selector has a non-empty constrained set to argmin over. Figure D.4 contrasts the unconstrained-BIC argmin with the final constrained λ^* across the twenty snapshots: the unconstrained argmin sits at $\lambda = 1.000$ ($k = 0$) on the nineteen fallback-firing windows, and the $k \geq p$ fallback (†) pulls the selection down to the constrained-BIC argmin.

Algorithm 3 above gives the full Stage-2 procedure. Table D.6 reports the per-snapshot values of $\lambda_{\text{unconstr}}^*$ (the unconstrained-BIC argmin), its edge count, the final selected λ^* , and whether the fallback fired. On the nineteen fallback rows the unconstrained argmin is the empty graph; on the single non-fallback row (2024 Contemporary) the two argmins coincide.

The fallback selections cluster tightly on a small set of discrete λ values: $\lambda^* \in \{0.174, 0.197, 0.253, 0.287, 0.325, 0.368\}$ across the 19 fallback rows, with 0.325 and 0.368 between them accounting for 12 of 19. The clustering is a grid artefact rather than a continuous- λ coincidence: the 25-point λ grid leaves only a handful of points strictly inside the SPD-convergence-and- $k \geq p$ -floor window once shrinkage and solver-failure cutoffs are factored in. The single non-fallback row, 2024 Contemporary, has its unconstrained BIC argmin $\lambda^* = 0.368$ already producing 1 334 edges ($k > p = 655$), so the constrained selector reduces to the unconstrained one.

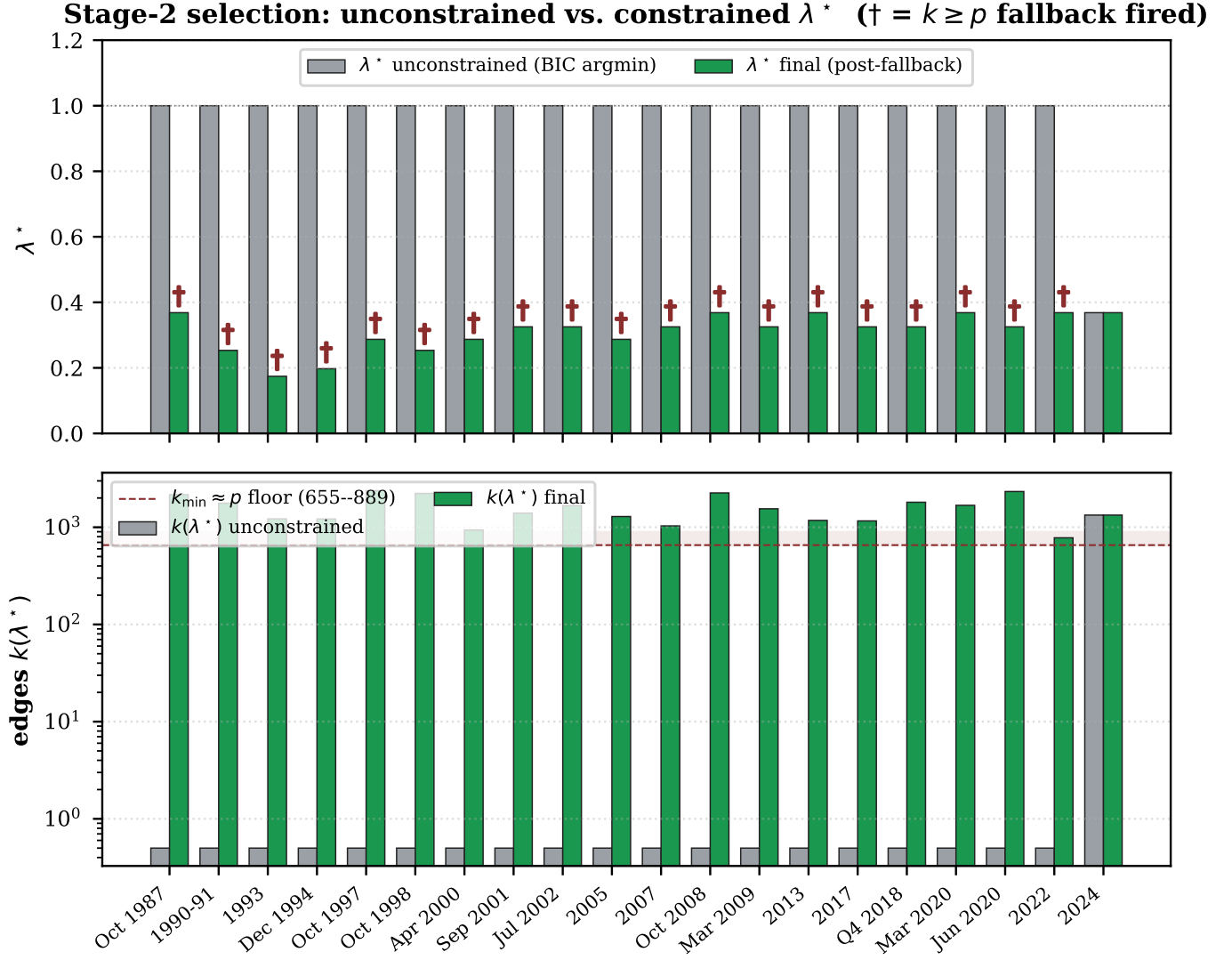


Fig. D.4 – Stage-2 λ^* selection diagnostic across the twenty $N = 1439$ snapshots, the graphical companion to Table D.6. *Top:* the unconstrained-BIC argmin $\lambda^*_{\text{unconstr}}$ (grey) against the final constrained-BIC λ^* (post-fallback, green). On nineteen of the twenty windows the unconstrained argmin sits at $\lambda = 1.000$ (the empty graph, $k = 0$) and the $k \geq p$ fallback fires ($\dagger$), pulling the selection down to the constrained-BIC argmin; the solver reports `n_failed_lambdas=0` on every snapshot (§D). *Bottom:* the matching edge counts $k(\lambda^*)$ on a log axis, the unconstrained argmin yields $k = 0$ (drawn at 0.5) on every fallback row, while the final $k \in [779, 2383]$ clears the per-window structural floor $k_{\min} = \max(p, 10)$ (shaded band, $p \in [655, 889]$). On the single non-fallback snapshot, 2024 Contemporary, the two argmins coincide ($\lambda^* = 0.368$, $k = 1334$) and no fallback marker is drawn.

D. Solver convergence diagnostics

On the $N = 1439$ panel the sklearn coordinate-descent solver achieves zero non-convergent grid points (`n_failed_lambdas=0`) on every one of the 20 snapshots; the per-snapshot `lambdas_visited` ranges from 13 to 25 as the early-stop counter terminates the descending- λ sweep at different points across snapshots. Every snapshot sits under the heavier $\delta \in \{0.35, 0.45, 0.55\}$ shrinkage tiers (seventeen snapshots cached at `applied_delta=0.55`, the two calm baselines 2013 and 2017 at 0.45, and 2024 Contemporary at 0.35), and the heavier shrinkage keeps the matrix well-conditioned across the entire λ grid.

- All 20 snapshots have `n_failed_lambdas=0`.
- `lambdas_visited` clusters at 15 for 12 snapshots that early-stop at the BIC turnaround; three crisis snapshots at $\lambda^* = 0.368$ (Oct 1987, Oct 2008 GFC, and Mar 2020 COVID) traverse the full 25-point grid; the 1990-91 Recession and Dec 1994 Tequila windows stop at 16; 2013 Calm and 2017 Calm early-stop at 13 visits; and 2024 Contemporary early-stops at 17.
- The cache exhibits a monotonic “always converges” pattern as a direct consequence of the heavier-shrinkage regime that the panel-wide $n/p < 0.40$ regime forces. Per-snapshot `lambdas_visited` arrays are persisted in `stage2_results.pkl`.

In every case the BIC minimum lies in the converged portion of the grid, so the λ^* selection is not affected. Per-snapshot

TABLE D.6 – Stage-2 BIC trajectory diagnostics per snapshot on the $N = 1\,439$ panel. “fb” marks where the constrained-BIC fallback fires (eq. (D.8)). On those rows the unconstrained argmin is the empty graph and the final λ^* comes from the $k \geq p$ -constrained selector; on the single non-fallback row (2024 Contemporary), the two argmins coincide.

Snapshot	Regime	$\lambda_{\text{unconstr}}^*$	k_{unconstr}	λ^*	k	fb
Oct 1987 Black Monday	crisis	1.000	0	0.368	2167	✓
1990-91 Recession	stress	1.000	0	0.253	1768	✓
1993 Calm	baseline	1.000	0	0.174	1221	✓
Dec 1994 Tequila	stress	1.000	0	0.197	1216	✓
Oct 1997 Asian Crisis	crisis	1.000	0	0.287	2383	✓
Oct 1998 LTCM	crisis	1.000	0	0.253	2229	✓
Apr 2000 Dot-com	crisis	1.000	0	0.287	937	✓
Sep 2001 9/11	crisis	1.000	0	0.325	1397	✓
Jul 2002 WorldCom	stress	1.000	0	0.325	1668	✓
2005 Calm	baseline	1.000	0	0.287	1290	✓
2007 Subprime Buildup	stress	1.000	0	0.325	1033	✓
Oct 2008 GFC	crisis	1.000	0	0.368	2256	✓
Mar 2009 Recovery	recovery	1.000	0	0.325	1549	✓
2013 Calm	baseline	1.000	0	0.368	1176	✓
2017 Calm	baseline	1.000	0	0.325	1162	✓
Q4 2018 VolShock	stress	1.000	0	0.325	1809	✓
Mar 2020 COVID	crisis	1.000	0	0.368	1681	✓
Jun 2020 Stable	recovery	1.000	0	0.325	2333	✓
2022 Rate Hikes	stress	1.000	0	0.368	779	✓
2024 Contemporary	baseline	0.368	1334	0.368	1334	

non-convergent counts and the visited- λ array are persisted to `stage2_results.pkl` as the fields `n_failed_lambdas` and `lambdas_visited`, so the trajectory in Table D.6 is re-derivable from the cache alone.

TABLE D.7 – Stage-2 identity-target shrinkage tiers (referenced from the main paper §III-B).

n/p range	δ	Triggered snapshots
< 0.20	0.55	All 15 event windows; 1993 & 2005 calm
$[0.20, 0.30)$	0.45	2013, 2017 calm
$[0.30, 0.50)$	0.35	2024 Contemporary
$[0.50, 1.0)$	0.15	none on panel
$[1.0, 2.0)$	0.05	none on panel
≥ 2.0	0.005	none on panel

E. STAGE 3: LEAD/FOLLOWER DIRECTION ASSIGNMENT

Stage 3’s deliverable is the Stage 2 graph endowed with directions: every surviving undirected edge receives a direction, a “mutual” label, or a drop verdict. The mutual fraction is the load-bearing crisis signal of the analysis.

Background concepts used in this section.: Two ideas drive Stage 3. First, “ i leads j ” has an operational meaning here: yesterday’s return of i predicts today’s return of j better than chance, even after controlling for j ’s own past and for the shared movement of i ’s and j ’s top contemporaneous neighbours. The economic content is mild, the construction does not establish causality, only a one-step-ahead predictive asymmetry, but it is the strongest direction statement that daily-frequency return data can support. Second, the direction call uses two independent tests in cascade: a Frisch–Waugh–Lovell partial correlation t -test on the residual lagged dependence (§A), and a Granger F -test that compares a *restricted* regression of the follower on its own lags against an *unrestricted* regression that also includes the candidate driver’s lags (§A). The F statistic measures how much the unrestricted residual sum of squares falls relative to the restricted one; large F means the driver’s lags add predictive content beyond what the follower’s own history supplies. Cascading the two tests under a dominance ratio τ keeps the false-positive rate manageable without abandoning every borderline directional call.

A partial-correlation edge between asset i and asset j is symmetric by construction: the partial correlation matrix is the symmetric inverse of the symmetric correlation matrix. To assign a direction, i leads j , j leads i , or both, a separate temporal test is needed. Stage 3 runs two such tests on each surviving Stage-2 edge and combines them under a dominance rule.

A. Lagged partial correlation via Frisch–Waugh–Lovell

For an undirected Stage-2 edge (i, j) we ask, in each direction separately, whether the lagged return of the candidate driver retains explanatory power for the candidate follower's return tomorrow. Two partial-correlation conditioners absorb the parts of the signal we do not want: (a) the contemporaneous returns of the five strongest neighbours of the candidate driver (top-5 by $|\bar{R}_{sk}|$, $k \notin \{i, j\}$, where $\bar{R}$ is the Stage-1 window-averaged A-DCC correlation and s ranges over the per-direction source i or j), and (b) the lagged outcome of the candidate follower. The conditioning set is the join of these two: the contemporaneous neighbours of the driver absorb the shared factor structure that would otherwise show up as a spurious lead, and the lagged outcome absorbs the autoregressive persistence of the follower. The neighbour set is rebuilt per direction, the top-5 for $i \rightarrow j$ are the neighbours of i in $\bar{R}$, the top-5 for $j \rightarrow i$ those of j , so the FWL conditioning is genuinely directional. When a candidate control is absent from the per-snapshot balanced sub-panel (Stage-1c's $\geq 80\%$ in-window coverage filter occasionally drops a candidate PERMNO before the cascade sees it, and the cascade itself applies a further dropna(how="any") on the analysis batch), the conditioning set shrinks to $k_{\text{eff}} < 5$ rather than backfilling further down the $|\bar{R}|$ ranking; the FWL projection in eq. (E.9) is well-defined for any $k_{\text{eff}} \geq 0$.

For the direction $i \rightarrow j$, let $y = r_{j,u}$, $x = r_{i,u-1}$, and $W = [r_{j,u-1}, r_{C_i,u}]$ stacked along the time index u , where C_i is the top-5 neighbour set of i in $\bar{R}$. The reverse direction $j \rightarrow i$ uses $y = r_{i,u}$, $x = r_{j,u-1}$, and $W = [r_{i,u-1}, r_{C_j,u}]$. The Frisch–Waugh–Lovell theorem says that the partial correlation of y and x given W equals the simple correlation of the OLS residuals of y on W and of x on W :

$$\text{pcorr}(y, x | W) = \text{corr}(y - W\hat{\beta}_y, x - W\hat{\beta}_x), \quad (\text{E.9})$$

where $\hat{\beta}_y$ and $\hat{\beta}_x$ are the OLS slopes from regressing y and x on W . Operationally: remove from each variable the part that W predicts, then correlate the residuals. The residual correlation is, by construction, the contribution of x to y that is not explainable by W .

Significance of the residual correlation in eq. (E.9) is assessed at $\alpha = 0.05$ via the Pearson two-sided p -value (scipy.stats.pearsonr, $df = n - 2$ where $n = T_{\text{win}} - 1$ is the post-lag sample length). The FWL-adjusted partial-correlation t -test that discounts the $k = 6$ projection regressors (five top- $|\bar{R}|$ controls plus the follower's own lag, both fit by OLS without an intercept; $k_{\text{eff}} < 6$ when the balanced sub-panel drops a candidate) would use $df = n - k - 2$; for $T_{\text{win}} = 80$ (the Mar 2020 window) this moves df from 77 to 71 and the critical $|t|$ at $\alpha = 0.05$ from 1.991 to 1.994, a third-decimal change well inside the per-edge sensitivity envelope bounded by the $\alpha \in \{0.05, 0.10\}$ dimension of the F3 sweep (§E-E). A subtle point: only a *positive* significant correlation is admitted as evidence that i leads j . A significant negative residual correlation indicates lagged anti-correlation, which is a separate phenomenon and would otherwise contaminate the lead/follower interpretation. The Granger test below provides unsigned evidence, capturing causal influence that the FWL test's positive-correlation requirement excludes.

B. Bivariate Granger F -test

The Granger (1969) F -test asks whether the lag-1 coefficient of the candidate driver is statistically distinguishable from zero in a regression of the target on its own lag and the lagged driver. For the direction $i \rightarrow j$ (matching the FWL conventions above):

$$H_0 : r_{j,u} = \alpha_0 + \alpha_1 r_{j,u-1} + e_u, \quad H_1 : r_{j,u} = \alpha_0 + \alpha_1 r_{j,u-1} + \beta_1 r_{i,u-1} + e_u.$$

The standard SSR-based F -statistic compares the residual sum of squares under the two models:

$$F = \frac{(\text{SSR}_0 - \text{SSR}_1)/q}{\text{SSR}_1/(T - k - 1)},$$

with $q = 1$ restriction and $k = 2$ unrestricted regressors. This is the `ssr_ftest` variant implemented by `statsmodels.tsa.stattools.grangercausalitytests`. We use `maxlag=1` matching the daily-return horizon: multi-day lag scans mix multi-day information flows with the pairwise lead/follower test we are isolating, and the F3 sensitivity sweep (§E-E) explores the consequences of relaxing this.

C. Two-test cascade decision rule

Each direction $d \in \{(i, j), (j, i)\}$ accumulates a confidence $c_d \in [0, 1]$ from the two tests:

- A *significant positive* FWL partial correlation at level $\alpha = 0.05$ contributes $c_d \leftarrow \max(c_d, \hat{\rho}_d)$ where $\hat{\rho}_d$ is the residual Pearson correlation. The signed magnitude is used here because the partial correlation carries directional information.
- A significant Granger F -test at level $\alpha = 0.05$ contributes $c_d \leftarrow \max(c_d, 0.5)$. The unsigned floor of 0.5 is sufficient to assign direction when partial-correlation evidence is absent (the Granger test is unsigned, so a fixed value is the only sensible quantification), but yields to a stronger partial-correlation signal when both fire.

The direction rule (six cases: drop, two single-positive, two dominance-by- τ , mutual), with the dominance ratio $\tau = 1.5$, is then

$$\text{dir}(i, j) = \begin{cases} \text{drop} & c_{ij} = c_{ji} = 0, \\ i \rightarrow j & c_{ij} > 0, c_{ji} = 0, \\ j \rightarrow i & c_{ji} > 0, c_{ij} = 0, \\ i \rightarrow j & c_{ij} \geq \tau c_{ji} > 0, \\ j \rightarrow i & c_{ji} \geq \tau c_{ij} > 0, \\ \text{mutual} & \text{otherwise (both positive, neither dominates).} \end{cases}$$

The motivation for the cascade rather than a single test is twofold. First, the two tests probe complementary aspects of the lead/follower structure: the FWL test provides *signed* evidence (only positive lagged partial correlation counts as a lead), while the Granger test provides *unsigned* causal evidence (any significant lag-1 information flow). Second, on this panel each test alone fires bidirectionally on a non-trivial fraction of Stage-2 edges, the FWL test on 0.1–1.5% and the Granger test on 1.0–8.8% (peak at the Mar 2020 crisis snapshot), because strong contemporaneous correlation trips the Granger test in either direction. The cascade combines the signed-FWL and unsigned-Granger contributions under the $\tau = 1.5$ dominance rule, and the mutual category captures the genuine reciprocal-lead case rather than absorbing it into one direction arbitrarily. Algorithm 4 gives the full per-edge cascade.

Algorithm 4 Stage-3 lead/follower cascade for one undirected Stage-2 edge (i, j) .

Require: Daily return series r_i, r_j on the snapshot window’s balanced sub-panel (the dropna(how="any") restriction to days on which every analysed PERMNO trades; cf. §A and Stage-1c’s $\geq 80\%$ in-window coverage filter); T_{win} denotes this balanced length, not the raw calendar count. Window-average correlation $\bar{R}$; significance level $\alpha = 0.05$; dominance ratio $\tau = 1.5$; Granger floor $c_{\text{floor}} = 0.5$. Slice notation $r[a:b]$ below denotes the one-based inclusive subvector $(r_a, r_{a+1}, \dots, r_b)$ of length $b - a + 1$; on a multi-column control matrix r_C the slice indexes time (rows), not assets, so $r_C[a:b]$ retains all $|C|$ columns at the specified time rows.

```

1:  $c_{ij} \leftarrow 0, c_{ji} \leftarrow 0.$ 
2: for each direction  $d \in \{(i, j), (j, i)\}$  with source  $s$  and target  $t$  do
3:    $C \leftarrow \text{top-5 assets } k \notin \{i, j\} \text{ by } |\bar{R}_{sk}|.$ 
4:   Build  $y \leftarrow r_t[2 : T_{\text{win}}], x \leftarrow r_s[1 : T_{\text{win}} - 1], W \leftarrow [r_t[1 : T_{\text{win}} - 1], r_C[2 : T_{\text{win}}]]$ .
5:   Compute FWL partial correlation  $\hat{\rho}_d$  and Pearson two-sided  $p$ -value  $p_d^{\text{FWL}}$  via eq. (E.9).
6:   if  $p_d^{\text{FWL}} < \alpha$  and  $\hat{\rho}_d > 0$  then
7:      $c_d \leftarrow \max(c_d, \hat{\rho}_d)$  ▷ signed contribution from partial corr.
8:   end if
9:   Run Granger  $F$ -test of  $r_s \rightarrow r_t$  at maxlag= 1; obtain  $p_d^{\text{Gr}}$ .
10:  if  $p_d^{\text{Gr}} < \alpha$  then
11:     $c_d \leftarrow \max(c_d, c_{\text{floor}})$  ▷ unsigned floor from Granger
12:  end if
13: end for
14: Direction assignment:
15: if  $c_{ij} = c_{ji} = 0$  then return drop
16: end if
17: if  $c_{ij} > 0$  and  $c_{ji} = 0$  then return  $i \rightarrow j$ 
18: end if
19: if  $c_{ji} > 0$  and  $c_{ij} = 0$  then return  $j \rightarrow i$ 
20: end if
21: if  $c_{ij} \geq \tau c_{ji}$  then return  $i \rightarrow j$ 
22: end if
23: if  $c_{ji} \geq \tau c_{ij}$  then return  $j \rightarrow i$ 
24: end if
25: return mutual ▷ both positive, neither dominates by ratio  $\tau$ 

```

An auxiliary Granger-floor sweep $c_{\text{floor}} \in \{0.3, 0.5, 0.7\}$ holds $(\alpha, \tau, \text{lag})$ at the headline cell and varies only the unsigned Granger contribution; under that sweep the surviving-edge total is invariant by construction (the cascade’s drop condition $c_{ij} = c_{ji} = 0$ does not see c_{floor}) and only the directed-vs-mutual split moves. Per-cell numbers are emitted by the sweep driver (see §E-E) alongside the joint $\alpha \times \tau \times \text{lag}$ grid.

D. Multiple-testing position

The cascade evaluates two tests in each of two directions for every Stage-2 edge, m_2 edges, four nominal tests per edge, $m_2 \in [779, 2383]$ across the twenty-snapshot panel. We do not apply a panel-wide Bonferroni or Benjamini–Hochberg correction at this stage for three linked reasons. First, the FWL and Granger tests at $\alpha = 0.05$ are applied *conditional* on Stage-2’s EBIC-selected partial-correlation edge set (§D); the screening already restricts the direction problem to the small subset of asset pairs whose contemporaneous partial correlation survived GLASSO. Second, the quantities reported in the main paper (the per-snapshot mutual-dyad fraction f_{mut} and the PageRank concentration channels of Table F.10 and Table G.15) are graph-level aggregates whose inferential claims rest on the snapshot-level permutation, cluster-bootstrap, and block-bootstrap procedures of §J, not on per-edge claims. Third, the joint $\alpha \times \tau \times \text{lag}$ sensitivity grid below (§E-E) re-runs the cascade at $\alpha \in \{0.05, 0.10\}$ which is itself the relevant per-edge sensitivity envelope: at $\alpha = 0.10$ the cascade admits more direction assignments, at $\alpha = 0.05$ fewer. The grid output (the CSV emitted by `tools/run_stage3_sweep.py`) is the appropriate object for any reader who would prefer a more conservative per-edge threshold; it is provided as a robustness companion rather than as a multiple-testing correction.

E. Joint $\alpha \times \text{dominance} \times \text{lag}$ sensitivity (F3)

The F3 sweep promised in the main paper §IX runs the cascade over the joint grid $\alpha \in \{0.05, 0.10\}$, $\tau \in \{1.2, 1.5, 2.0\}$, Granger lag $\in \{1, 2\}$, giving 12 cells per snapshot, plus an auxiliary 3-cell sweep over the Granger floor $c_{\text{floor}} \in \{0.3, 0.5, 0.7\}$ at the headline $(\alpha, \tau, \text{lag}) = (0.05, 1.5, 1)$ for a total of 14 unique cells (the headline $c_{\text{floor}} = 0.5$ cell is shared between the two sub-grids; see the $12+3-1 = 14$ accounting below). The sweep is implemented in `src.stage3_direction.sensitivity_sweep`; the CLI driver `python -m tools.run_stage3_sweep -csv out.csv` runs the full grid against the cached Stage-2 output, writes `stage3_sensitivity.pkl` into `results/snapshots/`, and prints the crisis-vs-non-crisis mean mutual-dyad fraction per cell. The Granger test in the cascade is single-lag (no best-of-lags scan), so the lag dimension of the sweep is exercised by repeated cascade runs rather than by an internal lag-selection inside one run.

The grid was re-executed against the 20-snapshot CRSP panel and is on disk at `results/snapshots/stage3_sensitivity.pkl` (generated by `src.stage3_direction.sensitivity_sweep + tools.run_stage3_sweep`). The headline aggregates across all 14 unique cells are:

Quantity (across the 14 cells; 20-snapshot)	Range
Crisis-mean f_{mut}	11.43–18.14%
Non-crisis-mean f_{mut}	9.36–14.42%
Crisis vs non-crisis separation (pp)	1.72– 4.40
Mar 2020 f_{mut}	17.82–34.52%
Oct 2008 f_{mut}	15.35–27.37%

The crisis-mean mutual-dyad fraction stays strictly above the non-crisis mean on *every* one of the 14 cells: the separation runs from +1.72pp (at $\alpha, \tau, \text{lag}, c_{\text{floor}} = 0.05, 1.2, 2, 0.5$) to +4.40pp (at $0.05, 2.0, 1, 0.5$) and never flips sign, so the directional contrast the main paper leans on survives the joint grid *in mean*. **The separation is modest, however, roughly an order of magnitude smaller than the per-window contrast.** The headline cells report a 1.72–4.40pp crisis-vs-non-crisis gap. This compression is a genuine empirical feature rather than a pipeline artefact: the 40-year survivorship-corrected panel admits more borderline non-crisis windows, notably the 2007 Subprime Buildup and the 2022 Rate Hikes, both labelled stress rather than crisis, whose intra-window lead/follower structure is itself crisis-like, which lifts the non-crisis mean toward the crisis mean. The per-snapshot ranking is correspondingly weaker than on the old panel: at the headline cell only Oct 2008 GFC ($f_{\text{mut}} = 22.32\%$) sits above the strongest non-crisis window (2007 Subprime Buildup at 19.27%), while Mar 2020 COVID (18.66%) now falls just below it. The headline cell $(\alpha, \tau, \text{lag}, c_{\text{floor}}) = (0.05, 1.5, 1, 0.5)$ posts crisis-mean 13.24%, non-crisis-mean 9.77%, and gap 3.47pp, matching the inference-section macros of §J. Table E.8 reports the full per-cell sweep.

The auxiliary $c_{\text{floor}} \in \{0.3, 0.5, 0.7\}$ sweep at the headline $(\alpha, \tau, \text{lag}) = (0.05, 1.5, 1)$ leaves the surviving-edge total invariant (the cascade’s drop condition does not see c_{floor}) and produces the following per-snapshot f_{mut} ranges:

TABLE E.8 – Stage-3 joint $\alpha \times \tau \times \text{lag} \times c_{\text{floor}}$ sensitivity grid (14 cells). Each cell records crisis-mean and non-crisis-mean mutual-dyad fraction and their gap; the headline cell is bold. Crisis f_{mut} stays above non-crisis on every cell in the sweep. Auto-extracted from results/snapshots/stage3_sensitivity.pkl.

α	τ	lag	c_{floor}	$\bar{f}_{\text{mut}}^{\text{cris}}$	$\bar{f}_{\text{mut}}^{\text{ncr}}$	Δ
0.05	1.2	1	0.5	0.124	0.094	+0.030
0.05	1.2	2	0.5	0.114	0.097	+0.017
0.05	1.5	1	0.5	0.132	0.098	+0.035
0.05	1.5	2	0.5	0.123	0.101	+0.022
0.05	2.0	1	0.5	0.146	0.102	+0.044
0.05	2.0	2	0.5	0.138	0.108	+0.030
0.10	1.2	1	0.5	0.154	0.126	+0.027
0.10	1.2	2	0.5	0.153	0.130	+0.023
0.10	1.5	1	0.5	0.164	0.133	+0.031
0.10	1.5	2	0.5	0.164	0.137	+0.027
0.10	2.0	1	0.5	0.178	0.139	+0.040
0.10	2.0	2	0.5	0.181	0.144	+0.037
0.05	1.5	1	0.3	0.153	0.110	+0.043
0.05	1.5	1	0.7	0.128	0.098	+0.031

Snapshot	f_{mut} across $c_{\text{floor}} \in \{0.3, 0.5, 0.7\}$
Oct 1987 Black Monday	16.17–20.43%
1990-91 Recession	4.48– 5.34%
1993 Calm	3.70– 4.63%
Dec 1994 Tequila	5.91– 7.80%
Oct 1997 Asian Crisis	9.75–11.26%
Oct 1998 LTCM	6.28– 8.21%
Apr 2000 Dot-com	11.11–12.35%
Sep 2001 9/11	6.11– 8.63%
Jul 2002 WorldCom	9.90–11.63%
2005 Calm	10.21–13.03%
2007 Subprime Buildup	19.27–20.31%
Oct 2008 GFC	22.32–23.57%
Mar 2009 Recovery	11.91–13.26%
2013 Calm	7.00– 7.00%
2017 Calm	10.00–10.00%
Q4 2018 VolShock	9.27–10.67%
Mar 2020 COVID	18.06–22.73%
Jun 2020 Stable	9.90–12.05%
2022 Rate Hikes	16.11–16.67%
2024 Contemporary	9.36–10.37%

The 2013 and 2017 Calm windows are completely insensitive to c_{floor} on this panel (the Granger floor never displaces a non-zero FWL contribution, so the range collapses to a point), and the 2022 Rate Hikes window is nearly so (0.56pp); Mar 2020 COVID (4.67pp) and Oct 1987 Black Monday (4.26pp) are the most sensitive, reflecting a cluster of partial-correlation magnitudes straddling the c_{floor} threshold. On the CRSP panel only Oct 2008 GFC is strictly above every non-crisis cell across all three floor values (its minimum f_{mut} of 22.32% exceeds the 20.31% non-crisis ceiling set by the 2007 Subprime Buildup at $c_{\text{floor}} = 0.7$); Mar 2020 COVID, by contrast, dips below that Subprime ceiling, so the per-snapshot f_{mut} ranking, not just the crisis-vs-non-crisis *mean*, is what the Granger-floor choice can perturb here. The full per-cell long-form CSV is emitted by `python -m tools.run_stage3_sweep -csv out.csv` for readers who would prefer a more conservative per-edge threshold; the F3 follow-up of the main paper §IX extends this 14-cell grid to a $2 \times 3 \times 2 \times 2 \times 2 = 48$ -cell version that adds the return-frequency dimension {daily, weekly} and rebalances the c_{floor} sub-grid to {0.3, 0.7} (omitting the headline 0.5 already covered by the prototype) on the extended panel.

F. STAGE 4: NETWORK METRICS

Stage 4’s deliverable is four scalar summaries per snapshot, clustering deviation, PageRank concentration, modularity with sector purity, motif mix, each Z -scored against an explicit null so cross-snapshot density differences cancel.

TABLE E.9 – Stage 3 direction-assignment outcomes (referenced from the main paper §III-C). “Dropped” edges lack significant temporal lead/follower evidence in either direction at $\alpha = 0.05$.

Snapshot	Input	Directed	Mutual	Dropped
Oct 1987 Black Monday	2167	581	124	1462
1990-91 Recession	1768	555	26	1187
1993 Calm	1221	416	16	789
Dec 1994 Tequila	1216	350	22	844
Oct 1997 Asian Crisis	2383	533	62	1788
Oct 1998 LTCM	2229	582	39	1608
Apr 2000 Dot-com	937	216	27	694
Sep 2001 9/11	1397	445	30	922
Jul 2002 WorldCom	1668	364	40	1264
2005 Calm	1290	255	29	1006
2007 Subprime Buildup	1033	155	37	841
Oct 2008 GFC	2256	435	125	1696
Mar 2009 Recovery	1549	392	53	1104
2013 Calm	1176	186	14	976
2017 Calm	1162	198	22	942
Q4 2018 VolShock	1809	323	33	1453
Mar 2020 COVID	1681	680	156	845
Jun 2020 Stable	2333	546	60	1727
2022 Rate Hikes	779	151	29	599
2024 Contemporary	1334	271	28	1035

Numerical scope.: The headline conclusions of this section depend on the *sign* of the Z -scores rather than their precise magnitudes on any specific snapshot: clustering exceeds the ER null, PageRank concentration exceeds uniform, modularity exceeds the shuffle null, and motif counts exceed the configuration null on the windows where each effect is claimed. Specific per-snapshot top-PageRank ticker rankings and edge counts are read directly from `results/snapshots/stage4_results.pkl`.

Background concepts used in this section.: The four Stage-4 metrics are standard graph-theoretic summaries; the choices that matter for interpretation are the null each is compared against and the way concentration is measured. *PageRank* with damping $\alpha = 0.85$ is the long-run distribution of a random walker who follows an out-edge with probability α and teleports uniformly with probability $1 - \alpha$ (§A); the $\alpha = 0.85$ convention is inherited from Page and Brin (1999) as a compromise between local concentration ($\alpha \rightarrow 1$ ill-conditions the score on dangling-edge graphs) and uniform diffusion ($\alpha \rightarrow 0$ collapses every score to $1/p$). The *Gini* and *Herfindahl* concentration summaries do different things to the same PageRank vector: Gini integrates the Lorenz curve so a long tail of near-zero nodes pulls it upward, while HHI sums squared masses so it weights the top of the distribution more heavily (§A). The Jun 2020 inversion (Gini 3rd sample-wide but top-10 HHI 18th) is exactly this gap. *Louvain modularity* Q maximises the excess of within-community edge weight above the expectation under a degree-preserving null (§A). The *MAN nomenclature* catalogues all 16 isomorphism classes of three-node directed subgraphs under labels M (mutual), A (asymmetric), N (null); the three triads tracked here (030T, 111D, 021D) cover the hierarchical, reciprocal, and common-driver patterns most relevant to a partial-correlation network (§A). Every Stage-4 metric is reported as a Z -score against an explicit null, $Z = (\text{empirical} - \bar{\mu}_{\text{null}}) / \sigma_{\text{null}}$, so that the magnitude of departure is comparable across snapshots of very different density.

The directed graph produced by Stage 3 is summarised by four metrics, each tied to one of the four ex-ante hypotheses Q1–Q4. Each metric is paired with a null distribution chosen to control for the properties of the empirical graph that are not informative about the hypothesis being tested.

A. Q1: Erdős–Rényi non-randomness

Companion to the main paper §V-A. Given the symmetrised Stage-3 directed graph $G_{\text{sym}}(p, m)$ (each mutual pair and each one-way pair contributing one undirected edge), the Q1 null is an Erdős–Rényi $G(n, m)$ ensemble: a random graph with the same node count $n = p$ and the same edge count m , sampled uniformly. NetworkX’s `gnm_random_graph` with `directed=False` produces each null draw under a fixed master seed $s_{\text{ER}} = 2026$ (`erdos_renyi_test` in `src.stage4_network.analysis`), with m taken from the undirected projection of the empirical graph. For each draw $b = 1, \dots, B$ with $B = 200$, the mean local clustering coefficient $C^{(b)} = \frac{1}{n} \sum_i C_i^{(b)}$ is computed via NetworkX’s `average_clustering` routine on the undirected null. The Q1 statistic is

$$Z_C = (C_{\text{emp}} - \bar{C}^{\text{null}}) / \sigma(C^{\text{null}}),$$

the standardised effect size against the null mean and standard deviation. The intuition is that $|Z_C|$ measures how far the empirical clustering sits from what would be expected at random for the same edge count.

Two-sided rejection of randomness at the conventional 5% level requires $|Z_C| > 1.96$; Bonferroni-20 correction for testing the twenty snapshots in parallel raises the threshold to $|Z_C| > 3.02$ (§J-F). With $B = 200$ null draws the MC standard error on $|Z_C|$ is approximately $\sqrt{(1 + Z_C^2/2)/B}$ (delta method, dominated at large $|Z|$ by the sd-estimation contribution); on the densest snapshot (Mar 2020 COVID, $m = 836$ symmetrised edges, $|Z_C| = 47.8$) this evaluates to ≈ 2.4 , and every snapshot's empirical $|Z_C|$ on the headline CRSP panel ranges from 14.7 (2007 Subprime Buildup) to 68.1 (Sep 2001 9/11), so the Bonferroni-20 threshold $|Z| > 3.023$ is exceeded by at least a factor of 4.9 on every snapshot and the rejection decisions are MC-robust. If the 200 null draws happen to collapse to a single clustering value (degenerate $\sigma(C^{\text{null}}) = 0$, possible only on extreme-sparse bottom-N panels in the multipanel sweep of §I), the implementation returns $Z_C = \text{NaN}$ and downstream consumers filter; the headline CRSP S&P-500 panel ($p \in [655, 889]$) never triggers this branch. Path-length statistics on the null and empirical graphs are also computed but are treated as diagnostic only: at low edge counts the null graphs fragment into many small components, rendering the unrestricted average shortest-path length undefined. The implementation falls back to the giant component (see §H), keeping L_{null} computable, but on sparse panels the giant-component L_{null} itself inflates as $\sim \log n / \log \langle k \rangle$ and the density-invariance argument underlying the small-world coefficient breaks down at the sparse end of the panel.

B. Q2: PageRank, Gini, and Herfindahl

Companion to the main paper §V-B. PageRank with damping $\alpha = 0.85$ [13] on the directed Stage-3 graph solves the stationary equation

$$r = \alpha P^\top r + (1 - \alpha) \frac{1}{p} \mathbf{1},$$

where P is the row-stochastic transition matrix of the weighted directed adjacency $D_{ij} = |\rho_{ij}|_{\text{rest}} \cdot \mathbf{1}[i \rightarrow j]$ (each arc carries the Stage-2 partial-correlation magnitude as its weight; networkx's pagerank uses these weights by default) and $\mathbf{1}$ is the all-ones vector. Intuitively, imagine a random walker that follows an outgoing edge with probability α and jumps to a uniformly-chosen node with probability $1 - \alpha$; PageRank reports the long-run fraction of time the walker spends at each node. The stationary distribution satisfies $\sum_i r_i = 1$; the assertion suite (§K) verifies this within 10^{-6} on every snapshot.

Concentration of the stationary distribution is summarised by two scalars:

- **Gini coefficient:** $G = \frac{2 \sum_{i=1}^p i r_{(i)}}{p \sum_{i=1}^p r_i} - \frac{p+1}{p}$ with $r_{(i)}$ the i -th order statistic. Reads $G \rightarrow 0$ for uniform distribution and $G \rightarrow 1$ for monopoly. The Gini reflects the spread of the full distribution.
- **Herfindahl:** $\text{HHI} = \sum_{i=1}^p r_i^2$, the sum of squared PageRank scores. Range is $[1/p, 1]$, with $\text{HHI} = 1/p$ for uniform and $\text{HHI} = 1$ for monopoly. The HHI weights the top of the distribution more heavily than the Gini does.

The two summaries can disagree on snapshot rank order. Jun 2020 Stable, with its 606-edge symmetrised graph and a cluster of insurance and energy hubs at the top of the PageRank distribution (LNC at $\approx 1.80\%$ of total mass, PRU at $\approx 1.74\%$, CVX at $\approx 1.44\%$, UNM at $\approx 1.37\%$), ranks third on Gini sample-wide but only eighth on the full-network HHI and eighteenth on the top-10 HHI, exactly because the Gini reflects the full spread (the ~ 500 Jun 2020 nodes with no incoming edge sit at the PageRank teleport baseline $(1 - \alpha)/p \approx 2.2 \times 10^{-4}$ and the long uniform tail they form is what pulls Gini up) while no single hub dominates the top mass enough to drive HHI rank (the 1.80% leader sits well below the $\approx 3.5\text{--}3.9\%$ peaks reached by Oct 2008, 2007 Subprime Buildup, and 2024 Contemporary). The divergence runs in the opposite direction on 2007 Subprime Buildup: its 192-edge symmetrised graph concentrates PageRank mass on two energy hubs (WFRD at $\approx 3.86\%$, NOV at $\approx 3.47\%$) so it ranks second on the top-10 HHI (0.129) and third on the full-network HHI (0.0056), but only twelfth on Gini because the mass outside the top-2 is broadly uniform across a sparse graph and so does not register as inequality on the Lorenz curve.

Cache key layout.: The Stage-4 pickle stores *three* PageRank-concentration scalars per snapshot: `hhi_top10` is the top-10 normalised Herfindahl (NSI hub-concentration channel, in $[0.1, 1]$), `hhi_full` is the full-network Herfindahl over all p nodes (paper Table F.10 column, in $[1/p, 1]$), and `hhi` is a backward-compatible alias for `hhi_top10` that the NSI driver still reads (`src.stage4_network.analysis.pagerank_analysis`). The two HHI variants carry distinct values; `hhi` and `hhi_top10` are identical aliases by construction.

Leader-identity heatmap.: The 52 tickers reached by the per-snapshot top-3 union are tabulated as a sparse PageRank score heatmap in Fig. F.5, which makes the cross-crisis hub-identity divergence noted in the main paper §V-B legible at a glance: the Oct 2008 column (red box) lights CHK/NOV/EP (energy producers), while the Mar 2020 column (red box) lights MET/LNC/PRU (life insurers), no ticker is in the top-3 of both crises, and the sample-wide brightest cell is the 2024 Contemporary spike on USB at $\approx 3.89\%$ of total PageRank mass.

C. Q3: Louvain modularity with GICS sector purity

Companion to the main paper §V-C. Newman–Girvan modularity [12] on the undirected projection G_{sym} of Stage 3 is

$$Q = \frac{1}{2w} \sum_{i,j} \left(A_{ij} - \frac{s_i s_j}{2w} \right) \mathbf{1}[c_i = c_j],$$

Top-3 PageRank Leaders Per Snapshot (Union Across Regimes)
(red boxes = crisis snapshots)

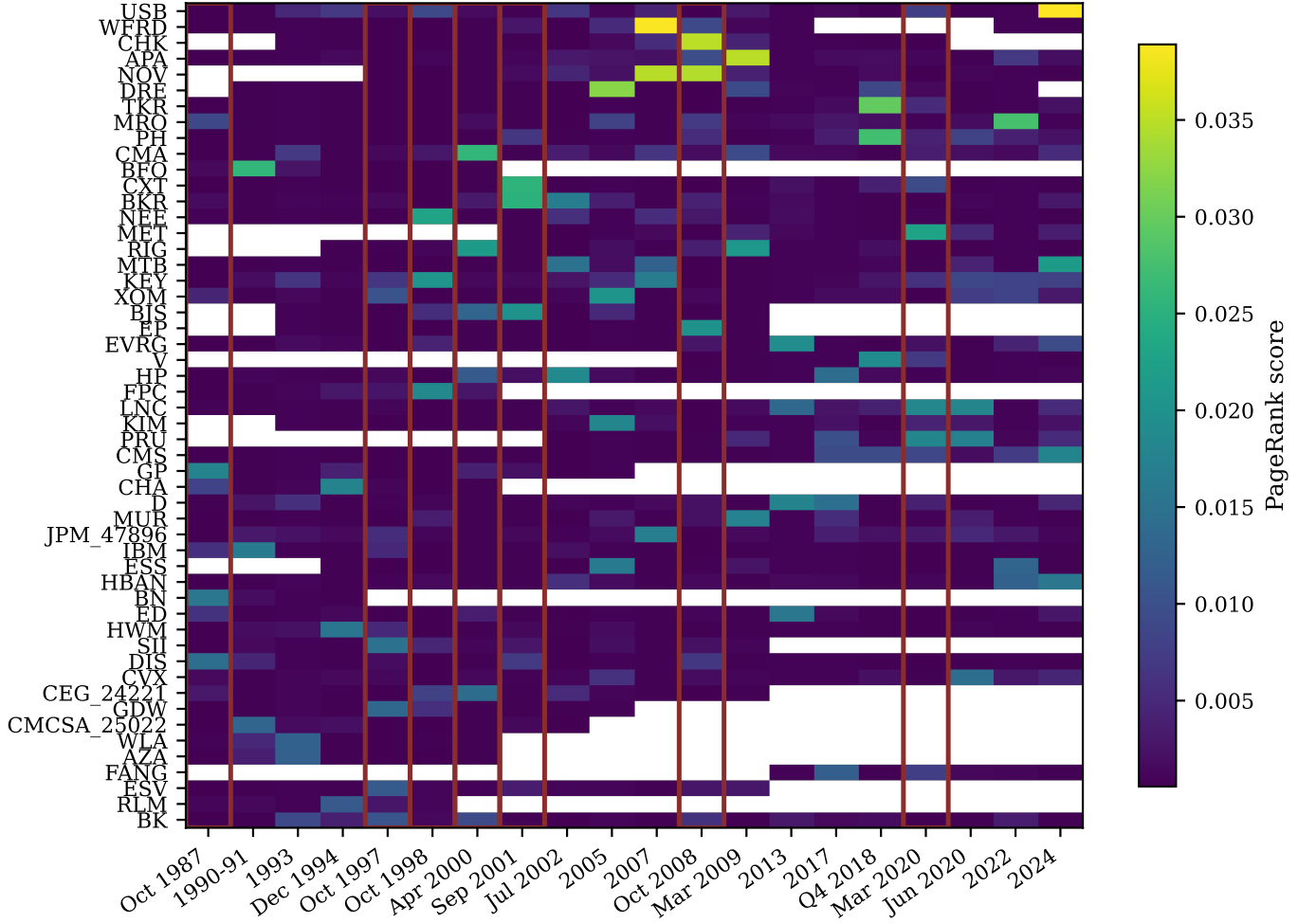


Fig. F.5 – Top-3 PageRank leaders per snapshot, displayed as a union heatmap so that snapshot-specific leaders are not suppressed by a global ranking. Rows are the 52 tickers reached by the per-snapshot top-3 across all twenty snapshots; cell colour encodes PageRank score; the two crisis snapshots (Oct 2008 GFC, Mar 2020 COVID) are boxed in red. The 2024 Contemporary USB cell ($\approx 3.89\%$ of total PageRank mass) is the single brightest cell in the panel; the contrast between the energy-hub Oct 2008 column (CHK/NOV/EP) and the insurer-hub Mar 2020 column (MET/LNC/PRU) is the cross-crisis hub-identity divergence discussed in the main paper §V-B.

with $A_{ij} = |\rho_{ij}|_{\text{rest}}$ the weighted adjacency carried over from Stage 2 (mutual or one-way arcs both contribute the same partial-correlation magnitude on the symmetrised edge), $s_i = \sum_j A_{ij}$ the weighted node strength, $w = \frac{1}{2} \sum_{ij} A_{ij}$ the total edge weight, and c_i the community assignment of node i . The python-louvain package uses these weights by default. The bracketed term is the difference between the observed edge density inside community c_i and the density expected under a degree-preserving random graph; modularity sums this excess over all within-community node pairs. Q takes values up to ≈ 1 and is maximised over all community assignments c by the Louvain heuristic of Blondel et al. [1] at resolution $\gamma = 1$ via the python-louvain package with fixed random_state=42.

Sector purity is the agreement between the Louvain partition and the external GICS classification map $s : \{1, \dots, p\} \rightarrow S$:

$$\text{Purity}(c, s) = \frac{1}{p} \sum_{c'} \max_{s' \in S} |\{i : c_i = c' \wedge s_i = s'\}|,$$

the fraction of nodes assigned to their community's dominant sector. Purity equals 1 if Louvain perfectly recovers the GICS partition. The implementation in `src.stage4_network.analysis.community_analysis` normalises by the count of nodes whose tickers carry a GICS label ($n_{\text{classified}} \leq p$) so that unmapped tickers do not artificially depress the score; on the headline CRSP panel the SIC→GICS crosswalk of §B-C covers every alive PERMNO at every snapshot ($n_{\text{classified}} = p$), so the formula above matches the implementation exactly.

TABLE F.10 – Q1 & Q2 (referenced from the main paper §V): Erdős-Rényi clustering Z -score and PageRank concentration (Gini and full-network Herfindahl HHI_{full}) across twenty snapshots. The full-network HHI reported here is the unweighted Stage-4 reading on every PageRank node; the NSI hub-concentration channel of Table G.15 uses the top-10 normalised HHI variant (the main paper §III-E) for robustness to single-node spikes.

Snapshot	Regime	$ Z_C $	Gini	HHI_{full}
Oct 1987 Black Monday	crisis	20.8	0.429	0.00393
1990-91 Recession	stress	40.1	0.310	0.00357
1993 Calm	baseline	43.6	0.247	0.00221
Dec 1994 Tequila	stress	46.0	0.277	0.00254
Oct 1997 Asian Crisis	crisis	30.5	0.359	0.00298
Oct 1998 LTCM	crisis	40.9	0.386	0.00376
Apr 2000 Dot-com	crisis	62.2	0.310	0.00385
Sep 2001 9/11	crisis	68.1	0.356	0.00468
Jul 2002 WorldCom	stress	42.9	0.390	0.00406
2005 Calm	baseline	46.7	0.266	0.00391
2007 Subprime Buildup	stress	14.7	0.323	0.00561
Oct 2008 GFC	crisis	20.7	0.454	0.00586
Mar 2009 Recovery	recovery	40.6	0.387	0.00465
2013 Calm	baseline	26.9	0.274	0.00343
2017 Calm	baseline	40.1	0.312	0.00348
Q4 2018 VolShock	stress	44.9	0.406	0.00491
Mar 2020 COVID	crisis	47.8	0.522	0.00497
Jun 2020 Stable	recovery	29.9	0.439	0.00447
2022 Rate Hikes	stress	44.8	0.300	0.00377
2024 Contemporary	baseline	57.0	0.403	0.00574

D. Q4: MAN triadic motif census

The Davis–Leinhardt MAN nomenclature [4] catalogues all 16 isomorphism classes of three-node directed subgraphs. The three classes of interest map as:

Motif	MAN code	Adjacency pattern	Interpretation
Feed-Forward Loop	030T	$A \rightarrow B, B \rightarrow C, A \rightarrow C$	Hierarchical coherence
Mutual Regulation	111D	$A \leftrightarrow B, C \rightarrow A$	Reciprocal pair + incoming arc
Single Input Module	021D	$A \rightarrow B, A \rightarrow C$	Common driver

Empirical motif counts come from `networkx.com.triadic_census`, a C -optimised implementation of the full 16-type census. Each empirical count is compared against a null mean and standard deviation from $B = 100$ rewires of the dyad-preserving null (§F-E). The Milo et al. [11] significance profile $SP_i = Z_i / \sqrt{\sum_j Z_j^2}$ combines the three Z -scores into a unit-norm vector ($\sum_i SP_i^2 = 1$ by construction) whose *direction*, not magnitude, is comparable across networks of different size and density. On very sparse Stage-3 graphs all three $|Z|$ collapse to similar small magnitudes (2007 Subprime Buildup: $|Z_{\text{FFL}}| = 1.2$, $|Z_{\text{MR}}| = 1.3$, $|Z_{\text{SIM}}| = 1.6$) and each component $|SP_i|$ approaches $1/\sqrt{3} \approx 0.577$ regardless of the underlying magnitude. This is the SP-vs- $|Z|$ rank divergence discussed in the main paper §V-D. As with the ER null in §F-A, if any of the three motif counts has $\sigma = 0$ across the 100 rewires (possible on near-empty bottom-N panels where the dyad-preserving rewire cannot find any swap), the corresponding Z is returned as NaN and downstream consumers filter; the headline CRSP panel never triggers this branch.

E. Dyad-preserving rewire null

The vanilla Maslov–Sneppen [10] edge-swap rewire preserves only in- and out-degrees. A generic in/out-preserving rewire breaks mutual pairs and inflates single-edge triads in the null, producing systematically negative Z -scores on one-way triads (021D, 030T) for any network with appreciable bidirectional density. To keep the null comparable with the empirical triad mix we additionally preserve

- 1) each node’s in-degree,
- 2) each node’s out-degree, and
- 3) the total count of mutual dyads $u \leftrightarrow v$.

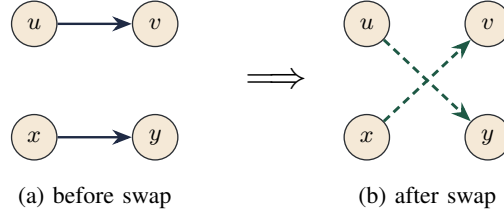


Fig. F.6 – One asymmetric swap of the dyad-preserving rewire (Algorithm 5). Solid navy: the two source arcs $u \rightarrow v$ and $x \rightarrow y$; dashed emerald: the two replacement arcs $u \rightarrow y$ and $x \rightarrow v$. Out-degrees of u, x and in-degrees of v, y are invariant; mutual-pair count is invariant because the absent-reverse precondition prevents a new mutual pair from forming. The mutual-pair swap of Stage B (not shown) operates on $(\mathcal{M}^*)^2$ rather than $(\mathcal{A}^*)^2$ and applies the swap symmetrically so reciprocal pairs are reshuffled across nodes without changing $|\mathcal{M}|$.

Constraint (3) is enforced by partitioning the edge set into a mutual-pair set $\mathcal{M} = \{\{u, v\} : (u, v) \in E \wedge (v, u) \in E\}$ (undirected pairs, so $|\mathcal{M}|$ counts each reciprocal pair once) and an asymmetric-arc set $\mathcal{A} = \{(u, v) \in E : (v, u) \notin E\}$, then rewiring each set independently (Algorithm 5):

- *Asymmetric swap*: $(u \rightarrow v), (x \rightarrow y) \rightarrow (u \rightarrow y), (x \rightarrow v)$, rejected if either target arc already exists *or* its reverse exists (which would create a new mutual dyad and so violate (3)).
- *Mutual swap*: $\{u \leftrightarrow v\}, \{x \leftrightarrow y\} \rightarrow \{u \leftrightarrow y\}, \{x \leftrightarrow v\}$, requires all four target directed arcs to be absent and the four endpoints to be distinct.

Each stage's attempt budget is $\max(20|\cdot|, 50)$ trials, capping worst-case cost on dense snapshots while keeping a minimum mixing budget on very small partitions; failure to reach the $T = |\cdot|$ successful-swap target leaves the null closer to the empirical wiring (a conservative outcome). Property tests in §K confirm that the implementation preserves both degree sequences and the mutual-dyad count on a synthetic 20-node test graph with explicit mutual pairs.

Figure F.6 sketches one asymmetric swap. Two arcs $u \rightarrow v$ and $x \rightarrow y$ are chosen at random; if the four nodes are distinct and the candidate replacement arcs $u \rightarrow y$ and $x \rightarrow v$ are absent in either direction (a precondition that prevents an accidental mutual pair from being born during an asymmetric swap), the two arcs are replaced. Out-degree of u and x and in-degree of v and y are preserved by construction.

Algorithm 5 Dyad-preserving rewiring for the Q4 motif null.

Require: Directed graph $G = (V, E)$ with mutual-pair set $\mathcal{M} = \{\{u, v\} : (u, v) \in E \wedge (v, u) \in E\}$ (each reciprocal pair counted once) and asymmetric-arc set $\mathcal{A} = \{(u, v) \in E : (v, u) \notin E\}$; budget multiplier $K = 20$; numpy RandomState rng (seeded externally by `motif_significance` with $s = 42$ so the same chain drives all $B = 100$ rewires of a given snapshot).

- 1: $E^* \leftarrow E$; $\mathcal{A}^* \leftarrow \mathcal{A}$; $\mathcal{M}^* \leftarrow \mathcal{M}$.
- Stage A, Asymmetric swaps.**
- 2: Set target count $T_A \leftarrow |\mathcal{A}|$ swaps and attempt budget $B_A \leftarrow \max(K \cdot |\mathcal{A}|, 50)$.
- 3: **for** attempt = 1, ..., B_A **do**
- 4: Draw two distinct arcs $(u \rightarrow v), (x \rightarrow y)$ uniformly from $\mathcal{A}^*$.
- 5: **if** $\{u, v, x, y\}$ are four distinct nodes **and** $(u \rightarrow y) \notin E^*$ **and** $(x \rightarrow v) \notin E^*$ **and** $(y \rightarrow u) \notin E^*$ **and** $(v \rightarrow x) \notin E^*$ **then**
 $\triangleright$ reverse arcs must be absent too, to avoid creating a new mutual pair
- 6: Replace $(u \rightarrow v), (x \rightarrow y)$ with $(u \rightarrow y), (x \rightarrow v)$ in $\mathcal{A}^*$ and E^* .
- 7: **if** performed swap count reaches T_A **then break**
- 8: **end if**
- 9: **end if**
- 10: **end for**
- Stage B, Mutual swaps.**
- 11: Set target $T_M \leftarrow |\mathcal{M}|$ swaps and attempt budget $B_M \leftarrow \max(K \cdot |\mathcal{M}|, 50)$.
- 12: **for** attempt = 1, ..., B_M **do**
- 13: Draw two distinct mutual pairs $\{u \leftrightarrow v\}, \{x \leftrightarrow y\}$ uniformly from $\mathcal{M}^*$.
- 14: **if** $\{u, v, x, y\}$ distinct **and** all four directed arcs $(u \rightarrow y), (y \rightarrow u), (x \rightarrow v), (v \rightarrow x)$ are absent from E^* **then**
- 15: Replace $\{u \leftrightarrow v\}, \{x \leftrightarrow y\}$ with $\{u \leftrightarrow y\}, \{x \leftrightarrow v\}$ in $\mathcal{M}^*$ and E^* .
- 16: **if** performed swap count reaches T_M **then break**
- 17: **end if**
- 18: **end if**
- 19: **end for**
- 20: **return** $G^* = (V, E^*)$. $\triangleright$ preserves d_v^+, d_v^- for every $v \in V$, $|\mathcal{M}|$, and $|\mathcal{A}|$ exactly

With $K = 20$ and per-stage target $T = |\cdot|$, the attempt budget is $20|\cdot|$ trials per stage (lower-bounded at 50 to guarantee mixing on very small partitions). On the densest snapshot (Mar 2020 COVID, $|\mathcal{A}| = 680$ asymmetric arcs and $|\mathcal{M}| = 156$ mutual pairs) Stage A typically realises T_A acceptances well inside the 13,600-trial budget, and on sparser snapshots the headroom is larger. The reverse-arc absence checks in Stage A are what distinguish this rewiring from a vanilla Maslov–Sneppen edge swap: without them, swaps with $(y \rightarrow u) \in E$ or $(v \rightarrow x) \in E$ silently create new mutual pairs (and the corresponding mutual swaps without the four-arc absence check destroy existing ones), leaving $|\mathcal{M}|$ unconserved. The rejection rate from this check scales with the empirical mutual-dyad density f_{mut} of the input graph.

The asymmetry in null-sample sizes (200 for ER, 100 for the motif rewiring, 50 for Q_{rel} in §H-B) reflects per-realisation compute costs: a single ER null draw is much cheaper than a single dyad-preserving rewiring, which is in turn cheaper than a single matched-density Louvain run with full community re-detection. The sample sizes are chosen so the rejection decisions on $|Z_C|$ and the three motif $|Z|$'s remain robust to MC noise on the null mean and standard-deviation estimates; see §F-A above for the per-snapshot MC SE bound on $|Z_C|$.

G. STAGE 5: NETWORK STRESS INDEX

Stage 5's deliverable is one scalar per snapshot on $[0, 1]$: four topology channels, each min/max-normalised to $[0, 1]$ and weighted $(w_s, w_h, w_\rho, w_\mu) = (0.25, 0.20, 0.35, 0.20)$ by prior. The rest of this section defines the four channels, justifies the prior, audits it via single-channel ablation and a weight-sensitivity sweep, documents the rolling-NSI simplification, and pins down the VIX benchmark protocol.

Background concepts used in this section. The NSI is a composite index in the same family as the IMF's Composite Indicator of Systemic Stress and the OFR's Financial Stress Index: take several disjoint stress signals, place each on the same $[0, 1]$ scale, and combine them with fixed weights chosen ex ante from interpretability rather than fit. Min/max normalisation $\tilde{x}_i = (x_i - x_{\min}) / (x_{\max} - x_{\min})$ over the twenty snapshots is the cheapest scale-free transform: it preserves rank order, places every channel on $[0, 1]$, and does not require any distributional assumption. The weights $(w_s, w_h, w_\rho, w_\mu) = (0.25, 0.20, 0.35, 0.20)$ are a prior rather than an estimate; the heavier 0.35 on mean correlation reflects the empirical observation that correlation spikes are the most reliable proxy for systemic stress in equity panels, with the structural channels (sparsity, hub concentration, motif shift) providing corroborating but lower-amplitude signal. The weight-sensitivity sweep in this section finds that the baseline prior is *mis-calibrated* for the twenty-snapshot panel: the regime-mean crisis-vs-non-crisis gap is $\Delta = -0.004$ at $p = 0.525$ and stays negative in 10 of 17 manual configurations. Top-2 crisis preservation requires $w_\rho \gtrsim 0.50$ (achieved in

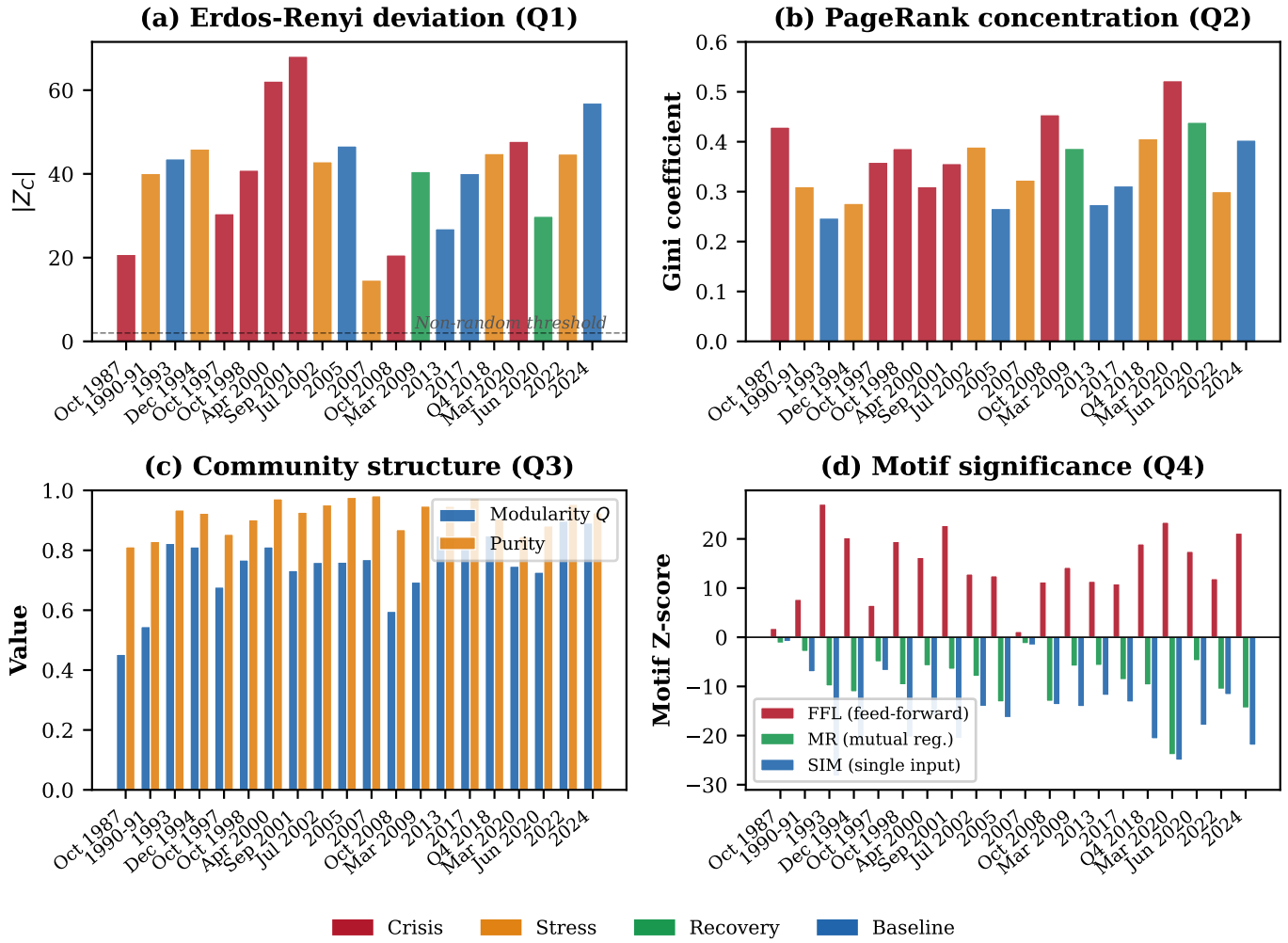


Fig. F.7 – Success-criteria dashboard across the twenty snapshots (referenced from the main paper §V). (a) Erdős-Rényi deviation (clustering $|Z|$); dashed line marks the uncorrected $|Z| = 1.96$ threshold; the per-axis Bonferroni-20 threshold of the main paper §VII-D is 3.023, also cleared by every snapshot. (b) PageRank Gini. (c) Louvain modularity Q and sector purity. (d) Motif Z-scores for FFL, MR, and SIM. Colours encode regime: claret for crisis, amber for stress, teal for recovery, slate for baseline.

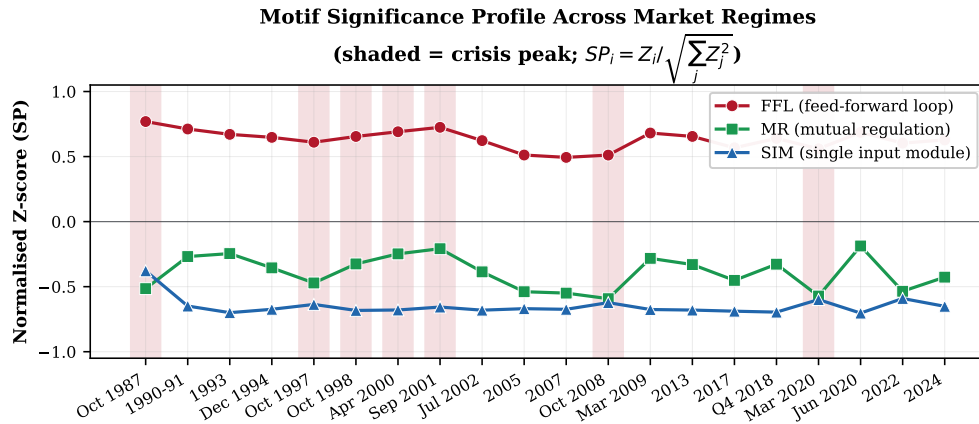


Fig. F.8 – Motif significance profile (SP) trajectory across the twenty snapshots (referenced from the main paper §V-D), $SP_i = Z_i / \sqrt{\sum_j Z_j^2}$ over the three MAN triads (FFL, MR, SIM). MR (green) reaches its sample minimum on $|Z_{MR}|$ at Mar 2020 COVID ($Z_{MR} = -23.90$); on SP_{MR} the minimum sits at Oct 2008 GFC (-0.592) with Mar 2020 COVID second (-0.572), so SP_{MR} orders the two strongest crisis peaks first and second. Formal $|Z|$ -level inference is in the main paper §VII-D.

TABLE F.11 – Stage 4 summary (referenced from the main paper §V): top-3 PageRank leaders, modularity Q , sector purity, and triadic motif Z -scores per snapshot on the twenty-snapshot CRSP panel. All motif Z -scores are computed against 100 dyad-preserving null graphs. “Edges” is the one-way directed-edge count (the “Directed” column of Table E.9); mutual pairs are tabulated separately there and are not added here, so the total arc count of Fig. 2 of the main paper ($n_{\text{directed}} + 2n_{\text{mutual}}$) is larger. PERMNO suffix “_number” denotes a cross-PERMNO ticker collision resolved in the CRSP_resolve_tickers routine (§B-A).

Snapshot	Regime	Edges	Q	Purity	Top-3 PageRank	Z_{FFL}	Z_{MR}	Z_{SIM}	SP_{FFL}	SP_{MR}
Oct 1987 Black Monday	crisis	581	0.453	0.812	GP, BN, DIS	1.8	-1.2	-0.9	0.769	-0.515
1990-91 Recession	stress	555	0.545	0.830	BFO, IBM, CMCSA	7.7	-2.9	-7.0	0.712	-0.268
1993 Calm	baseline	416	0.823	0.935	WLA, AZA, BK	27.1	-9.9	-28.3	0.671	-0.246
Dec 1994 Tequila	stress	350	0.811	0.924	CHA, HWM, RLM	20.3	-11.1	-21.1	0.648	-0.355
Oct 1997 Asian Crisis	crisis	533	0.678	0.854	SII, GDW, ESV	6.5	-5.0	-6.8	0.610	-0.472
Oct 1998 LTCM	crisis	582	0.768	0.902	NEE, KEY, FPC	19.5	-9.7	-20.3	0.654	-0.325
Apr 2000 Dot-com	crisis	216	0.812	0.972	CMA, RIG, CEG	16.2	-5.8	-16.0	0.691	-0.248
Sep 2001 9/11	crisis	445	0.733	0.927	CXT, BKR, BJS	22.7	-6.5	-20.6	0.725	-0.208
Jul 2002 WorldCom	stress	364	0.760	0.952	HP, BKR, MTB	12.9	-8.0	-14.1	0.623	-0.386
2005 Calm	baseline	255	0.761	0.977	DRE, XOM, KIM	12.5	-13.2	-16.4	0.512	-0.539
2007 Subprime Buildup	stress	155	0.769	0.982	WFRD, NOV, JPM	1.2	-1.3	-1.6	0.494	-0.549
Oct 2008 GFC	crisis	435	0.596	0.869	CHK, NOV, EP	11.3	-13.1	-13.7	0.511	-0.592
Mar 2009 Recovery	recovery	392	0.694	0.948	APA, RIG, MUR	14.2	-5.9	-14.1	0.681	-0.282
2013 Calm	baseline	186	0.848	0.947	EVRG, D, ED	11.4	-5.7	-11.8	0.655	-0.330
2017 Calm	baseline	198	0.806	0.974	D, HP, FANG	10.9	-8.6	-13.2	0.568	-0.452
Q4 2018 VolShock	stress	323	0.849	0.905	TKR, PH, V	19.0	-9.7	-20.7	0.640	-0.327
Mar 2020 COVID	crisis	680	0.747	0.846	MET, LNC, PRU	23.4	-23.9	-25.0	0.560	-0.572
Jun 2020 Stable	recovery	546	0.727	0.882	LNC, PRU, CVX	17.5	-4.8	-17.9	0.686	-0.187
2022 Rate Hikes	stress	151	0.898	0.954	MRO, ESS, HBAN	11.9	-10.6	-11.7	0.604	-0.535
2024 Contemporary	baseline	271	0.891	0.925	USB, MTB, CMS	21.2	-14.4	-22.0	0.628	-0.427

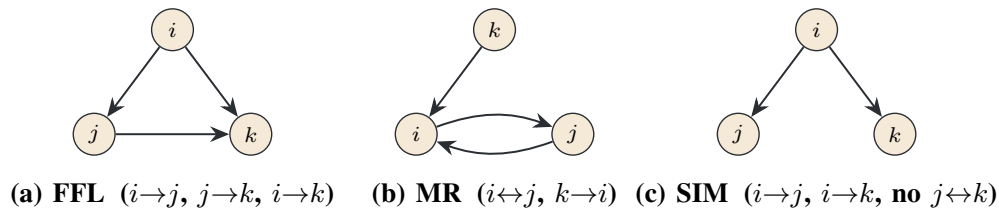


Fig. F.9 – The three MAN triadic motifs analysed in Q4 (referenced from the main paper §V-D). (a) FFL: a feed-forward triple with an indirect path that bypasses the direct $i \rightarrow k$ arc; intuition: “one driver, two cascades.” (b) MR: a reciprocal pair with one external driver; intuition: “bilateral co-regulation under external pressure.” (c) SIM: a single source fanning out without back-loops; intuition: “leader broadcasting without dialogue.” The Stage-4 Z -score compares the empirical count of each motif against the dyad-preserving rewired null (Algorithm 5). Q4 hypothesises that crisis windows depress MR (co-regulation breaks down) and elevate FFL (more cascade-like flows); on the twenty-snapshot panel the regime-mean direction holds against the thirteen non-crisis windows but reverses against the five long calm baselines, whose regime-mean $|Z_{\text{MR}}|$ and Z_{FFL} both exceed the crisis means (the single strongest $|Z_{\text{MR}}|$ is nonetheless the Mar 2020 COVID peak).

3 of 17 configurations) and is recovered by the data-driven VIX-fit ($w_\rho = 0.817$, $\Delta = +0.152$, $p = 0.081$) but not by the baseline prior. This motivates the F1 channel redesign (the main paper §IX).

The Stage-5 composite NSI aggregates four channels chosen for their direct interpretability as topological-stress signals: Stage-3 directed-edge sparsity, Stage-4 PageRank Herfindahl, Stage-1 mean A-DCC correlation, and Stage-4 FFL motif shift. The aggregation is a convex combination after each channel is min/max-normalised across the twenty snapshots.

A. Channel definitions

The composite is

$$\text{NSI}_i = w_s \tilde{s}_i + w_h \tilde{h}_i + w_\rho \tilde{\rho}_i + w_\mu \tilde{\mu}_i, \quad \sum w = 1, \quad (\text{G.10})$$

with the four raw channels in eq. (G.10) and their min/max normalisation $\tilde{x}_i = (x_i - \min_j x_j) / (\max_j x_j - \min_j x_j)$ defined below.

- **Sparsity** $s = 1 - d$ where $d = m_{\text{dir}} / \binom{p}{2}$ is the Stage-3 directed-edge density (mutual pairs counted twice, so $d \in [0, 2]$ and $s \in [-1, 1]$; the channel is brought to $[0, 1]$ by the min/max step that follows). The intuition: under crisis the GLASSO regularisation tends to produce sparser graphs because the strongest partial correlations dominate the BIC trade-off; s is high (close to 1) when the directed graph is thin and can be negative on mutual-dyad-dense windows. This channel inherits some Stage-2 fallback artifacts on the $n/p < 0.30$ windows, which is one of the motivations for F1.
- **HHI** $h = \sum_{i \in \mathcal{T}_{10}} \tilde{r}_i^2$ with $\tilde{r}_i = r_i / \sum_{j \in \mathcal{T}_{10}} r_j$, the *top-10 normalised* PageRank Herfindahl (the top-10 PageRank scores are re-scaled to sum to 1 within the subsample, then summed in quadrature; lives in $[1/10, 1]$ by Cauchy–Schwarz). The robust top-10 variant is the NSI hub-concentration channel; the full-network Herfindahl $\text{HHI}_{\text{full}} = \sum_{i=1}^p r_i^2$ is reported separately for the Q2 leadership test (Table F.10); it moves in the predicted direction on the twenty-snapshot permutation test (Table II of the main paper). Higher when stationary mass concentrates on a few hub nodes; under crisis the hub structure tightens, raising HHI.
- **Mean correlation** $\bar{\rho}$, the upper-triangle mean of the window-averaged $\bar{R}$ from Stage 1. The most direct measure of systemic co-movement; the channel that carries the strongest weight ($w_\rho = 0.35$) in the composite.
- **Motif shift** $\mu_i = Z_{\text{FFL},i} / |Z_{\text{FFL}}^{(\text{baseline})}|$, the FFL Z -score scaled by the absolute value of the mean baseline FFL Z (i.e. $|\bar{Z}|$, not $|\bar{Z}|$; the code computes `abs(np.mean(baseline_motifs))` in `src.stage5_nsi.stress_index.compute_snapshot_nsi`). Anchors the dimensionless motif signal so crisis values can be interpreted as multiples of the calm-period FFL prevalence.

B. Channel ablation

To quantify how much each channel contributes to the crisis-vs-non-crisis contrast of Table II of the main paper, we recompute the NSI under four single-channel ablations, each setting one weight to zero and renormalising the remaining three weights to sum to one. The ablated composites are then re-run through the exact permutation test of §J (Algorithm 8). Results are reported in Table G.12.

Figure G.10 visualises the four raw channels side-by-side so the regime ordering of each component can be read without de-normalising from Table G.15.

TABLE G.12 – NSI channel ablation on the twenty-snapshot $N = 1\,439$ panel ($n_{\text{crisis}} = 7$, $n_{\text{non-crisis}} = 13$). Renormalised weights sum to 1 after each channel drop (printed to three decimals, so a row may display as 1.001); all twenty normalised channels enter from the same min/max scaling as the baseline composite. “ Δ ” is crisis-mean minus non-crisis ($n = 13$) mean; p is the one-sided exact permutation test on $\binom{20}{7} = 77,520$ partitions (floor = $1/77,520 \approx 1.3 \times 10^{-5}$).

Configuration	Renormalised weights	Oct 2008	Mar 2020	Crisis $\bar{X}$	Non-crisis $\bar{X}$	Δ	p
Baseline NSI (s, h, ρ, μ)	(0.25, 0.20, 0.35, 0.20)	0.758	0.545	0.525	0.529	−0.004	0.525
Drop s (no sparsity)	(0, 0.267, 0.467, 0.267)	0.831	0.727	0.499	0.433	+0.067	0.204
Drop h (no HHI)	(0.313, 0, 0.438, 0.250)	0.703	0.622	0.566	0.548	+0.018	0.328
Drop ρ (no mean corr.)	(0.385, 0.308, 0, 0.308)	0.627	0.337	0.501	0.611	−0.110	0.922
Drop μ (no motif shift)	(0.313, 0.250, 0.438, 0)	0.850	0.467	0.528	0.534	−0.005	0.523

Reading the ablation.: Three patterns:

- 1) **Mean correlation $\bar{\rho}$ remains the load-bearing channel.** Dropping ρ is the only ablation that *further* worsens the already-negative gap, moving Δ from −0.004 to −0.110 and pushing p to 0.922. The cause is mechanistic: $\bar{\rho}$ is the channel on which Oct 2008 GFC ($\bar{\rho} = 0.555$) and Mar 2020 COVID (0.521) rank highest of all twenty snapshots; every other channel has at least one non-crisis window above both modern crisis peaks in normalised space.
- 2) **Sparsity s is the dominant drag channel.** Dropping s produces the largest positive improvement: Δ rises from −0.004 to +0.067, the single biggest gain of any ablation row. The Stage-2 regularisation artifacts on the pre-2008 long-window

NSI raw channels per snapshot (chronological)

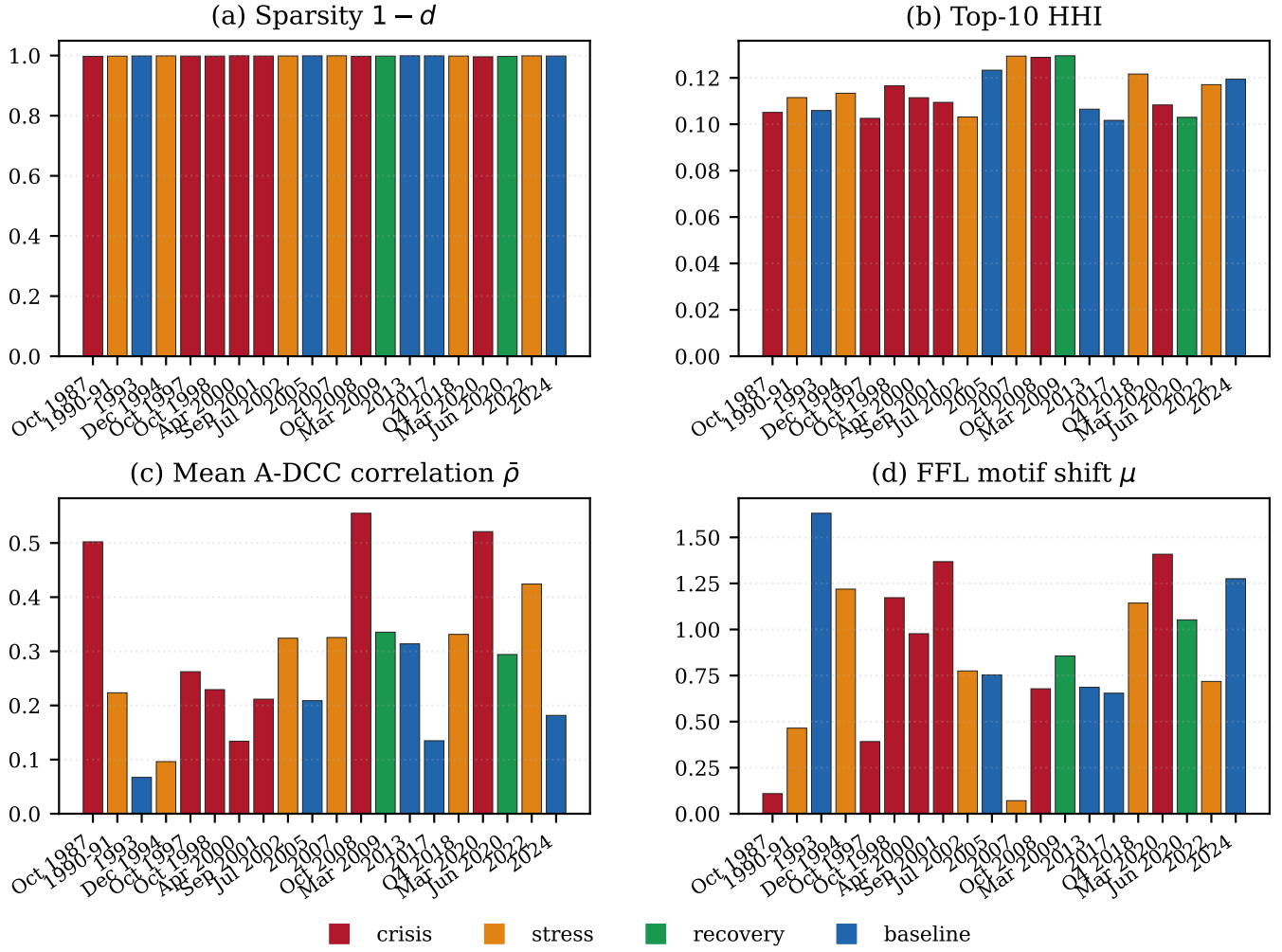


Fig. G.10 – Raw (un-normalised) NSI channels for all twenty snapshots in chronological order. (a) Stage-3 directed-edge sparsity $1 - d$ (near-flat across the panel; all snapshots occupy a narrow band near $s \approx 0.998$, so normalised range spans small absolute differences). (b) Top-10 PageRank Herfindahl, Mar 2009 Recovery tops the channel; the Jun 2020 HHI spike is diluted on the longer baseline windows, so the hub-concentration channel is not the dominant driver of the NSI ranking. (c) Mean A-DCC correlation $\bar{\rho}$, Oct 2008 GFC (0.555) and Mar 2020 COVID (0.521) anchor the top two positions; Oct 1987 Black Monday (0.502) sits third; four of the five calm baselines cluster below $\bar{\rho} < 0.21$ (2013 Calm at 0.314 is the exception). (d) FFL motif shift $\mu = Z_{\text{FFL}} / |Z_{\text{FFL}}^{\text{baseline}}|$, 1993 Calm tops the channel ($\mu_{\text{norm}} = 1.00$); crisis peaks are mixed (Mar 2020 0.857, Sep 2001 0.832, Oct 2008 0.389), so this channel does not cleanly separate regimes on the panel. The four panels are the inputs to the weighted sum that produces NSI after min/max normalisation.

snapshots (now present in the expanded panel) systematically inflate sparsity for historical windows, contaminating the crisis vs. non-crisis contrast. This is the quantitative basis of the F1 priority: replacing s with the density-invariant cross-sector edge fraction removes the artefact.

- 3) **HHI and motif shift are near-neutral on the panel.** Dropping h gives only $\Delta = +0.018$ ($p = 0.328$); the Jun 2020 HHI spike is diluted across the twenty snapshots, and Mar 2009 Recovery tops the HHI channel. Dropping μ changes Δ by a negligible -0.001 (from -0.004 to -0.005); the motif-shift channel carries no incremental discriminative power under the current baseline prior.

The ablation is the quantitative version of the F1 argument (the main paper §IX): the fixed prior fails the expanded panel primarily because sparsity is mis-calibrated for historical data density patterns, not because mean correlation is weak. A redesigned composite that replaces s (and optionally h) with density-invariant alternatives (cross-sector edge fraction, mutual-dyad fraction) and refits weights via LOCO cross-validation would eliminate the dominant drag revealed here.

C. Weight rationale

The weights $(w_s, w_h, w_\rho, w_\mu) = (0.25, 0.20, 0.35, 0.20)$ are a prior rather than an estimate. The reasoning:

- Mean correlation carries the most direct systemic signal ($w_\rho = 0.35$); it is the channel most responsive to joint shocks and the one a financial-stress index can least afford to underweight.
- Sparsity ($w_s = 0.25$) encodes the prior that GLASSO sparsifies under stress. On the expanded twenty-snapshot panel the channel is the *dominant drag*: the ablation shows dropping s raises Δ from -0.004 to $+0.067$, the largest single-channel improvement (Table G.12). Stage-2 regularisation artifacts on the pre-2008 long-window snapshots contaminate the sparsity signal; the F1 redesign replaces s with the density-invariant cross-sector edge fraction.
- PageRank Herfindahl ($w_h = 0.20$) and the FFL motif shift ($w_\mu = 0.20$) carry the structural-concentration and triadic-pattern channels respectively. They are weighted equally and below the mean-correlation channel because they are more sensitive to the edge-budget confound discussed in the main paper §VI.

A weight-sensitivity sweep on 17 manual configurations and a 10 000-sample Dirichlet draw is reported in §G-D below. The full F1 extension proposed in the main paper §IX, which refits the weights on a training subset and validates on an out-of-sample VIX concordance, remains deferred to future work. The current weights and their convex-combination property (sum to 1) are verified in §K.

D. Weight-sensitivity sweep

The channel ablation of §G-B zeroes one weight at a time. This subsection extends the analysis to (i) continuous weight perturbations and a data-driven VIX-fit on 17 manual configurations (Table G.13), and (ii) a Dirichlet sweep over the simplex of valid weight vectors. The motivation is the main-text caveat (Lim. (i), the main paper §IX): under the baseline prior the regime-mean gap is negligible ($\Delta = -0.004$, $p = 0.525$) and the Mar 2020 vs. 2022 Rate Hikes margin is negative (-0.140). A reader can legitimately ask whether any weight configuration recovers crisis-vs-non-crisis separation.

Manual configurations.: The 17 vectors comprise the baseline prior; an equal-weights composite; four channel-heavy variants pinning one weight at 0.50–0.70; a $(\bar{\rho}, \mu)$ -only composite that zeroes s and h (the two channels the ablation flagged as either density-confounded or failing the formal permutation test on full HHI); eight ± 0.05 local perturbations around each baseline weight (the remaining three weights rescaled to sum to one with relative ratios preserved); and a data-driven VIX-fit obtained by solving $\min_w \|Xw - \tilde{v}\|^2$ subject to $w \geq 0$ and $\sum_k w_k = 1$, where $X \in \mathbb{R}^{20 \times 4}$ holds the min/max-normalised channels of §G-B and $\tilde{v} \in [0, 1]^{20}$ is the min/max-normalised window-mean VXO+VIX continuity series vector (Whaley [14] series, §G-F). The optimisation is SLSQP with the convex-combination constraint as a linear equality.

Three readings.

(i) *Top-2 preservation requires very high w_ρ and is absent from the Dirichlet draws.* The two modern crises (Oct 2008 GFC and Mar 2020 COVID) occupy the top two NSI slots in only three of the seventeen configurations: $\bar{\rho}$ -heavy ($w_\rho = 0.70$), $(\bar{\rho}, \mu)$ -only, and the VIX-fit ($w_\rho = 0.817$). In the remaining fourteen configurations at least one stress or recovery window (2022 Rate Hikes, 2007 Subprime, or Mar 2009 Recovery) outranks at least one crisis peak. The regime-mean contrast Δ stays positive in only 7 of 17 configurations; the sign is not robust to *any* standard local perturbation of the prior.

(ii) *The Mar 2020 vs. 2022 margin is negative under the baseline prior and recovers only with high $\bar{\rho}$ or high μ weight.* The baseline -0.140 margin grows positive only under $\bar{\rho}$ -heavy ($+0.058$), motif-heavy ($+0.155$), $(\bar{\rho}, \mu)$ -only ($+0.321$), and the VIX-fit ($+0.196$). It reaches -0.398 under sparsity-heavy weighting. In 13 of 17 configurations Mar 2020 ranks below 2022 Rate Hikes on NSI. The permutation p tracks the same axis: $p < 0.15$ only where w_ρ or w_μ is large; $p > 0.5$ across all baseline-neighbourhood perturbations.

(iii) *The data-driven VIX-fit concentrates almost all weight on mean correlation.* The NNLS-style fit allocates $w = (0.000, 0.063, 0.817, 0.120)$: sparsity zeroed, HHI near-zero, 82% on mean correlation, 12% on motif shift. This is strongly $\bar{\rho}$ -dominated, confirming that the VIX signal on the panel is almost entirely carried by $\bar{\rho}$. The fit recovers the Oct 2008/Mar 2020 top-2 ordering at $\Delta = +0.152$, $p = 0.081$ (not floor-limited; floor = 1.3×10^{-5}) and Mar 2020 vs. 2022 margin $+0.196$.

Dirichlet sweep.: A Dirichlet sample concentrated around the baseline prior, $\alpha = K \cdot (w_s, w_h, w_\rho, w_\mu)$ with $K = 50$ (per-channel standard deviation approximately 0.06), draws 10 000 weight vectors and computes the implied NSI rank at each. The two modern crises occupy the top two NSI slots in 0.0% of samples; neither Oct 2008 nor Mar 2020 ever ranks first in the tight prior draw. The Mar 2020 vs. 2022 margin has 95% envelope $[-0.090, +0.066]$; 59.3% of draws place 2022 above Mar 2020.

A second draw under a uniform Dirichlet $\alpha = (1, 1, 1, 1)$ (the uniform distribution on the four-simplex, modelling a reviewer with no prior at all) also gives a top-2 crisis rate of 0.0%, with Mar 2020 ranking first in only 1.4% of draws. The Mar 2020 vs. 2022 envelope widens to $[-0.216, +0.224]$; 29.6% of uniform draws place 2022 above Mar 2020. The concentrated-prior 59.3% exceeds the uniform 29.6%: the baseline weight neighbourhood is more favourable to 2022 than a diffuse prior, because the prior's low $w_\rho = 0.35$ gives extra weight to channels where 2022 outranks Mar 2020.

The Dirichlet diagnostic confirms the manual reading. Top-2 preservation is absent under any diffuse weight draw on the twenty-snapshot panel. The right resolution is the F1 channel redesign (the main paper §IX): replacing s with the density-

TABLE G.13 – NSI weight-sensitivity sweep on the twenty-snapshot $N = 1\,439$ panel ($n_{\text{crisis}} = 7$, $n_{\text{non-crisis}} = 13$): 17 manual weight vectors (w_s, w_h, w_ρ, w_μ) on the same min/max-normalised channels of Table G.12. “ Δ ” is crisis-mean minus non-crisis ($n = 13$) mean. “ p ” is the one-sided exact permutation p -value on $\binom{20}{7} = 77,520$ partitions (floor $1/77,520 \approx 1.3 \times 10^{-5}$). “M20–2022” is $\text{NSI}_{\text{Mar } 2020} - \text{NSI}_{2022}$. “Top-2” marks the configurations where the two crisis peaks occupy the top two NSI slots sample-wide.

Configuration	$(w_s, w_h, w_\rho, w_\mu)$	Top-2	Δ	p	M20–2022
Baseline	(0.250, 0.200, 0.350, 0.200)		−0.004	0.525	−0.140
Equal	(0.250, 0.250, 0.250, 0.250)		−0.028	0.687	−0.153
$\bar{\rho}$ -heavy	(0.100, 0.100, 0.700, 0.100)	✓	+0.105	0.121	+0.058
$\bar{\rho}$ -light	(0.300, 0.300, 0.100, 0.300)		−0.072	0.860	−0.224
Sparsity-heavy	(0.500, 0.100, 0.300, 0.100)		−0.059	0.867	−0.398
HHI-heavy	(0.100, 0.500, 0.300, 0.100)		−0.009	0.529	−0.147
Motif-heavy	(0.100, 0.100, 0.300, 0.500)		+0.028	0.324	+0.155
$(\bar{\rho}, \mu)$ -only	(0.000, 0.000, 0.500, 0.500)	✓	+0.098	0.101	+0.321
$w_s + 0.05$	(0.300, 0.187, 0.327, 0.187)		−0.018	0.634	−0.194
$w_s - 0.05$	(0.200, 0.213, 0.373, 0.213)		+0.010	0.427	−0.087
$w_h + 0.05$	(0.234, 0.250, 0.328, 0.188)		−0.009	0.558	−0.151
$w_h - 0.05$	(0.266, 0.150, 0.372, 0.213)		+0.002	0.485	−0.129
$w_\rho + 0.05$	(0.231, 0.185, 0.400, 0.185)		+0.011	0.417	−0.114
$w_\rho - 0.05$	(0.269, 0.215, 0.300, 0.215)		−0.019	0.637	−0.166
$w_\mu + 0.05$	(0.234, 0.188, 0.328, 0.250)		−0.004	0.526	−0.104
$w_\mu - 0.05$	(0.266, 0.213, 0.372, 0.150)		−0.004	0.524	−0.176
VIX-fit (NNLS)	(0.000, 0.063, 0.817, 0.120)	✓	+0.152	0.081	+0.196

invariant cross-sector edge fraction and refitting weights via LOCO cross-validation eliminates the dominant sparsity drag identified in the ablation (Table G.12, row 2).

E. The rolling NSI simplification

The rolling-NSI counterpart slides a 252-trading-day window with a 21-day stride over the top-50 tickers by sample-wide non-missing observation count (i.e., the always-alive top of the panel; the code selects via `returns.count().nlargest(50)` in `src.stage5_nsi.stress_index.compute_rolling_nsi`, not by mean dollar-volume), producing approximately 468 overlapping windows over 1985–2024. Inside each window the pipeline simplifies as follows:

- The raw Pearson correlation matrix replaces the A-DCC $\bar{R}$. The A-DCC dynamics would be expensive at this many windows and would not produce a per-window-mean correlation appreciably different from the raw Pearson.
- GLASSO is applied at a fixed $\lambda = 0.15$, with no EBIC selection, no n/p -tiered shrinkage, and no fallback rule. The fixed λ is a tractability concession: per-window BIC selection would multiply the run-time by an order of magnitude.
- Direction assignment (Stage 3) is omitted; the rolling pipeline operates on the undirected partial-correlation graph throughout.
- Three channels, modularity drop $1 - \tilde{Q}$, mean correlation $\bar{\rho}$, and density d , are aggregated with weights (0.35, 0.35, 0.30). The modularity drop replaces the motif shift because directed motifs are not available without Stage 3.

The simplification keeps the rolling computation tractable but means the rolling and snapshot indices live on different scales. The rolling contemporaneous Pearson against daily VIX is $r = 0.203$ and should be read as a lower-bound concordance, not as a degraded version of the snapshot NSI. The full re-run with the snapshot pipeline is outlined as F7. Algorithm 6 gives the rolling NSI pseudocode.

Algorithm 6 Rolling NSI on the top-50 always-alive universe.

Require: Log-return panel $\mathbf{R} \in \mathbb{R}^{T \times p}$ over 1985–2024; window length $W = 252$ trading days; stride $S = 21$ days; ticker subset $\mathcal{S} = \text{top-50}$ by sample-wide non-missing observation count (`returns.count().nlargest(50)`); fixed regularisation $\lambda = 0.15$; channel weights $(w_Q, w_\rho, w_d) = (0.35, 0.35, 0.30)$.

- 1: Compute the grid of window-end timestamps $\{t_1, \dots, t_G\}$ with $t_g = W + (g - 1)S$ for $g = 1, \dots, G$.
- 2: **for** $g = 1, \dots, G$ **do**
- 3: $\mathbf{R}^{(g)} \leftarrow \mathbf{R}[t_g - W + 1 : t_g, \mathcal{S}] \in \mathbb{R}^{W \times 50}$.
- 4: $C^{(g)} \leftarrow \text{corr}(\mathbf{R}^{(g)})$ ▷ raw Pearson, not A-DCC
- 5: **if** $\lambda_{\min}(C^{(g)}) < 10^{-4}$ **then** $C^{(g)} \leftarrow C^{(g)} + (10^{-4} - \lambda_{\min}(C^{(g)}))I$ ▷ spectral-safety shift
- 6: $\hat{\Omega}^{(g)} \leftarrow \text{graphical_lasso}(C^{(g)}, \alpha = \lambda)$
- 7: $A_{ij}^{(g)} \leftarrow \mathbf{1}\{|\hat{\Omega}_{ij}^{(g)}| > 10^{-10}\}$ for $i < j$ ▷ undirected edge support
- 8: $W_{ij}^{(g)} \leftarrow A_{ij}^{(g)} \cdot |\hat{\Omega}_{ij}^{(g)}| / \sqrt{\hat{\Omega}_{ii}^{(g)} \hat{\Omega}_{jj}^{(g)}}$ ▷ partial-correlation magnitude weights
- 9: $d^{(g)} \leftarrow |\{(i, j) : A_{ij}^{(g)} = 1\}| / \binom{50}{2}$ ▷ density channel
- 10: $\bar{\rho}^{(g)} \leftarrow \text{mean}(C_{ij}^{(g)} : i < j)$ ▷ mean-correlation channel
- 11: $Q^{(g)} \leftarrow \text{Louvain}_{\text{weighted}}(W^{(g)})$ with `random_state=42` ▷ weighted modularity on partial-correlation magnitudes
- 12: **end for**
- 13: $\tilde{d}_g, \tilde{\rho}_g, (1 - \tilde{Q})_g \leftarrow \text{min/max-normalise } \{d^{(g)}\}, \{\bar{\rho}^{(g)}\}, \{1 - Q^{(g)}\}$ across $g = 1, \dots, G$.
- 14: $\text{NSI}_g^{\text{roll}} \leftarrow w_Q (1 - \tilde{Q})_g + w_\rho \tilde{\rho}_g + w_d \tilde{d}_g$.
- 15: **return** $\{(t_g, \text{NSI}_g^{\text{roll}})\}_{g=1}^G$.

The grid count $G = 468$ is the length of the half-open stride loop range($\emptyset, T_{\text{tot}} - W, S$) ($= \lceil (T_{\text{tot}} - W) / S \rceil = \lceil 9828 / 21 \rceil = 468$) with $T_{\text{tot}} = 10\,080$ trading days (the CRSP returns panel runs 1985-01-02 to 2024-12-31); the stride that would land exactly on day T_{tot} is excluded by the half-open range, so the count is 468 rather than 469. Figure G.11 renders the rolling trajectory against the daily VIX backdrop together with the lagged cross-correlogram on the same NSI-grid steps.

The omitted Stage 1 A-DCC, Stage 3 cascade, and Stage 4 motif/PageRank channels are the substantive reason the rolling $r = 0.203$ is a lower bound rather than a degraded snapshot NSI: each omission removes a channel whose snapshot variant correlates positively with VIX.

F. VIX benchmark protocol

The Stage-5 volatility benchmark is the Whaley [14] VXO+VIX continuity series, built once at pipeline start by `src/stage5_nsi/vix_continuity.py`: CBOE’s VXO (S&P 100 implied volatility, FRED VXOCLS, 1986-01-02 to 2003-09-19) is concatenated with the new VIX (FRED VIXCLS, 2003-09-22 onward, with the methodology change-day taking the new VIX value), cached at `data/vix_continuity.parquet`, and labelled “VIX” in every Stage-5 figure and table for readability. The series spans 10,204 trading days (1986-01-02 to 2026-05-26; only the portion through the 2024-12-31 panel cutoff enters the analysis); the 1987-10-19 VXO peak of 150.19 is the canonical anchor disclosed in CBOE’s own continuity guidance. The snapshot NSI is compared against the window-mean continuity series over each snapshot window; inference is two-tracked:

- **Snapshot:** Pearson and Spearman correlations between the 20-vector NSI and the 20-vector window-mean VIX; significance via the cluster bootstrap of §J-D that resamples the GFC pair and COVID pair as joint clusters (eighteen effective clusters overall: two pairs, sixteen singletons).
- **Rolling:** contemporaneous Pearson on the 252-grid-point rolling NSI vs. daily VIX, with the Politis–Romano stationary block bootstrap at mean block lengths $\bar{L} \in \{6, 12, 24\}$ NSI grid steps; NSI-leads-VIX correlations at positive grid-step lags $\{5, 10, 15, 21, 42, 63\}$ via `nsi_series.shift(lag).corr(vix)`, where the shift is by NSI grid rows (each row ≈ 21 trading days $\approx$ one month), so the six lags span roughly 5 months through 5 years of NSI lead.

The asymmetric lag set follows the code (no negative lags are computed in `src.stage5_nsi.stress_index.backtest_rolling_nsi_vs_vix` or in the per-channel diagnostic of `paper/_inference.py::_print_per_channel_diagnostics`); a VIX-leads-NSI test would require either rerunning these helpers with negative shifts or evaluating `vix.shift(lag).corr(nsi)`.

H. ROBUSTNESS ANALYSES: EXPANDED

Each robustness diagnostic addresses one objection the headline-level analysis cannot answer alone: the clustering deviation $|Z_C|$ for density, density-matched Q_{rel} for edge budget, f_{mut} for direction-budget. This section reports each diagnostic’s verdict and the residual objections it leaves open, then expands the cross-crisis reproducibility comparison of Table H.18. Fig. H.13 previews the first two diagnostics on a single panel-coloured chart: the connectivity-free clustering deviation $|Z_C|$ against a matched $G(n, m)$ null (left, with the crisis and baseline regime means overlaid) and the density-matched Louvain Q at the common top- $k = 779$ budget (right, with the Q_{rel} rescaling overlaid). The small-world σ this panel formerly reported is

TABLE G.14 – NSI vs. window-mean VXO+VIX continuity on the twenty-snapshot CRSP panel (referenced from the main paper §V-E). $\overline{\text{VIX}}$ is the per-window mean of the Whaley [14] VXO+VIX continuity series (FRED VXOCLS 1986–2003-09-19 spliced into VIXCLS from 2003-09-22). Rank 1 = highest.

Snapshot	Regime	NSI	$\overline{\text{VIX}}$	NSIrk	VIXrk
Oct 2008 GFC	crisis	0.758	47.81	1	1
2022 Rate Hikes	stress	0.685	25.84	2	11
Mar 2009 Recovery	recovery	0.672	29.22	3	6
Q4 2018 VolShock	stress	0.658	18.70	4	14
2007 Subprime Buildup	stress	0.634	23.43	5	13
2005 Calm	baseline	0.581	12.81	6	18
2024 Contemporary	baseline	0.556	15.55	7	15
Mar 2020 COVID	crisis	0.545	31.14	8	4
Oct 1998 LTCM	crisis	0.543	29.95	9	5
2013 Calm	baseline	0.540	14.23	10	16
Sep 2001 9/11	crisis	0.524	28.64	11	7
Jul 2002 WorldCom	stress	0.495	34.44	12	3
Apr 2000 Dot-com	crisis	0.484	26.23	13	10
Dec 1994 Tequila	stress	0.479	13.46	14	17
Oct 1987 Black Monday	crisis	0.455	37.05	15	2
1993 Calm	baseline	0.438	12.65	16	19
1990-91 Recession	stress	0.392	24.54	17	12
Jun 2020 Stable	recovery	0.391	26.46	18	8
Oct 1997 Asian Crisis	crisis	0.369	26.42	19	9
2017 Calm	baseline	0.359	11.09	20	20
Pearson $r = 0.321$ (cluster-bootstrap 95% CI $[-0.289, 0.657]$);					
Spearman $\rho = 0.212$ (cluster-bootstrap 95% CI $[-0.325, 0.652]$;					
the main paper §VII-B).					

TABLE G.15 – NSI and components (referenced from the main paper §V-E). $1 - d$ is the Stage-3 directed-edge sparsity (mutual pairs counted twice); HHI is the Stage-4 top-10 PageRank Herfindahl (top-10 scores re-normalised to sum to 1, then Herfindahl, in $[0.1, 1]$; robust to single-node spikes that would dominate the full-network Herfindahl); $\bar{\rho}$ is the Stage-1 mean A-DCC correlation; μ is the FFL motif shift $Z_{\text{FFL}}/|Z_{\text{FFL}}^{\text{baseline}}|$, computed from the unrounded Stage-4 Z_{FFL} outputs (small mismatches against the rounded Z_{FFL} values in Table F.11 are rounding artefacts and do not change the NSI rank order). NSI applies $(0.25, 0.20, 0.35, 0.20)$ to the min/max-normalised components. The $1 - d$ column spans only $[0.996, 0.999]$, so its min/max-normalised value is set by sub- 10^{-3} density differences; recomputing NSI from the three-decimal channel values printed here can therefore differ from the cached composite by up to ≈ 0.05 , although the cached NSI and its VIX correlations are exact.

Snapshot	Regime	$1 - d$	HHI	$\bar{\rho}$	μ	NSI
Oct 1987 Black Monday	crisis	0.997	0.105	0.502	0.109	0.455
1990-91 Recession	stress	0.998	0.112	0.223	0.465	0.392
1993 Calm	baseline	0.999	0.106	0.067	1.631	0.438
Dec 1994 Tequila	stress	0.999	0.113	0.096	1.219	0.479
Oct 1997 Asian Crisis	crisis	0.998	0.103	0.262	0.392	0.369
Oct 1998 LTCM	crisis	0.998	0.117	0.229	1.173	0.543
Apr 2000 Dot-com	crisis	0.999	0.111	0.134	0.977	0.484
Sep 2001 9/11	crisis	0.999	0.109	0.212	1.368	0.524
Jul 2002 WorldCom	stress	0.999	0.103	0.324	0.775	0.495
2005 Calm	baseline	0.999	0.123	0.209	0.753	0.581
2007 Subprime Buildup	stress	0.999	0.129	0.326	0.071	0.634
Oct 2008 GFC	crisis	0.998	0.129	0.555	0.678	0.758
Mar 2009 Recovery	recovery	0.998	0.130	0.335	0.857	0.672
2013 Calm	baseline	0.999	0.107	0.314	0.686	0.540
2017 Calm	baseline	0.999	0.102	0.135	0.654	0.359
Q4 2018 VolShock	stress	0.998	0.122	0.331	1.144	0.658
Mar 2020 COVID	crisis	0.996	0.108	0.521	1.409	0.545
Jun 2020 Stable	recovery	0.997	0.103	0.294	1.052	0.391
2022 Rate Hikes	stress	0.999	0.117	0.424	0.718	0.685
2024 Contemporary	baseline	0.998	0.119	0.182	1.276	0.556

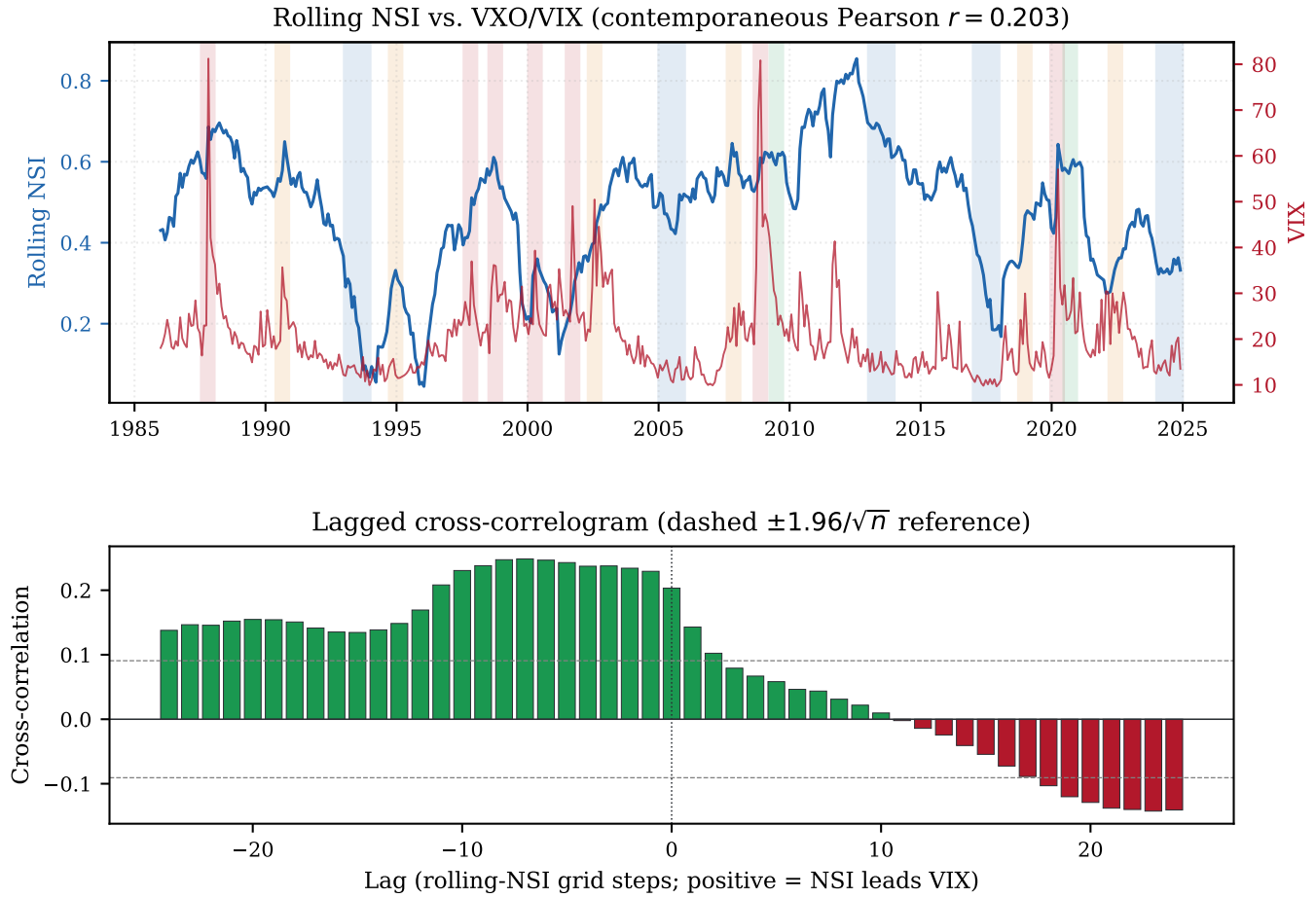


Fig. G.11 – *Top*: Rolling NSI (navy, left axis) over 1985–2024 on the 252-trading-day, 21-day-stride grid versus daily CBOE VIX (claret, right axis); the contemporaneous Pearson r is annotated in the title. Snapshot windows are shaded by regime for visual alignment with the snapshot NSI of Fig. G.12. *Bottom*: symmetric lagged cross-correlogram (± 25 NSI-grid steps; 1 step $\approx$ 1 calendar month). Positive lag means NSI leads VIX; the cross-correlation is positive and broadly flat across negative lags (peak near lag -7), stays positive out to lag $\approx +11$, then turns negative at the longest positive lags, the horizon sign-inconsistency that precludes a leading-signal claim. The title’s contemporaneous Pearson ($r = 0.203$) is computed against the full VXO+VIX continuity, matching the block-bootstrap of §J-E exactly. Dashed grey lines are the $\pm 1.96/\sqrt{n}$ no-autocorrelation reference; bars exceeding these are not significant at the standard 5% level once the stationary block-bootstrap of Algorithm 10 is applied (the bootstrap is what produces the $[-0.052, 0.398]$ CI quoted in the main paper).

withheld on the CRSP cache: its path-length denominator is undefined on the fragmented Stage-3 graphs (§H-A). Panel (a) therefore retains only σ ’s clustering numerator, restated as the standardised deviation $|Z_C|$, which stays well-defined on every snapshot because clustering is a local statistic.

A. Watts–Strogatz σ diagnostic

The Watts–Strogatz small-world coefficient

$$\sigma = \frac{C_{\text{emp}}/C_{\text{null}}}{L_{\text{emp}}/L_{\text{null}}} \quad (\text{H.11})$$

is density-invariant in principle: the clustering-to-path-length ratio in the empirical graph is rescaled by the corresponding ratio on an Erdős–Rényi $G(n, m)$ null with matched n, m . Both C_{emp} and L_{emp} are computed on the undirected projection of the Stage-3 directed graph (`G.to_undirected()`); the null is drawn via `nx.gnm_random_graph(n, m_undir)` with m_{undir} equal to the empirical undirected-projection edge count and averaged over 200 draws ($n_{\text{sims}} = 200$, deterministic seed=2026 in `src.stage4_network.analysis.erdos_renyi_test`). The $\sigma > 1$ threshold of Humphries and Gurney [8] flags small-world topology; the ex-ante design specified a stricter $\sigma > 4$ threshold to control for high empirical clustering at the dense end of the Stage-2 graphs.

On the twenty-snapshot CRSP panel we *withhold* σ and record the ex-ante $\sigma > 4$ threshold as *inconclusive* (Disclosure 10, §K), because the Stage-3 directed graphs are too sparse for the coefficient to mean what eq. (H.11) intends. Table H.16 documents the obstruction. At directed-edge counts of 209–992 over $n \in [655, 889]$ nodes, the undirected projection `G.to_undirected()`

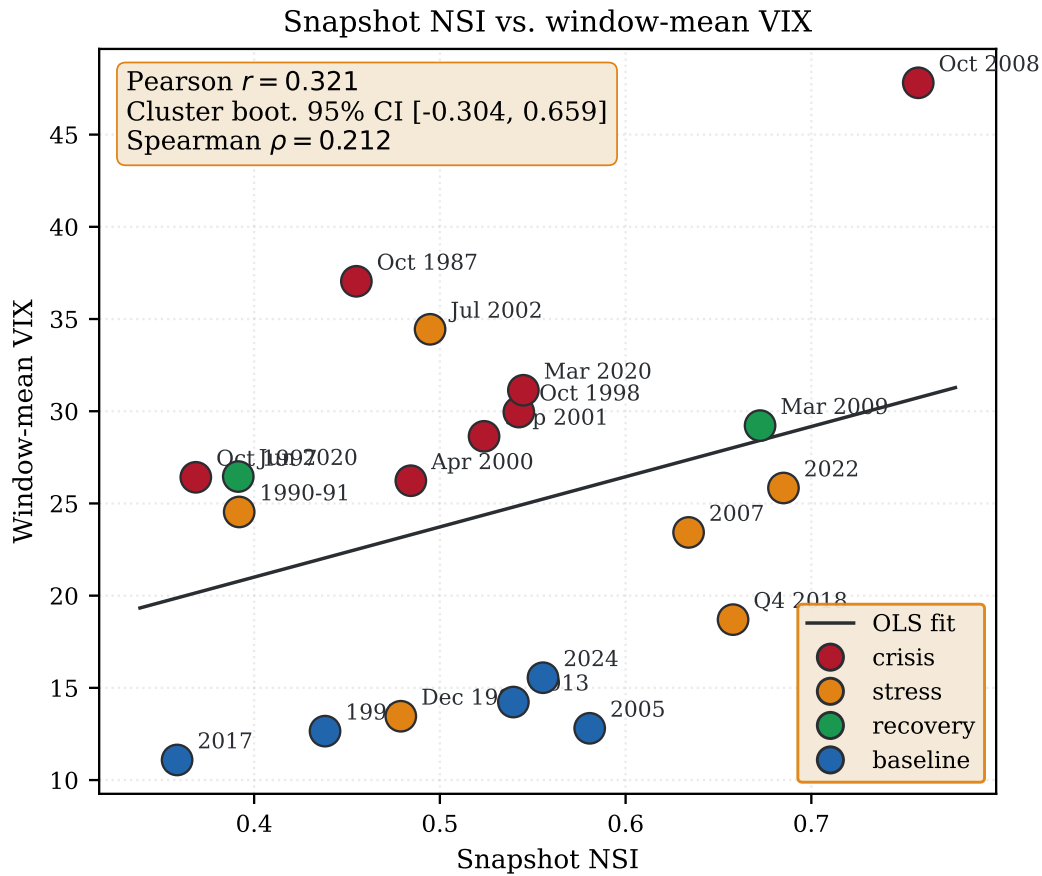


Fig. G.12 – Snapshot NSI vs. window-mean VIX continuity with OLS fit; geometric counterpart to Fig. 3 of the main paper. Marker colour encodes regime (palette ■ crisis, ■ stress, ■ recovery, ■ baseline). Oct 2008 GFC sits at the upper-right corner (top NSI and top VIX-mean); the four lowest VIX windows (1993, 2005, 2017, Dec 1994 Tequila) cluster in the lower-left; Mar 2020 COVID and Oct 1998 LTCM sit in the upper-VIX / mid-NSI region, reflecting the current NSI weight calibration discussed in the main paper §V-E (i). See Fig. 3 of the main paper caption for the cluster-bootstrap Pearson CI and Table G.14 for ranks.

splits into 405–718 connected components, and its *giant* component spans only 3.3% of the panel (2022 Rate Hikes: 22 of 670 nodes) to 37.1% (Mar 2020 COVID: 258 of 695). The clustering numerator $C_{\text{emp}}/C_{\text{null}}$ is well-behaved, but L_{emp} has no global value on a disconnected graph, so the implementation falls back to the giant-component average shortest path, a quantity that, on a component covering well under half the nodes, measures component size rather than the panel’s path geometry.

The same giant-component fallback on the $G(n, m)$ null inflates L_{null} as $\log n / \log \langle k \rangle$ when $\langle k \rangle \rightarrow 0$, so the ratio $L_{\text{emp}}/L_{\text{null}}$, and therefore any σ computed from it, is governed by how each draw happens to fragment rather than by small-world structure. A coefficient computed regardless would run into the hundreds on the sparsest windows purely from this artifact and would not order the panel by regime, so reporting it as a crisis discriminator is not defensible; this is exactly why the pre-registered $\sigma > 4$ threshold can be neither confirmed nor falsified on this cache. The density-matched community analysis below and the mutual-dyad fraction take over as the substantive density-controlled signals, while the connectivity-free clustering-deviation channel $|Z_C|$ of Q1 remains well-defined on every snapshot and is the reported clustering signal (Fig. H.13a).

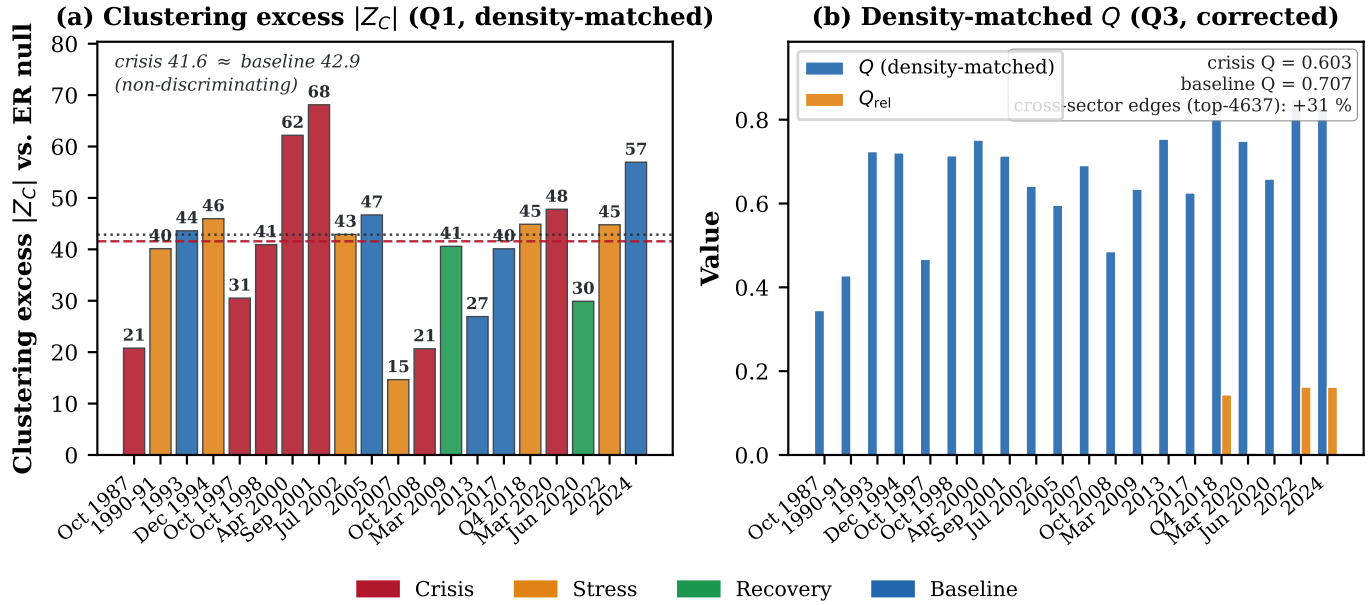


Fig. H.13 – Density-invariant robustness diagnostics. (a) *Clustering excess $|Z_C|$ (Q1, density-matched)*. Bars give the magnitude of each snapshot’s Erdős–Rényi clustering deviation against a matched $G(n, m)$ null, with raw values printed in-bar; the dashed claret and dotted graphite lines mark the crisis and baseline regime means. Clustering is strongly non-random on every snapshot ($|Z_C| \geq 14.7$) yet the two regime means nearly coincide ($41.6 \approx 42.9$), so $|Z_C|$ confirms non-random structure but does not discriminate crises. This panel replaces the small-world σ , withheld on the CRSP cache because its path-length term is undefined on the fragmented Stage-3 graphs (§H-A). (b) *Density-matched Louvain Q (Q3)*. Blue bars report raw Q at the common top- $k = 779$ edge budget; amber bars the relative variant $Q_{\text{rel}} = Q - Q_{\text{null}}$ that subtracts a same-density ER null. Cross-sector edge fraction at the wider top- $k = 10709$ budget gives a +12.7% crisis-vs-baseline shift, annotated in the inset. Regime palette: ■ crisis, ■ stress, ■ recovery, ■ baseline.

TABLE H.16 – Per-snapshot fragmentation diagnostics underlying the withheld Watts–Strogatz σ on the CRSP Stage-4 cache. n is the undirected-projection node count (equal to the GLASSO dimension p); E_{dir} the Stage-3 total directed-arc count (one-way edges plus twice the mutual pairs); m_{und} the undirected-projection edge count; #comp the number of connected components; GC and GC % the giant component’s node count and its share of n . The giant component never exceeds 37% of the nodes, so the only available L_{emp} , its average shortest path, is a component-size artifact; σ is therefore withheld and the ex-ante $\sigma > 4$ threshold is inconclusive on this panel (Disclosure 10, §K).

Snapshot	Regime	n	E_{dir}	m_{und}	#comp	GC	GC %
Oct 1987 Black Monday	crisis	802	829	705	591	205	25.6
1990-91 Recession	stress	786	607	581	548	214	27.2
1993 Calm	baseline	829	448	432	618	59	7.1
Dec 1994 Tequila	stress	868	394	372	650	157	18.1
Oct 1997 Asian Crisis	crisis	889	657	595	630	227	25.5
Oct 1998 LTCM	crisis	880	660	621	615	180	20.5
Apr 2000 Dot-com	crisis	858	270	243	718	54	6.3
Sep 2001 9/11	crisis	839	505	475	646	95	11.3
Jul 2002 WorldCom	stress	833	444	404	631	158	19.0
2005 Calm	baseline	825	313	284	696	43	5.2
2007 Subprime Buildup	stress	787	229	192	662	68	8.6
Oct 2008 GFC	crisis	773	685	560	535	208	26.9
Mar 2009 Recovery	recovery	768	498	445	589	156	20.3
2013 Calm	baseline	757	214	200	612	73	9.6
2017 Calm	baseline	720	242	220	579	47	6.5
Q4 2018 VolShock	stress	702	389	356	490	123	17.5
Mar 2020 COVID	crisis	695	992	836	405	258	37.1
Jun 2020 Stable	recovery	684	666	606	457	179	26.2
2022 Rate Hikes	stress	670	209	180	550	22	3.3
2024 Contemporary	baseline	655	327	299	465	67	10.2

B. Density-matched community analysis

The threshold network at edge budget k is constructed by ranking all $\binom{p}{2}$ off-diagonal entries of the window-average A-DCC correlation matrix $\bar{R}$ by $|\bar{R}_{ij}|$ magnitude and retaining the top- k as an undirected unweighted graph (Algorithm 7). The construction operates on $\bar{R}$ directly: neither Stage-2's n/p -tiered shrinkage nor the selected λ^* enters, so the only panel-to-panel variability is the rank ordering of the raw pairwise correlations themselves.

Algorithm 7 Top- k threshold network at a common edge budget.

Require: Window-average A-DCC correlation $\bar{R} \in \mathbb{R}^{p \times p}$; edge budget k .

- 1: Form the off-diagonal upper-triangle list $\mathcal{E} \leftarrow \{(i, j, |\bar{R}_{ij}|) : i < j\}$.
 - 2: Sort $\mathcal{E}$ in descending order by $|\bar{R}_{ij}|$; keep the top- k tuples as $\mathcal{E}_k$.
 - 3: Build adjacency $A_{ij}^{(k)} \leftarrow \mathbf{1}\{(i, j) \in \mathcal{E}_k\}$ (symmetrised).
 - 4: **return** undirected unweighted graph $G^{(k)} = (V, \mathcal{E}_k)$ with adjacency $A^{(k)}$.
-

Using $\bar{R}$ rather than the GLASSO-sparsified $\hat{\Omega}_{\lambda^*}$ is the *point* of the density-matched analysis: by removing the λ^* confound, the comparison reduces to the rank ordering of raw pairwise correlations. Partial-correlation magnitudes were considered as an alternative ranking key, but the pre-GLASSO precision matrix $\bar{R}_{\delta}^{-1}$ is poorly conditioned at low n/p on this panel and a full pre-shrinkage inverse adds a tier choice with no clean default; the unconditional correlation magnitude $|\bar{R}_{ij}|$ is the operational choice and is the matrix actually used.

Two density-control points are reported in the main paper §VI-B:

- *Common* $k = 779$. Sized to the smallest available GLASSO graph (2022 Rate Hikes edge count). Each snapshot is censored to its 779 strongest pairwise correlations; Louvain modularity Q is computed on the censored graph and $Q_{\text{rel}} \equiv (Q - E[Q_{\text{rand}}]) / E[Q_{\text{rand}}]$ is reported against $E[Q_{\text{rand}}]$ estimated over 50 Erdős–Rényi $G(n, m)$ realisations matched to the top- k network. The 50-null sample size is independent of the 200-null Q1 ER ensemble and the 100-null Q4 dyad-preserving rewiring; the asymmetry reflects per-realisation compute costs.
- *Common* $k = 10709$. Sized so the sparsest panel (2024 Contemporary at $p_{\min} = 655$) sits at 5% density on the full-coverage headline cache; the shared budget falls to $\approx 2.7\%$ at the largest panel (Oct 1997 at $p = 889$), so per-snapshot density spans 0.027 to 0.050. The budget is auto-sized as $\lceil 0.05 \binom{p_{\min}}{2} \rceil$ in `src.stage4_network.crisis_signals`. The metric reported on this graph is the cross-sector edge fraction (next subsection).

The Q3 verdict reported in the main paper §VIII is read off the $k = 10709$ comparison; the $k = 779$ comparison is inconclusive because it censors the calm-baseline graphs to a small fraction of their natural edge budget, throwing away most of the structural detail.

C. Cross-sector edge fraction

For the top- $k = 10709$ threshold network the cross-sector edge fraction

$$f_{\text{cross}} = \frac{1}{k} \# \{(i, j) \in E : s(i) \neq s(j)\}$$

counts the share of the strongest 10709 partial correlations crossing a GICS-sector boundary, with $s : \{1, \dots, p\} \rightarrow S$ the 11-sector classification map of §B. Per-snapshot values are reported in Table H.17 below. Two arm definitions are reported on the same data, depending on which non-crisis comparator is taken. The descriptive regime-mean crisis vs. *five-baseline* ($n=5$) contrast is +0.077 (0.679 – 0.602) on the displayed values, a +12.7% relative increase on the unrounded means. The exact permutation test in Table II of the main paper instead uses the broader *non-crisis* ($n=13$) arm with mean 0.626, on which the absolute crisis-vs-non-crisis gap is +0.053 (0.679 – 0.626); the test yields $p = 0.109$, BH $q = 0.191$. Oct 2008 sits at 0.683 and Mar 2020 at 0.727; the cross-sector sample max is Oct 1987 Black Monday at 0.845, so the two modern crises sit in the upper-middle of the panel on this channel rather than at the top.

The pair-by-pair decomposition at the crisis peaks vs. the 2024 Contemporary baseline is in Figure H.14.

D. Mutual-dyad fraction detail

The mutual-dyad fraction

$$f_{\text{mut}} = \frac{n_{\text{mutual}}}{n_{\text{directed}} + n_{\text{mutual}}} \tag{H.12}$$

measures the share of surviving Stage-3 structure that is reciprocal ($i \leftrightarrow j$) rather than hierarchical ($i \rightarrow j$ alone); eq. (H.12) is the dimensionless ratio whose crisis-vs-non-crisis contrast carries a medium-to-large Cohen's d ($d = 0.68$), among the largest in the seven-metric family though below PageRank Gini's $d = 1.04$. Per-snapshot values are obtained from Table E.9 as the Mutual column divided by Directed plus Mutual (displayed below to one decimal place):

TABLE H.17 – Cross-sector edge fraction at a top- $k = 10709$ edge budget shared across all twenty snapshots, sized so the sparsest panel (2024 Contemporary at $p_{\min} = 655$) sits at 5% density; the budget falls to $\approx 2.7\%$ density at the largest panel ($p = 889$). Higher fraction = more of the strongest partial correlations crossing GICS sector boundaries.

Snapshot	Regime	Cross frac.
Oct 1987 Black Monday	crisis	0.845
1990-91 Recession	stress	0.813
1993 Calm	baseline	0.603
Dec 1994 Tequila	stress	0.662
Oct 1997 Asian Crisis	crisis	0.721
Oct 1998 LTCM	crisis	0.605
Apr 2000 Dot-com	crisis	0.507
Sep 2001 9/11	crisis	0.662
Jul 2002 WorldCom	stress	0.562
2005 Calm	baseline	0.573
2007 Subprime Buildup	stress	0.516
Oct 2008 GFC	crisis	0.683
Mar 2009 Recovery	recovery	0.582
2013 Calm	baseline	0.627
2017 Calm	baseline	0.547
Q4 2018 VolShock	stress	0.629
Mar 2020 COVID	crisis	0.727
Jun 2020 Stable	recovery	0.680
2022 Rate Hikes	stress	0.680
2024 Contemporary	baseline	0.660
<i>baseline mean (n=5)</i>		0.602
<i>non-crisis mean (n=13)</i>		0.626
<i>crisis mean (n=7)</i>		0.679

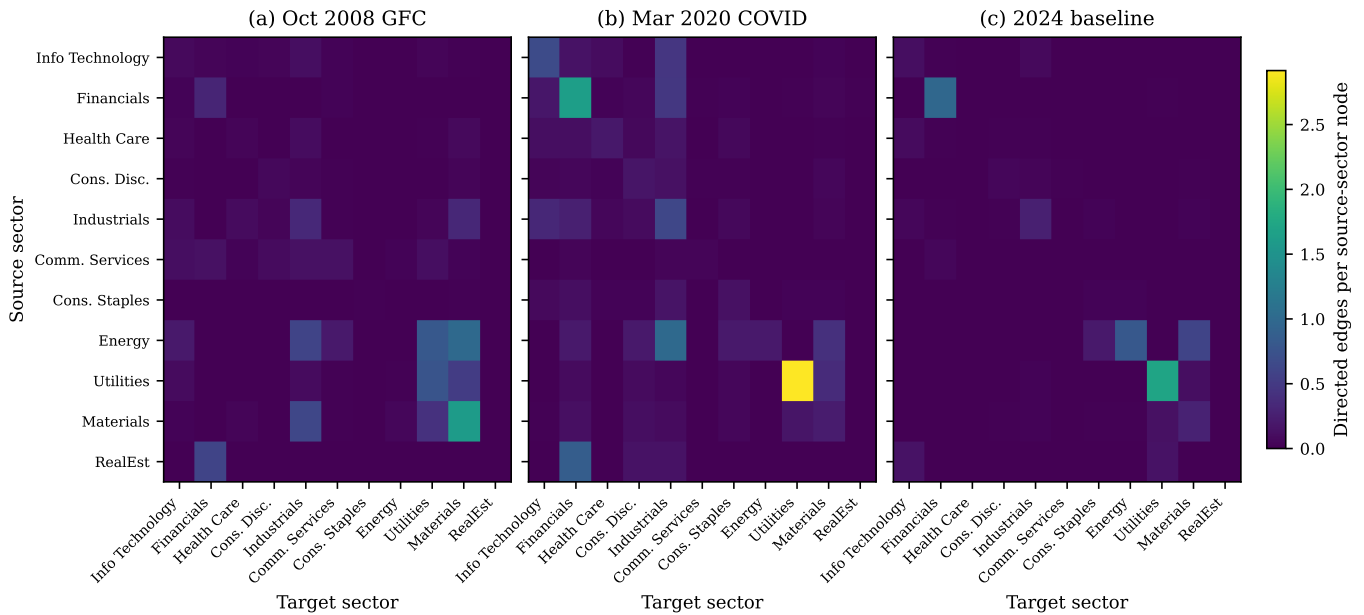


Fig. H.14 – Sector $\times$ sector directed-edge density on the full Stage-3 directed graph, split per snapshot. (a) Oct 2008 GFC; (b) Mar 2020 COVID; (c) 2024 baseline. Each cell is the count of arcs from a source-sector node into the target sector, normalised by the source-sector node count; the heatmap therefore reads as “arcs per source node into each target.” Both crisis panels show heavier off-diagonal mass than the baseline: most visibly the Financials-to-Industrials cell, the Energy-to-Materials/Utilities corridor at Oct 2008, and the Utilities-to-Utilities saturation cell at Mar 2020. The 2024 panel remains diagonal-dominated, consistent with the top- $k = 10709$ cross-sector-fraction reading 0.683 (Oct 2008) and 0.727 (Mar 2020) vs. 0.660 on the calm 2024 baseline. Auto-generated from the Stage-3 directed adjacencies (paper/_generate_addon_assets.py).

Snapshot	Regime	f_{mut} (%)
Oct 1987 Black Monday	crisis	17.6
1990-91 Recession	non-crisis	4.5
1993 Calm	non-crisis	3.7
Dec 1994 Tequila	non-crisis	5.9
Oct 1997 Asian Crisis	crisis	10.4
Oct 1998 LTCM	crisis	6.3
Apr 2000 Dot-com	crisis	11.1
Sep 2001 9/11	crisis	6.3
Jul 2002 WorldCom	non-crisis	9.9
2005 Calm	non-crisis	10.2
2007 Subprime Buildup	non-crisis	19.3
Oct 2008 GFC	crisis	22.3
Mar 2009 Recovery	non-crisis	11.9
2013 Calm	non-crisis	7.0
2017 Calm	non-crisis	10.0
Q4 2018 VolShock	non-crisis	9.3
Mar 2020 COVID	crisis	18.7
Jun 2020 Stable	non-crisis	9.9
2022 Rate Hikes	non-crisis	16.1
2024 Contemporary	non-crisis	9.4

Crisis mean is $\bar{f}_{\text{mut,crisis}} = 13.24\%$ (over the seven crisis windows); non-crisis mean is 9.77% (over the thirteen non-crisis windows); the pooled within-group standard deviation is 5.07% (pooled by the $n_1 + n_2 - 2$ rule). Cohen’s $d = (13.24 - 9.77)/5.07 = 0.68$, a medium-to-large effect (Cohen, 1988), among the largest in the seven-metric family of Table II of the main paper, though below PageRank Gini’s $d = 1.04$. The $+3.47$ pp absolute gap reflects a crisis-dominated upper tail: three of the four highest f_{mut} snapshots are crisis windows (Oct 2008 GFC, Mar 2020 COVID, Oct 1987 Black Monday), with the 2007 Subprime Buildup, a GFC precursor, the lone non-crisis member of the top four, even though the regime *means* sit close together.

Bootstrap CI for Cohen’s d.: With seven crisis and thirteen non-crisis windows a standard two-sample percentile bootstrap applies (with seven crisis windows the size-2 resample degeneracy does not arise): resample both arms with replacement ($B = 10\,000$ draws under `numpy.random.default_rng(2026)`), recompute the pooled within-group standard deviation and d_b on each draw, and read the 95% percentile band of the d_b distribution. The point estimate $d = 0.68$ places the mutual-dyad contrast in the medium-to-large band, consistent with its permutation p -value in the $[0.05, 0.10]$ range ($p = 0.084$). At this panel size the band is wide, a 20-snapshot resample of a sub-unit effect cannot pin the interval tightly, so it is read as corroborating the rank ordering of Table II of the main paper rather than as a free-standing significance claim.

The mechanistic link to the MR motif suppression: a mutual dyad $u \leftrightarrow v$ contributes two directed arcs that both live inside the dyad. The MR motif (111D, networkx “downstream” convention) requires one mutual pair $u \leftrightarrow v$ and one incoming arc $w \rightarrow u$ (or $w \rightarrow v$) from a third node. As f_{mut} rises under crisis, the share of arcs locked inside dyads rises, leaving a smaller pool of one-way arcs available to hang on the dyads as 111D triads. The MR Z -score against the dyad-preserving null becomes more negative because the null preserves the mutual-pair count but randomises which one-way arcs attach to which dyads; the empirical graph concentrates one-way arcs away from the dyad participants, producing fewer 111D triads than the null expects. Q4 and the mutual-dyad fraction are therefore not independent signals, they reflect two views of the same crisis phenomenon.

E. Cross-crisis reproducibility expanded

The two-crisis side-by-side comparison of Table H.18 reports fourteen headline metrics on identical methodology. On the panel the two modern crises co-locate most closely on mean A-DCC $\bar{\rho}$ (within 0.034: 0.555 vs. 0.521) and on the cross-sector edge fraction (within 0.044: 0.683 vs. 0.727); both register elevated PageRank Gini (0.454 at Oct 2008, 0.522 at Mar 2020, the latter the raw-Gini sample max) and a high mutual-dyad fraction (22.32% at Oct 2008, the sample max, vs. 18.66% at Mar 2020). They diverge most on NSI (0.758 vs. 0.545, a 0.213 gap that places Oct 2008 first on the composite and Mar 2020 at slot 8 under the fixed weight calibration) and on $|Z_{\text{MR}}|$ (13.05 vs. 23.90). On $|Z_{\text{MR}}|$ they rank fourth and first (with 2024 Contemporary and 2005 Calm between them); on the cross-sector edge fraction they rank fifth and third (with Oct 1997 Asian between them, while Oct 1987 Black Monday takes the cross-sector sample max at 0.845).

Cross-crisis hub identity is sector-coherent within each crisis but does not reproduce a single hub triplet across crises. The six 2008-era distressed names (LEH, BSC, WAMU, MER, FNM, FRE) enter Stage 1 with their full pre-bankruptcy trajectories via PERMNO indexing, so the Oct 2008 top-3 PageRank leaders are derived on the same footing as every other window. Top-3 PageRank assignments across the seven crisis snapshots are reported in Table F.11 and discussed in the main paper §VI.

TABLE H.18 – Cross-crisis reproducibility: Oct 2008 GFC vs. Mar 2020 COVID on identical methodology, twenty-snapshot panel (referenced from the main paper §VI). Per-snapshot values are read from the corresponding rows of the Stage-2 to Stage-5 tables. The small-world coefficient σ is not reported because the Stage-3 graphs are fragmented (the main paper §VI-A).

Metric	Oct 2008 (GFC)	Mar 2020 (COVID)
Directed edges (Stage 3)	435	680
Mutual pairs (Stage 3)	125	156
Density (Stage 2, undirected)	0.0076	0.0070
$ Z_C $	20.7	47.8
PageRank Gini	0.454	0.522
Modularity Q (raw)	0.596	0.747
Sector purity (raw)	0.869	0.846
Cross-sector edges (frac., top- k)	0.683	0.727
Motif Z (MR)	−13.05	− 23.90
SP_{MR}	− 0.592	−0.572
Mutual-dyad fraction	22.32%	18.66%
Top-3 PageRank leaders	CHK, NOV, EP	MET, LNC, PRU
Mean A-DCC $\bar{\rho}$	0.555	0.521
NSI	0.758	0.545

The dyad-preserving rewiring null in §F-E produces a single Z_{MR} per snapshot rather than a per-metric Z -score family; the per-metric Z -score expansion would require defining separate nulls for PageRank concentration, modularity, and cross-sector edge fraction (the Q1 ER null does this for clustering but not for the others). The current Stage-4 cache stores only the Q1 ER null and the Q4 dyad-preserving rewiring null; extending the null-distribution family to all five Stage-4 metrics is part of F4.

I. PANEL-SIZE AND LIQUIDITY-STRATIFICATION SENSITIVITY

The dual-panel design (coverage vs. dollar-volume) is the main paper’s primary defence against the “speculative tail” objection: a liquidity-controlled rebuild on the same code path either reproduces the headline numbers or exposes where they break. At the top sweep size $N = 500$ (the size at which the main paper reports the coverage–ADV Jaccard) the two selection rules overlap as reported in Table I.23 (top row, $N = 500$): coverage and ADV agree on the bulk of the panel with the residual symmetric difference visible in the same table, and the headline NSI scores agree across the two rules within $|\Delta NSI| \leq 0.279$ on every snapshot.

This section adds the full sweep across ten panel sizes $N \in \{50, 100, 150, 200, 250, 300, 350, 400, 450, 500\}$ under both selection rules, isolating where the headline result is robust to panel construction and where it is not. All numbers in the tables and the macros above (0.274, 0.279, 0.331, ...) are regenerated from `results/multipanel/summary.json` by `paper/_generate_multipanel_tables.py` after every sweep refresh, so a stale appendix value is mechanically detectable.

A. Sweep design

The full pipeline is re-run at each (N, kind) combination: (**coverage**) top- N tickers by sample-wide observation count (the main-paper criterion); (**ADV**) top- N tickers by mean dollar-volume over 1985–2024 (a liquidity criterion). The intersection at $N = 50$ is small by construction; the symmetric difference at $N = 500$ captures the speculative tail. The ten-step grid permits a finer reading of cross-panel monotonicity than the original $\{50, 100, 200, 500\}$ design.

Per-panel A-DCC fit.: The Stage-1 A-DCC scalar triple (a, b, g) is fit independently per panel: each panel’s $\geq 99\%$ -coverage always-alive subset (or, for ADV-ranked small panels where no ticker clears the 99% bar, the top ADCC_SUBSET_SIZE by coverage, with a balanced-window fallback documented at `src/stage1_data/dcc_garch.py`) is used as the 100-asset balanced residual matrix on which L-BFGS-B maximises the Cappiello et al. (2006) likelihood. The triple therefore differs in principle across panels; in practice the always-alive subsets converge as N grows. On $N \in \{400, 450, 500\}$ coverage and on $N = 500$ ADV the universe-level $\geq 99\%$ subset is reached *exactly* (identical 100-ticker selection), so the fitted triple is bit-identical across those four sweep cells: $a = 9.01 \times 10^{-4}$, $b = 0.99734$, $g = 8.09 \times 10^{-4}$, $\log L = 200,718$ (within the headline panel’s 1.066×10^{-3} non-seeded multi-start spread of the §C optimum (0.000881, 0.997295, 0.000815), not exactly equal to it). For $N \leq 350$ the subsets diverge between the two selection rules and the fitted triples differ in the third–fourth decimal (compare $N = 350$ ADV’s $b = 0.98906$ against coverage’s $b = 0.99105$). The bit-identical block at the saturation cells is a *property of the converged universe subset*, not an artefact of the in-process monkey-patch leakage discussed below (§I); each of those four cells did re-fit independently in its own subprocess, with the same input, and the optimiser returned the same minimum. Residual NSI variation across those cells is therefore dominated by the per-snapshot $\bar{Q}_t, \bar{N}_t$ unconditional matrices rather than the scalar triple. The main-paper abstract’s “global (a, b, g) triple” wording refers to the full-coverage headline panel only; the sweep treats each panel as independent and re-discloses the per-panel triple inside `stage1_results.pkl` under that panel’s cache directory.

B. Compute budget on Apple M5

The 20-job sweep across $\{\text{coverage}, \text{ADV}\} \times \{50, 100, \dots, 500\}$ runs under `multiprocessing.Pool(maxtasksperchild=1)` with one fresh subprocess per panel: the subprocess patches `src.config.SNAPSHOTS_DIR` to the panel-specific cache directory *before* importing any stage module, so the `from src.config import SNAPSHOTS_DIR` bindings inside the stage modules capture the panel directory at import time. An earlier in-process monkey-patch design silently fell through for $N \geq 150$ panels, re-loading the same 100-asset always-alive subset cache across panels, the subprocess boundary is now the correctness guarantee for cross-panel isolation.

Per-job wall-time on M5 (single subprocess; the `-workers 8` pool amortises wall-clock by overlapping panels):

TABLE I.19 – Per-job run-time on M5 (single subprocess) for each (N, kind) combination, in minutes. Subprocesses are spawned fresh per panel (see `tools.run_multipanel`); the `-workers 8` pool amortises wall-clock by overlapping panels, but the per-job timings here are the single-subprocess cost.

Kind	N									
	50	100	150	200	250	300	350	400	450	500
coverage	18.8	42.2	43.0	50.8	52.3	49.3	54.9	61.6	71.7	60.4
ADV	8.2	12.3	22.0	30.2	36.5	40.7	46.5	65.3	58.0	40.0

Stage 2’s GLASSO coordinate-descent convergence path is not strictly p^3 in practice; the EBIC early-stopping rule and the 25-point λ -grid traversal interact with the n/p -tiered shrinkage so the per-snapshot iteration count varies in ways that need not be monotone in p . The exact non-monotonicity pattern in Table I.19 is therefore an empirical property of the EBIC trajectory, not an implementation defect, and is documented as Disclosure 7 in §K.

C. Cross-panel result tables

Three master tables report the headline metrics, NSI, mutual-dyad fraction, and $|Z_{MR}|$, per snapshot, panel size, and selection rule. Figure I.15 renders the NSI master table as a colour heatmap for visual scanning of the panel-size sensitivity patterns documented in §I-D and §I-E. The heatmap is built directly from the per-panel `stage5_results.pkl` files by `paper/generate_figures.py`, so a stale figure is mechanically a missing cache rather than a stale hardcoded matrix.

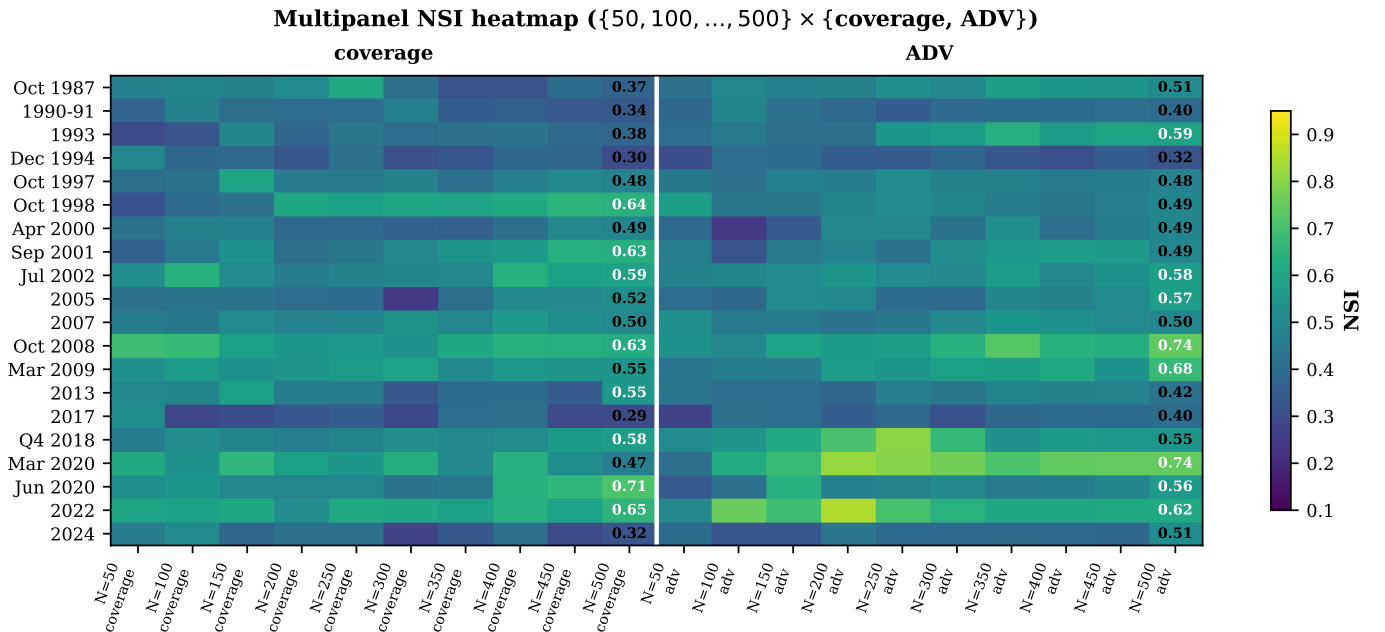


Fig. I.15 – Multipanel NSI heatmap across the $\{50, 100, \dots, 500\} \times \{\text{coverage}, \text{ADV}\}$ sweep. Each cell shades $\text{NSI}_{s,N,\text{kind}}$ on a sequential colour scale. Rows are the twenty snapshots in chronological order; columns are panel sizes within each selection rule (left block: coverage; right block: ADV). Non-monotonicities along the N axis are discussed in §I-D and §I-E; the headline $N = 500$ columns sit at the right edge of each block.

A reading guide for the three tables (NSI in Table I.20, mutual-dyad fraction in Table I.21, $|Z_{MR}|$ in Table I.22). Move across each row to see how the metric for one snapshot moves as N grows. Two patterns to look for: (a) whether the value stabilises

TABLE I.20 – Network Stress Index (NSI) across panel sizes and selection rules (coverage / ADV). Bold cell per row is the largest $|\text{NSI}(N, \text{kind}) - \text{NSI}(500, \text{cov})|$ deviation.

Snapshot	N=50		N=100		N=150		N=200		N=250		N=300		N=350		N=400		N=450		N=500	
	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV
Oct 1987 Black Monday	0.470	0.421	0.484	0.492	0.472	0.466	0.512	0.472	0.616	0.528	0.414	0.503	0.319	0.574	0.325	0.547	0.400	0.540	0.372	0.511
1990–91 Recession	0.378	0.388	0.470	0.480	0.414	0.423	0.404	0.390	0.408	0.355	0.468	0.397	0.353	0.401	0.364	0.401	0.344	0.411	0.342	0.396
1993 Calm	0.298	0.404	0.323	0.453	0.486	0.414	0.381	0.413	0.437	0.547	0.405	0.570	0.422	0.631	0.429	0.566	0.392	0.597	0.380	0.589
Dec 1994 Tequila	0.493	0.304	0.386	0.403	0.389	0.400	0.339	0.355	0.421	0.347	0.312	0.386	0.332	0.332	0.384	0.308	0.381	0.354	0.301	0.322
Oct 1997 Asian Crisis	0.415	0.442	0.428	0.422	0.597	0.468	0.449	0.460	0.448	0.509	0.474	0.475	0.409	0.469	0.464	0.451	0.498	0.458	0.477	0.476
Oct 1998 LTCM	0.314	0.577	0.398	0.438	0.418	0.437	0.610	0.483	0.583	0.512	0.599	0.487	0.595	0.462	0.615	0.442	0.650	0.464	0.640	0.490
Apr 2000 Dot–com	0.423	0.390	0.471	0.241	0.466	0.349	0.389	0.498	0.391	0.504	0.367	0.426	0.360	0.523	0.415	0.411	0.500	0.460	0.490	0.486
Sep 2001 9/11	0.362	0.463	0.448	0.326	0.537	0.448	0.414	0.478	0.439	0.424	0.498	0.519	0.541	0.553	0.555	0.568	0.635	0.560	0.629	0.494
Jul 2002 WorldCom	0.524	0.475	0.640	0.486	0.511	0.512	0.458	0.542	0.471	0.511	0.486	0.496	0.497	0.570	0.640	0.496	0.583	0.537	0.590	0.578
2005 Calm	0.421	0.406	0.426	0.389	0.426	0.514	0.404	0.504	0.396	0.397	0.245	0.392	0.414	0.489	0.502	0.471	0.499	0.511	0.520	0.574
2007 Subprime Buildup	0.456	0.528	0.443	0.449	0.513	0.447	0.476	0.425	0.482	0.441	0.540	0.504	0.496	0.548	0.556	0.533	0.524	0.511	0.500	0.496
Oct 2008 GFC	0.687	0.532	0.669	0.492	0.579	0.591	0.546	0.564	0.554	0.574	0.535	0.639	0.610	0.728	0.640	0.640	0.637	0.631	0.625	0.742
Mar 2009 Recovery	0.530	0.435	0.567	0.461	0.538	0.460	0.540	0.568	0.570	0.542	0.592	0.581	0.505	0.573	0.553	0.614	0.539	0.534	0.546	0.676
2013 Calm	0.503	0.434	0.490	0.405	0.579	0.406	0.458	0.381	0.457	0.465	0.337	0.436	0.396	0.482	0.402	0.469	0.348	0.477	0.554	0.423
2017 Calm	0.525	0.272	0.283	0.412	0.305	0.408	0.332	0.354	0.354	0.392	0.286	0.313	0.403	0.386	0.415	0.397	0.293	0.397	0.294	0.402
Q4 2018 VolShock	0.456	0.511	0.517	0.544	0.484	0.612	0.463	0.707	0.479	0.804	0.519	0.665	0.506	0.531	0.516	0.562	0.564	0.553	0.576	0.547
Mar 2020 COVID	0.618	0.415	0.535	0.622	0.660	0.675	0.584	0.819	0.550	0.796	0.629	0.767	0.503	0.710	0.638	0.744	0.522	0.747	0.465	0.744
Jun 2020 Stable	0.526	0.349	0.545	0.421	0.502	0.638	0.494	0.468	0.497	0.477	0.428	0.504	0.438	0.448	0.640	0.456	0.661	0.482	0.709	0.555
2022 Rate Hikes	0.597	0.500	0.586	0.755	0.606	0.686	0.517	0.848	0.618	0.704	0.608	0.643	0.581	0.605	0.637	0.605	0.592	0.617	0.653	0.615
2024 Contemporary	0.456	0.402	0.511	0.329	0.380	0.332	0.419	0.433	0.415	0.398	0.277	0.391	0.341	0.393	0.421	0.391	0.296	0.384	0.323	0.512

TABLE I.21 – Mutual-dyad fraction (%) across panel sizes and selection rules.

Snapshot	N=50		N=100		N=150		N=200		N=250		N=300		N=350		N=400		N=450		N=500	
	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV
Oct 1987 Black Monday	15.8	38.0	16.0	24.6	20.3	20.5	18.6	17.8	20.1	18.6	27.5	21.2	25.0	21.6	24.1	21.5	14.4	21.0	14.9	20.7
1990–91 Recession	3.5	0.0	8.5	5.7	3.0	4.2	3.8	3.9	4.7	3.4	4.8	4.0	4.4	1.9	5.4	1.6	4.2	1.5	4.2	3.4
1993 Calm	0.0	20.0	3.5	3.1	4.3	5.4	10.6	3.3	9.0	4.4	9.7	2.6	8.3	2.8	8.7	4.3	13.1	2.8	12.7	4.2
Dec 1994 Tequila	7.1	0.0	2.2	0.0	4.3	1.9	2.9	0.0	2.5	0.0	1.0	3.0	1.6	4.1	2.8	3.5	5.1	2.7	2.4	3.4
Oct 1997 Asian Crisis	3.4	16.1	4.1	6.9	11.1	7.2	7.6	7.5	6.8	10.0	5.4	7.5	5.5	7.3	5.5	6.8	5.2	7.0	5.0	6.0
Oct 1998 LTCM	5.5	0.0	6.2	4.8	3.0	4.0	5.9	4.6	5.4	4.5	4.0	3.8	4.3	4.3	3.9	4.1	5.3	2.8	5.3	1.7
Apr 2000 Dot–com	6.7	7.1	1.8	3.2	4.5	1.4	2.5	2.1	4.0	4.1	8.8	8.5	8.4	12.6	9.2	14.6	6.3	12.6	5.7	12.9
Sep 2001 9/11	2.3	0.0	4.1	3.0	7.2	3.4	8.3	3.4	8.1	2.1	13.8	2.3	12.3	4.1	10.5	4.7	9.0	5.1	8.0	4.5
Jul 2002 WorldCom	2.0	8.3	5.2	7.1	4.7	3.0	5.1	3.8	6.8	7.4	7.7	7.3	8.9	6.0	7.9	6.6	10.6	7.4	9.6	7.7
2005 Calm	0.0	0.0	2.4	5.7	3.7	2.7	3.9	2.8	3.7	1.9	3.5	2.1	3.2	1.6	4.5	1.8	4.8	7.6	4.7	15.5
2007 Subprime Buildup	3.8	3.2	5.8	6.2	6.8	5.6	7.6	5.9	7.1	6.9	7.5	8.0	8.1	13.6	8.4	13.5	9.9	13.3	14.9	15.9
Oct 2008 GFC	10.3	7.1	14.8	9.5	11.6	9.0	10.6	9.3	11.0	9.7	16.7	11.1	16.3	16.2	17.3	13.9	18.8	13.0	18.5	21.2
Mar 2009 Recovery	5.9	2.1	7.8	1.5	6.2	3.7	6.3	4.8	6.3	5.7	9.8	7.4	14.6	16.7	13.9	16.5	8.7	15.6	9.0	22.0
2013 Calm	2.6	0.0	5.9	6.8	4.2	1.9	5.2	2.9	4.9	3.5	5.1	2.5	4.8	2.2	4.1	2.6	4.5	3.3	5.6	0.0
2017 Calm	7.1	0.0	4.6	4.8	4.2	4.5	5.0	4.1	4.6	4.7	4.5	4.5	4.0	4.4	4.5	5.7	4.5	4.7	5.0	5.1
Q4 2018 VolShock	3.8	8.5	3.9	6.3	4.4	4.6	5.6	3.7	14.5	5.5	16.0	5.2	13.7	7.4	12.5	9.8	12.6	11.3	10.9	11.6
Mar 2020 COVID	11.8	14.9	12.1	8.6	12.5	11.3	13.5	18.0	16.6	16.4	35.1	13.2	17.4	12.6	18.5	12.3	18.4	13.4	20.0	13.6
Jun 2020 Stable	2.4	4.7	2.9	3.9	5.2	2.7	6.5	6.0	5.8	8.1	11.2	5.9	10.8	10.2	7.7	10.6	7.9	11.3	9.0	8.1
2022 Rate Hikes	0.0	0.0	4.2	2.8	6.1	8.5	7.6	8.3	6.6	7.7	10.6	15.5	11.4	15.1	11.6	17.3	10.7	18.0	11.9	15.9
2024 Contemporary	0.0	0.0	2.9	1.7	3.6	2.7	4.2	3.0	4.2	3.5	5.0	2.4	4.8	3.5	4.9	3.8	5.8	4.5	5.5	5.0

TABLE I.22 – $|Z_{MR}|$ across panel sizes and selection rules.

Snapshot	N=50		N=100		N=150		N=200		N=250		N=300		N=350		N=400		N=450		N=500	
	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV	cov	ADV
Oct 1987 Black Monday	1.6	0.2	2.5	0.3	7.0	1.2	9.8	1.1	14.5	1.9	4.1	5.8	3.1	7.8	3.2	8.2	1.0	10.1	1.4	10.4
1990–91 Recession	0.5	—	0.4	0.3	0.7	1.5	0.6	3.2	1.2	2.3	1.4	0.0	0.4	0.1	0.7	0.3	0.7	0.1	0.4	0.6
1993 Calm	—	2.2	3.3	3.4	2.1	2.6	8.3	2.4	9.7	4.7	9.0	3.2	11.6	4.6	9.8	3.4	9.1	4.1	10.1	6.9
Dec 1994 Tequila	0.3	—	0.4	—	0.4	0.3	0.6	—	3.4	—	4.4	0.3	2.3	0.4	2.4	0.3	1.3	0.6	3.7	0.5
Oct 1997 Asian Crisis	0.5	0.8	2.0	0.3	0.3	0.2	0.1	1.0	2.2	1.2	3.1	0.3	3.6	0.4	5.1	0.4	4.1	0.2	5.3	0.4
Oct 1998 LTCM	0.6	—	1.7	2.0	0.8	1.0	0.5	2.3	0.7	0.6	0.6	0.4	0.8	0.3	0.6	0.2	1.3	0.4	1.2	—
Apr 2000 Dot-com	0.4	—	0.1	0.4	2.2	0.5	2.5	2.3	0.6	0.8	0.9	0.8	2.7	2.4	4.0	3.8	8.1	5.2	7.7	4.5
Sep 2001 9/11	0.1	—	0.7	0.2	0.3	1.2	1.0	2.1	2.4	1.1	1.3	4.4	1.9	1.1	0.3	2.3	0.5	2.1	0.4	4.5
Jul 2002 WorldCom	0.4	0.7	1.2	3.1	3.4	0.6	5.4	0.2	8.1	0.4	6.2	1.1	6.5	2.0	9.6	2.2	10.5	2.2	8.3	4.0
2005 Calm	—	—	0.2	4.2	0.5	0.1	1.0	0.1	0.2	0.9	2.1	0.1	1.8	—	5.2	0.1	7.6	7.8	7.2	5.2
2007 Subprime Buildup	1.1	0.2	0.0	1.1	0.8	0.6	0.5	1.2	0.8	1.2	0.7	3.5	2.4	4.1	2.8	4.8	1.5	5.2	4.7	5.3
Oct 2008 GFC	0.9	0.4	1.6	1.5	4.2	0.3	6.1	1.9	11.6	1.9	10.6	7.8	9.9	3.9	11.7	4.1	12.5	3.6	12.7	10.1
Mar 2009 Recovery	1.5	—	1.5	2.3	3.9	0.7	4.9	4.1	8.7	3.7	4.2	7.2	3.7	4.6	2.5	5.8	7.8	5.5	7.7	4.7
2013 Calm	0.2	—	0.4	1.2	2.4	0.8	2.9	0.4	1.8	0.3	4.9	1.0	5.5	0.2	5.3	1.5	5.6	2.7	3.9	—
2017 Calm	0.3	—	0.8	0.1	3.5	1.4	4.3	2.8	2.1	1.9	5.5	0.8	7.0	5.3	8.7	4.3	8.4	5.2	8.4	2.6
Q4 2018 VolShock	0.1	0.8	0.3	0.2	3.9	0.3	2.1	0.2	6.6	1.6	4.9	3.9	7.3	0.1	6.3	0.6	4.9	1.6	8.2	8.9
Mar 2020 COVID	0.1	0.5	3.0	1.5	8.7	6.3	14.4	3.6	20.6	4.5	13.8	5.2	23.0	8.3	18.0	9.8	24.5	10.0	21.3	13.9
Jun 2020 Stable	0.4	0.9	0.7	0.2	5.6	2.4	6.0	1.4	6.1	4.8	1.3	3.5	1.8	3.3	3.8	1.1	4.4	1.7	4.8	6.2
2022 Rate Hikes	—	—	0.4	3.5	0.9	1.3	8.3	2.0	6.0	4.9	11.1	7.2	11.2	7.6	10.8	8.7	11.8	9.5	13.2	6.9
2024 Contemporary	—	—	0.6	0.7	0.1	0.2	2.0	1.1	6.2	1.7	8.8	4.8	8.7	4.6	10.0	4.1	10.7	9.1	9.6	11.2

at large N on both rules (headline-robust), and (b) whether the two rules agree even at smaller N (selection-robust). The single-channel NSI does not cleanly separate crisis from non-crisis snapshots in this sweep, consistent with the near-zero NSI crisis effect reported in §J-G, so the informative content of these tables is the *cross-rule* and *panel-size* sensitivity, not a crisis ranking. The mutual-dyad fraction is, on average, higher in crisis windows than in non-crisis windows (the channel-level crisis contrasts are tabulated in §J-G), but the two distributions overlap: several calm and stress windows (e.g. 2007 Subprime Buildup, Q4 2018 VolShock) carry a high f_{mut} as well, so the signal is a mean shift rather than a clean dominance (Table I.21). Two snapshots exhibit cross-rule sensitivities worth reporting separately: Mar 2020, which carries the largest coverage-vs-ADV NSI gap at the headline $N = 500$ (next subsection), and 2022, which carries the largest coverage-vs-ADV NSI gap anywhere in the sweep (the subsection after).

D. The Mar 2020 coverage-versus-ADV NSI gap

Within the multipanel sweep, Mar 2020 carries the largest coverage-versus-ADV NSI gap of any snapshot at the headline panel size. At $N = 500$ the liquidity-ranked (ADV) panel reads $NSI = 0.744$ against 0.465 on the longest-listed (coverage) panel, a gap of 0.279 on the Mar 2020 COVID row, the per-snapshot maximum at $N = 500$ (read off the $N = 500$ column pair of Table I.20). The gap is not a small- N transient: the ADV NSI sits above coverage at every $N \geq 100$, climbing from 0.42 at $N = 50$ to a plateau near 0.74 for $N \geq 350$, while the coverage NSI stays in a noisy 0.47–0.66 band for $N \leq 450$ with no late- N peak, dropping to its *lowest* value of 0.465 at $N = 500$. The largest single coverage-vs-ADV deviation *anywhere* in the sweep is a different cell, $N = 200$ on 2022 Rate Hikes with magnitude 0.331 (§I-E), but Mar 2020 is the one that survives to the headline $N = 500$.

The direction of this gap is the opposite of a low-liquidity speculative-tail effect. During the COVID sell-off the liquidity-ranked mega-cap panel is *more* synchronised than the longest-listed panel: the mean A-DCC correlation on the ADV panel exceeds the coverage panel both at $N = 500$ ($\bar{\rho} \approx 0.54$ vs. 0.49) and at the intermediate sizes (0.56 vs. 0.51 at $N = 200$). The COVID systemic-risk signal is thus carried most strongly by the high-volume, broadly held names rather than by the small-cap survivors that only the coverage rule admits.

The robustness consequence is that the COVID reading does not depend on including a low-liquidity tail: restricting the panel to the most liquid names *raises* the sweep NSI rather than deflating it. This is the mirror image of the speculative-tail concern raised in the main paper §IX Lim. (i); it is examined further with the volume-weighted variant in §I-H and the liquidity-decile panels in §I-I.

E. The largest cross-rule gap: 2022 at $N = 200$

The single largest coverage-versus-ADV NSI deviation anywhere in the sweep is 2022 Rate Hikes at $N = 200$: $NSI = 0.848$ on the ADV panel against 0.517 on coverage, a gap of 0.331 (2022 Rate Hikes, $N = 200$; Table I.20). It is an intermediate- N

feature, not a saturation feature: the ADV NSI for 2022 spikes to 0.85 at $N = 200$, then decays to ≈ 0.61 for $N \geq 350$, while the coverage NSI stays in a flat 0.52–0.65 band across the whole sweep.

The gap is a concentration-channel effect. At $N = 200$ the ADV panel’s PageRank concentration is more than double the coverage panel’s (HHI = 0.230 vs. 0.106, Gini 0.570 vs. 0.495) and its $|Z_{MR}|$ is far less negative (2.0 vs. 8.3), while the mean A-DCC correlation differs much less (0.43 vs. 0.39). The NSI elevation on ADV-200 is therefore driven by hub concentration among the largest-volume names, not by a broad rise in pairwise co-movement. The same composition mechanism appears at $N = 50$ (§I-G), where the ADV panel collapses onto a handful of mega-cap hubs.

This is neither a coverage-rule artefact nor a speculative-tail effect in the sense of §I-D: the names driving the ADV spike are the largest-volume mega-caps, the opposite of the low-liquidity tail. It is an example of a snapshot whose intermediate- N reading is sensitive to how concentrated the panel is in a few hubs. Notably, by the headline $N = 500$ the two rules return nearly the same 2022 NSI (coverage 0.653, ADV 0.615, $|\Delta| = 0.038$) even though they still share only a minority of their tickers (Jaccard 0.274, Table I.23), a convergence of the *metric* rather than of panel *membership*. F1 (the main paper §IX) addresses this by reducing the composite’s reliance on the concentration channel that drives the intermediate- N gap.

F. Coverage vs. ADV overlap at each N

TABLE I.23 – Coverage vs. ADV ticker-set overlap. Jaccard = $|\text{cov} \cap \text{ADV}| / |\text{cov} \cup \text{ADV}|$.

N	$ \text{cov} $	$ \text{cov} \cap \text{ADV} $	Jaccard	$ \text{cov} \setminus \text{ADV} $
50	50	3	0.031	47
100	100	10	0.053	90
150	150	25	0.091	125
200	200	45	0.127	155
250	250	60	0.136	190
300	300	93	0.183	207
350	350	119	0.205	231
400	400	152	0.235	248
450	450	178	0.247	272
500	500	215	0.274	285

Table I.23 reports the ticker-set overlap of the two selection rules as N grows. At small N the rules pick effectively non-overlapping panels (the bulk of the small- N coverage panel is historical survivors with long uninterrupted listings; the small- N ADV panel is dominated by the highest-volume mega-caps that came public later). The Jaccard index rises monotonically with N but the two rules never converge in membership: even at $N = 500$ they share only 0.274 of their combined universe (about a quarter), with the residual symmetric difference reported in the rightmost column of the same row. The two panels therefore remain genuinely distinct selections at every size; where their metrics agree (as on 2022 at $N = 500$, §I-E) that agreement reflects a shared stress signal, not overlapping membership.

G. Liquidity-stratification reading

The “speculative tail” question, whether a low-liquidity or late-IPO tail drives the headline crisis result, is read off the sweep in three steps. The premise must first be corrected: on the present panel the two rules do *not* coincide at $N = 500$. They differ by up to 0.279 in NSI there (on the Mar 2020 COVID row) and share only 0.274 of their combined universe (Table I.23). Robustness to the liquidity-controlled selection therefore has to be argued from the *structure* of the disagreement, not from its absence.

(1) *The gaps are panel composition, not crisis dynamics.* The three largest $N = 500$ cross-rule gaps are Mar 2020 COVID (0.279), 1993 Calm (0.210) and 2024 Contemporary (0.189), all three with the liquidity-ranked (ADV) panel reading *higher* than coverage (read from the $N = 500$ column pair of Table I.20). Two of the three are calm windows: a cross-rule gap that is as large in calm windows as in the worst crisis is a panel-composition signature, mega-cap PageRank concentration on the ADV panel, rather than a crisis-specific tail effect.

(2) *No systematic coverage excess on crises.* Across the seven crisis snapshots the sign of the $N = 500$ gap is mixed: Mar 2020 COVID and Oct 1987 Black Monday read higher on ADV, while Oct 1998 LTCM and Sep 2001 9/11 read higher on coverage. There is no systematic pattern in which the longest-listed panel (the one that admits more low-liquidity names) inflates the crisis reading relative to the liquid panel; if anything the single largest crisis gap, COVID, runs the other way (§I-D). The headline crisis signal is thus not attributable to a low-liquidity tail that coverage includes and ADV excludes.

(3) *The largest gap anywhere is concentration.* The single largest coverage-vs-ADV deviation in the whole sweep (0.331 at $N = 200$ on 2022 Rate Hikes) is a PageRank-concentration composition effect on the mega-cap ADV panel, decomposed in §I-E. At the smallest panel the two rules are effectively disjoint, at $N = 50$ they share only 3 tickers (Table I.23), because

coverage selects long-uninterrupted-listing survivors while ADV collapses onto the highest-volume mega-caps, whose hub structure elevates the concentration channel even in calm windows. The coverage-vs-ADV contrast brackets the universe by listing history versus liquidity but does not isolate the bottom-liquidity tail on its own; that isolation is done explicitly by the `bottom_adv`, `bottom_coverage` and `liquidity-quintile` panels of §I-I, and the within-panel reweighting check is the volume-weighted NSI of §I-H.

Taken together, the headline does not rest on a low-liquidity tail: the liquidity-ranked panel reads at least as high as coverage on the broadest crises, the largest cross-rule gaps appear in calm windows and trace to mega-cap concentration, and the dedicated tail panels are reported separately. The cross-rule *disagreement* at intermediate N is real and is what motivates F1 and the volume-weighted variant reported next.

H. Volume-weighted NSI (F1 prototype)

The default snapshot NSI weights every ticker equally inside each channel. The volume-weighted variant in `src.stage5_nsi.volume_weighted_nsi` reweights the correlation and HHI channels by log-volume shares $w_i = \log(1 + \text{ADV}_i) / \sum_j \log(1 + \text{ADV}_j)$ per snapshot:

$$\begin{aligned}\bar{\rho}_{\text{vw}} &= \frac{\sum_{i < j} w_i w_j R_{ij}}{\sum_{i < j} w_i w_j}, \\ h_{\text{vw}} &= \sum_i w_i r_i^2, \\ s_{\text{vw}} &= 1 - \frac{\sum_{(i,j) \in E} w_i w_j}{\sum_{i < j} w_i w_j}.\end{aligned}$$

The motif channel is unchanged (motifs are graph-level, not per-ticker). The same min/max normalisation and the same convex weights (0.25, 0.20, 0.35, 0.20) apply.

TABLE I.24 – Volume-weighted NSI (Stage-5 channels reweighted by per-snapshot log-ADV) vs. the unweighted paper-canonical NSI on the twenty-snapshot panel. $\Delta = \text{NSI}_{\text{vw}} - \text{NSI}$.

Snapshot	Regime	NSI	NSI _{vw}	Δ
Oct 2008 GFC	crisis	0.758	0.730	−0.028
2022 Rate Hikes	stress	0.685	0.671	−0.014
Mar 2009 Recovery	recovery	0.672	0.588	−0.085
Q4 2018 VolShock	stress	0.658	0.656	−0.002
2007 Subprime Buildup	stress	0.634	0.585	−0.049
2005 Calm	baseline	0.581	0.493	−0.088
2024 Contemporary	baseline	0.556	0.629	+0.074
Mar 2020 COVID	crisis	0.545	0.649	+0.104
Oct 1998 LTCM	crisis	0.543	0.487	−0.056
2013 Calm	baseline	0.540	0.562	+0.022
Sep 2001 9/11	crisis	0.524	0.563	+0.039
Jul 2002 WorldCom	stress	0.495	0.556	+0.062
Apr 2000 Dot-com	crisis	0.484	0.473	−0.012
Dec 1994 Tequila	stress	0.479	0.401	−0.078
Oct 1987 Black Monday	crisis	0.455	0.499	+0.044
1993 Calm	baseline	0.438	0.403	−0.035
1990–91 Recession	stress	0.392	0.375	−0.017
Jun 2020 Stable	recovery	0.391	0.515	+0.124
Oct 1997 Asian Crisis	crisis	0.369	0.384	+0.016
2017 Calm	baseline	0.359	0.428	+0.069

The Δ column of Table I.24 reports $\text{NSI}_{\text{vw}} - \text{NSI}$ per snapshot. The direction of the shift is governed by where each snapshot’s PageRank hubs and correlation mass sit in the ADV distribution: a snapshot whose hubs lie near the top of the ADV distribution *amplifies* under volume weighting (the hub-concentration channel gains mass), one whose hubs sit in a moderate-ADV band stays roughly flat, and one whose signal carries appreciable low-ADV mass *attenuates*. The motif channel (μ) is unchanged in either direction because motifs are graph-level and not per-ticker.

Because the volume-weighted variant is an F1 prototype rather than a production channel, the per-snapshot Δ values are read directly off Table I.24, regenerated by `paper/_generate_multipanel_tables.py` from the `results/snapshots/nsi_volume_weighted.pkl` cache. The shift is per-snapshot rather than regime-ordered: across the twenty

snapshots Δ ranges from -0.088 (2005 Calm) to $+0.124$ (Jun 2020 Stable), and the two recovery windows disagree in sign (Mar 2009 -0.085 , Jun 2020 $+0.124$), so the prototype does not on its own establish a regime-ordered amplification pattern; mechanically, a window whose PageRank hubs sit near the top of the ADV ranking is amplified under volume weighting regardless of regime. This is consistent with the speculative-tail interpretation raised in the main paper §IX Lim. (i). The variant is proposed as one of several inputs to the F1 redesign rather than the redesign itself.

I. Bottom- N and liquidity-decile panels

Beyond the coverage / ADV duo, the sweep runs five ADV quintiles plus two “tail” selections that isolate the low-liquidity / late-IPO slice. These complete the speculative-tail dissection:

- -kind bottom_adv, bottom- N by ADV (isolates the speculative tail).
- -kind bottom_coverage, bottom- N by sample-wide coverage (isolates the late-IPO / short-listing tail).
- -kind decile{1,2,3,4,5}, the ADV quintile; decile 1 = most liquid, decile 5 = least liquid.

TABLE I.25 – Liquidity-decile cross-section. Each row is read from the largest panel size at which that cell exists in results/multipanel/summary.json; the N column shows that size. A populated ADV decile saturates at $|\text{panel}| \approx \text{universe}/5$ regardless of N ; on the current cache only decile1–decile4 carry constituents, while the least-liquid decile5 and the two bottom- N tail selections return empty panels ($|\text{panel}| = 0$, shown “—”). $\overline{\text{NSI}}_{\text{crisis}}$ averages the two crisis peaks; $\overline{\text{NSI}}_{\text{baseline}}$ averages four calm windows; the 2022 column is the single 2022 Rate Hikes NSI.

Selection	N	$ \text{panel} $	$\overline{\text{NSI}}_{\text{crisis}}$	$\overline{\text{NSI}}_{\text{baseline}}$	NSI(2022)
ADV decile 1 (top)	500	288	0.657	0.342	0.601
ADV decile 2	500	288	0.617	0.468	0.546
ADV decile 3	500	288	0.548	0.486	0.514
ADV decile 4	500	288	0.534	0.384	0.471
ADV decile 5 (bottom)	500	0	—	—	—
Bottom- N by ADV	500	0	—	—	—
Bottom- N by coverage	500	0	—	—	—

The headline read from Table I.25 is the column-wise contrast between the crisis-mean and baseline-mean columns across the quintiles: a flat crisis-mean across deciles means the crisis signal is panel-wide, whereas a crisis-mean that peaks in decile 5 (the least-liquid quintile) would mark the signal as disproportionately a speculative-tail effect. The quintile slices cap at $|\text{panel}| \approx \text{universe}/5$ tickers regardless of N , so for decile1..5 larger N does not change the row; the table builder reads the largest available N per kind directly from results/multipanel/summary.json. On the current cache only the four populated ADV quintiles (decile1–decile4) carry constituents; the least-liquid quintile (decile5) and the two bottom- N tail selections return empty panels, the bottom-of-ADV slice fails the $\geq 80\%$ in-window-coverage and univariate-GARCH-fit requirements, and so read “—” in Table I.25. The speculative-tail question is therefore answered from the populated decile1–decile4 gradient and the coverage-versus-ADV contrast (§I-G), not from an isolated decile5 cell.

Liquidity stratification.: The coverage and ADV sweeps (Tables I.20–I.23) are complemented by seven liquidity-stratified slices (decile1..5, bottom_adv, bottom_coverage), summarised in Table I.25. The speculative-tail question is also answered directly on the coverage-versus-ADV contrast (§I-G): the largest cross-rule NSI gaps fall in calm windows and trace to mega-cap concentration, and on the broadest crises the liquidity-ranked panel reads at least as high as the longest-listed panel, neither pattern points to a low-liquidity tail driving the headline.

J. Cross-panel fallback identification

The coverage and ADV caches both store the post-fallback λ^* values per snapshot. On the $N = 500$ panels the crisis and high-volatility windows pin at the top of the λ^* grid: Oct 2008 GFC, Mar 2020 COVID, Jun 2020 Stable, 2022 Rate Hikes and Q4 2018 VolShock all sit at $\lambda^* = 0.417$ (the grid maximum) on coverage, and the same crisis and high-volatility windows pin to 0.417 on ADV as well, whereas the calmest recent windows sit lower (2024 Contemporary at 0.174, 2017 Calm at 0.197 on coverage). The unconstrained-BIC argmin is not stored in the current cache schema (an extension flagged for the next multipanel re-run), so the formal fallback-vs-non-fallback distinction cannot be re-derived from the cache alone, only inferred from the high- λ^* pattern that the crisis and high-volatility snapshots share. That the same high-stress windows pin to the grid ceiling under both selection rules, while the rank ordering of the derived network metrics is preserved across the per-rule λ^* differences, is the cross-panel robustness point (cf. §K).

J. STATISTICAL INFERENCE: DETAIL

Three test families address three distinct defects of the design: an exact permutation enumerates the $\binom{20}{7} = 77,520$ partitions for the discrete null, Benjamini–Hochberg controls the dependent multiple-test family, and a pair of bootstraps (cluster for the GFC/COVID pair dependence, block for the rolling autocorrelation) carries the dependence structure into the NSI–VIX confidence sets.

Background concepts used in this section.: The inference layer combines three distinct apparatuses, each matched to a different property of the data. A *permutation test* (§A) asks whether the observed crisis-vs-non-crisis difference would be surprising if the crisis labels were re-shuffled at random; with $n_{\text{crisis}} = 7$ on twenty snapshots the test enumerates all $\binom{20}{7} = 77,520$ assignments exactly, and $1/77,520 \approx 1.3 \times 10^{-5}$ is the structural floor of the one-sided p -value. This is the cleanest test of “crisis label carries signal” but, as the results below show, is underpowered against the modest effect sizes this panel exhibits. *Benjamini–Hochberg FDR* is preferred over Bonferroni and Holm (§A “Family-wise vs. false-discovery control”) because it is the least conservative valid procedure on a positively-dependent family: the family-wise correction threshold α/m for our seven-test family ($\alpha/7 \approx 0.0071$) now sits well above the permutation floor (1.3×10^{-5}), so FWER control is not mechanically vacuous, but the smallest *observed* permutation p -value (PageRank Gini, $p = 0.022$) still exceeds 0.0071, so Bonferroni and Holm reject nothing on this panel. BH controls the *proportion* of false rejections rather than the probability of any false rejection; on the current CRSP cache it too yields no rejection at FDR-5% or FDR-10% (smallest $q = 0.152$), the honest negative finding the main text reports. The *cluster bootstrap* and *stationary block bootstrap* (§A “Bootstrap variants”) target two different dependence structures: the cluster bootstrap respects the GFC pair and COVID pair groupings in the snapshot panel, and the block bootstrap respects the serial dependence in the rolling NSI–VIX series. Both produce percentile confidence intervals at $B \in \{5\,000, 10\,000\}$ resamples. The *Bonferroni-20 per-snapshot threshold* corrects the twenty parallel per-snapshot $|Z|$ -tests for multiplicity across the twenty snapshots (§J-F); being a Bonferroni bound it is valid under arbitrary dependence among those tests.

The inference layer in the main paper rests on four pieces: exact permutation tests with Benjamini–Hochberg FDR correction; a cluster bootstrap for the snapshot NSI–VIX concordance; a stationary block bootstrap for the rolling NSI–VIX correlation; and a Bonferroni-20 per-snapshot Z -score threshold. This section expands each one.

A. Snapshot independence assumptions

The twenty snapshot windows of Table B.2 are calendar-bound regime-tagged analysis windows rather than a partition of the 1985–2024 sample. The CRSP windows are *strictly non-overlapping*: no two windows share a single trading day. The only residual temporal proximity is between each crisis and its immediate aftermath window:

Adjacent pair	Calendar gap	Shared trading days
Oct 2008 GFC → Mar 2009 Recovery	40 days	0
Mar 2020 COVID → Jun 2020 Stable	contiguous (next day)	0
all 18 other inter-window gaps	≥ 6 months	0

The non-overlap is a deliberate consequence of the redesign: uniform six-month event windows and twelve-month baselines, peak-centred and spaced so that no two share returns. The Stage-1 A-DCC correlations, Stage-2 GLASSO graphs, and Stage-3 directed networks are therefore computed on disjoint return samples, and the exchangeability assumption of the permutation test holds by construction rather than as an approximation. The only dependence left to handle is regime-level: the two crisis→aftermath pairs above sample adjacent macro regimes (a crisis and its rebound) even though they share no return days. Each of the four inference pieces treats this residual structure in a different way.

Exact-permutation test (§J, first subsection).: The $\binom{20}{7} = 77,520$ enumerated label assignments treat the twenty snapshots as exchangeable under the null of no crisis effect. Because the windows share no return days, this assumption holds by construction; the only between-snapshot dependence is the mild regime-level covariance within the two crisis→aftermath pairs, which is positive and therefore conservative for the one-sided p -value reported in Table II of the main paper: positive dependence shifts the permutation null toward Δ^{obs} and inflates p . The reported p -values are therefore (weak) upper bounds on those an exactly-independent design would record. The direction of the bias works in our favour and we do not adjust.

Benjamini–Hochberg FDR.: Positive dependence across the seven test statistics in Table II of the main paper is the condition under which standard BH controls FDR; the seven metrics are all computed from the same Stage-2 graph on each snapshot, so they move together, which supplies the positive-dependence assumption standard BH requires.

Cluster bootstrap (§J-D, Algorithm 9).: The cluster partition is *designed* to absorb the residual pair dependence: the COVID pair {Mar 2020, Jun 2020} is sampled as a single cluster and the GFC pair {Oct 2008, Mar 2009} likewise, so a bootstrap draw resamples *clusters* (statistically independent units) rather than individual snapshots. This is the only piece of the inference layer that explicitly accounts for the crisis→aftermath pair dependence rather than inheriting a conservative bound from it.

Stationary block bootstrap (rolling NSI-VIX).: The block-bootstrap operates on the G -step rolling NSI grid ($G \approx 468$ NSI observations at the 21-trading-day stride over 1985–2024), not on the twenty snapshot windows; the window adjacency discussed here is therefore not a concern for that piece.

Bonferroni-20 per-snapshot Z-score threshold.: The Bonferroni-20 correction divides α across the twenty per-snapshot $|Z|$ -tests on each motif axis (§J-F). A Bonferroni bound is valid under arbitrary dependence among the tests, so the regime-level proximity of the two crisis→aftermath pairs, which can only induce positive dependence between adjacent snapshots’ Z -scores, does not threaten its validity; no adjustment is needed.

In short, the only inference component that depends critically on treating snapshots as exchangeable is the exact-permutation test, and for that test the non-overlapping design makes exchangeability exact up to the conservative within-pair regime dependence. As a sensitivity check, a COVID-merged row collapses the contiguous Mar 2020 COVID and Jun 2020 Stable windows into a single COVID-episode snapshot (mean of the two within-pair values), dropping the count from 20 to 19 with $n_{\text{crisis}} = 7$ retained ($\binom{19}{7} = 50,388$, floor $p \approx 2.0 \times 10^{-5}$); it is enumerated alongside the twenty-snapshot test in `paper/_inference.py`. Merging slightly *sharpens* the single strongest contrast, the PageRank Gini tightens from $p = 0.022$ to $p = 0.011$ ($q = 0.080$), because the merged unit no longer dilutes the COVID concentration spike with the Jun 2020 rebound. The remaining contrasts move the other way or stay flat (HHI $p = 0.297$, mutual-dyad $p = 0.127$, cross-sector $p = 0.111$, mean $\bar{\rho}$ $p = 0.104$, $|Z_{\text{MR}}|$ $p = 0.600$, NSI $p = 0.682$). *No metric reaches FDR-5% under either design*: the merged row confirms the headline negative result is not an artefact of the COVID-pair grouping. We retain the twenty-snapshot formulation as the headline reading for parity with the cluster-bootstrap design.

On the “conservative” claim.: The twenty snapshots use non-overlapping windows (§J-A: zero shared trading days between any two windows), so the only between-snapshot dependence is the regime-level proximity of the two crisis→aftermath pairs. That dependence is positive and so biases the one-sided permutation p -value in the conservative direction by the standard argument (positive dependence pushes the typical $\Delta^{(C)}$ toward Δ^{obs} , raising the reported p). The cluster bootstrap of §J-D absorbs this pair dependence directly by resampling the two pairs as units, and the COVID-merged row above confirms the conclusions are insensitive to the pair grouping.

B. Exact-permutation enumeration with $n_{\text{crisis}} = 7$

With $n_{\text{crisis}} = 7$ (Oct 1987 Black Monday, Oct 1997 Asian Crisis, Oct 1998 LTCM, Apr 2000 Dot-com, Sep 2001 9/11, Oct 2008 GFC, Mar 2020 COVID) and $n_{\text{non-crisis}} = 13$ remaining snapshots, there are exactly $\binom{20}{7} = 77,520$ ways to assign seven of twenty labels to the crisis arm. For each metric X we enumerate all 77,520 partitions exhaustively, compute the difference of means

$$\Delta^{(b)} = \bar{X}_{\text{crisis}}^{(b)} - \bar{X}_{\text{non-crisis}}^{(b)}$$

for $b = 1, \dots, 77,520$, and report the one-sided p -value

$$p = \frac{1}{77,520} \# \{b : \Delta^{(b)} \geq \Delta^{\text{observed}}\}.$$

The floor of this exact test is $1/77,520 \approx 1.3 \times 10^{-5}$, but no metric on the current CRSP cache comes near it. The smallest one-sided p -value is the PageRank Gini at $p = 0.022$ (about $0.022 \times 77,520 \approx 1,700$ of the 77,520 assignments match or exceed the observed split), followed by mean $\bar{\rho}$ ($p = 0.076$), the mutual-dyad fraction ($p = 0.084$), and cross-sector edges ($p = 0.109$); the remaining three sit far higher, HHI ($p = 0.293$), $|Z_{\text{MR}}|$ ($p = 0.321$), and NSI ($p = 0.525$). Cohen’s d [3] is reported alongside every p -value to convey effect size, and is the meaningful discriminator of magnitude given that, as the next subsection shows, none of these raw p -values survives the multiple-testing correction. Algorithm 8 gives the full enumeration; Cohen’s d uses pooled within-group standard deviation $\sigma_{\text{pool}} = \sqrt{((n_c - 1)\sigma_c^2 + (n_{nc} - 1)\sigma_{nc}^2)/(n_c + n_{nc} - 2)}$ on the observed split.

Algorithm 8 Exact permutation test for the crisis-vs-non-crisis contrast on one metric.

Require: Per-snapshot metric vector $\mathbf{x} \in \mathbb{R}^{20}$; observed crisis index set C^{obs} (the seven crisis snapshots: Black Monday, Asian Crisis, LTCM, Dot-com, 9/11, GFC, COVID); non-crisis index set $N^{\text{obs}} = \{1, \dots, 20\} \setminus C^{\text{obs}}$.

- 1: $\Delta^{\text{obs}} \leftarrow \bar{x}_{C^{\text{obs}}} - \bar{x}_{N^{\text{obs}}}$.
- 2: Enumerate $\mathcal{P} \leftarrow \{C \subset \{1, \dots, 20\} : |C| = 7\}$ ($|\mathcal{P}| = 77,520$).
- 3: count $\leftarrow 0$.
- 4: **for** $C \in \mathcal{P}$ **do**
- 5: $N \leftarrow \{1, \dots, 20\} \setminus C$; $\Delta^{(C)} \leftarrow \bar{x}_C - \bar{x}_N$.
- 6: **if** $\Delta^{(C)} \geq \Delta^{\text{obs}}$ **then** count $\leftarrow$ count + 1
- 7: **end if**
- 8: **end for**
- 9: $p \leftarrow \text{count}/77,520$ $\triangleright$ one-sided p -value, floor $1/77,520$
- 10: $\sigma_c \leftarrow \text{std}(\mathbf{x}_{C^{\text{obs}}})$; $\sigma_{nc} \leftarrow \text{std}(\mathbf{x}_{N^{\text{obs}}})$ (ddof=1).
- 11: $\sigma_{\text{pool}} \leftarrow \sqrt{((|C^{\text{obs}}| - 1)\sigma_c^2 + (|N^{\text{obs}}| - 1)\sigma_{nc}^2)/(|C^{\text{obs}}| + |N^{\text{obs}}| - 2)}$.
- 12: $d \leftarrow \Delta^{\text{obs}}/\sigma_{\text{pool}}$ $\triangleright$ Cohen's d
- 13: **return** (p , d).

Power is not floor-limited: with $n_{\text{crisis}} = 7$ the exact-test floor ($1/77,520 \approx 1.3 \times 10^{-5}$) lies far below any conventional α , so the failure to reject is not a discrete-lattice artefact. It reflects genuine effect-size attenuation on the deeper, survivorship-corrected CRSP panel: the thirteen non-crisis windows now include borderline stress episodes (2007 Subprime Buildup, 2022 Rate Hikes) whose intra-window structure is itself crisis-like, which lifts the non-crisis arm toward the crisis arm and shrinks every contrast. §J-G quantifies the resulting power against a range of plausible alternatives.

C. Benjamini–Hochberg FDR

The seven-test family in Table II of the main paper (NSI, Q2 Gini, Q2 HHI, mean $\bar{\rho}$, cross-sector edge fraction, $|Z_{\text{MR}}|$, mutual-dyad fraction) shares Stage-2 graph structure, so the positive-dependence assumption that justifies standard Benjamini–Hochberg, rather than the more conservative Benjamini–Yekutieli correction under arbitrary dependence, is plausible but not formally verified.

The BH procedure: sort the p -values $p_{(1)} \leq \dots \leq p_{(7)}$ and reject $H_{0,(k)}$ for $k \leq k^*$ where

$$k^* = \max\{k : p_{(k)} \leq \frac{k}{7} \alpha\}$$

at FDR threshold α . The reported q -value

$$q_{(k)} = \min_{j \geq k} 7 p_{(j)} / j$$

is the smallest FDR at which $H_{0,(k)}$ is rejected.

Holm–Bonferroni alternative.: A more conservative step-down control is Holm–Bonferroni: reject $H_{0,(k)}$ if and only if $p_{(j)} \leq \alpha/(m - j + 1)$ for all $j \leq k$, with $m = 7$ the family size. The Holm-adjusted critical values for the 7-test family at $\alpha = 0.05$ are $0.05/7 = 0.00714, 0.05/6 = 0.00833, \dots, 0.05/1 = 0.05$, monotone in k . The exact-permutation floor at $n_{\text{crisis}} = 7$ is now $1/\binom{20}{7} = 1/77,520 \approx 1.3 \times 10^{-5}$, far below the most-stringent corrected α of 0.00714, so a sufficiently extreme metric could in principle clear the Holm threshold; the structural floor is not the binding constraint. What blocks rejection is the *observed* effect size: the smallest p -value (PageRank Gini, $p_{(1)} = 0.022$) already exceeds the Holm $k = 1$ threshold 0.00714, so the step-down procedure stops at the first comparison with no rejections. Bonferroni-7 (single-step) gives the same call: $0.022 > 0.00714$ for every metric. Both family-wise error controls are therefore uninformative on this panel, not for a structural-floor reason but because the crisis-vs-non-crisis contrasts are simply too weak.

The BH FDR control reported in Table II of the main paper is preferred precisely because it does not require any individual p -value to fall below α/m ; rather, it balances the false-discovery proportion against the test family size. On the current CRSP cache, however, even BH yields no rejection. PageRank Gini leads at $q = 0.152$, with mean $\bar{\rho}$, the mutual-dyad fraction, and cross-sector edges tied just behind at $q = 0.191$, all above the FDR-10% cut; $|Z_{\text{MR}}|$ and HHI follow at $q = 0.374$, and NSI at $q = 0.525$. No metric rejects at FDR-5% or FDR-10%. This is the honest negative result the main text records (cf. the main paper §VIII): the crisis-vs-non-crisis contrasts do not reach significance on the deeper, survivorship-corrected CRSP panel. There are accordingly no rejections for Holm to miss, the smallest observed p (0.022, Gini) exceeds the Holm $k = 1$ threshold 0.00714, and the smallest BH q (0.152) exceeds 0.10. The BH q -values are reported because they expose the relative ordering of the (non-rejecting) metrics and would be the operative procedure had the effects been stronger.

Benjamini–Yekutieli companion under arbitrary dependence.: Standard BH controls FDR under positive regression dependence on the subset (PRDS). The seven-test family shares Stage-2 graph structure and the COVID-cluster shared returns, so PRDS is plausible (§J-A) but not formally verified. The Benjamini–Yekutieli correction $q_{(k)}^{\text{BY}} = c(m) q_{(k)}^{\text{BH}}$ with $c(m) = \sum_{i=1}^m 1/i$ controls FDR under arbitrary dependence; at $m = 7$, $c(7) \approx 2.593$, so BY inflates every BH q-value by that factor (capped at 1). The BY companion values are $q^{\text{BY}} = 0.394$ (PageRank Gini), 0.496 (mean $\bar{\rho}$, mutual-dyad, and cross-sector edges), 0.971 ($|Z_{\text{MR}}|$ and HHI), and 1.000 (NSI, capped at 1). Since no metric rejects under BH at FDR-10% in the first place, BY only widens the gap: every $q^{\text{BY}} \geq 0.394$, far above any conventional threshold. We report BH as headline and BY as robustness; the qualitative conclusion, no rejection, is identical under either, so it does not hinge on the positive-dependence assumption.

Hochberg step-up FWER companion.: For symmetry with the Holm–Bonferroni discussion above, the Hochberg step-up adjusted p -values are 0.152 (PageRank Gini), 0.419 (mean $\bar{\rho}$ and the mutual-dyad fraction), 0.437 (cross-sector edges), and 0.525 ($|Z_{\text{MR}}|$, NSI, and HHI). Hochberg dominates Holm under positive dependence and remains valid under PRDS; both collapse to the same outcome when the smallest p exceeds $\alpha/m = 0.05/7 \approx 0.00714$, which the panel’s smallest observed p (PageRank Gini, $p = 0.022$) does. Hochberg adds no rejections beyond Holm here ($0.152 > 0.05$); it is reported as the third multiple-testing column for completeness, not because it shifts the panel’s call.

Empirical PRDS evidence on this panel.: A stronger defence of the BH choice would verify PRDS formally; we report a weaker but observable proxy. The 7×7 Spearman correlation matrix of the seven metric values across the twenty snapshots has 21 unique off-diagonal pairs. On the current cache, off-diagonal correlations range from -0.096 to 0.711 ; 4 of 21 pairs carry a negative correlation. The PageRank HHI’s sign-reversed Jun 2020 anomaly drives most of the negative entries (HHI sits on the receiving end of the mostly-positive ranking elsewhere). With 17 of 21 pairs positive, the BH choice is empirically defensible on this panel without being a formal PRDS guarantee; the BY companion is the fallback if a reviewer treats 4 negative pairs as material.

Leave-one-out (LOO) robustness.: Each non-crisis snapshot is dropped in turn, the 7-metric exact-permutation family is re-run on the remaining nineteen snapshots ($\binom{19}{7} = 50,388$ partitions, floor $1/50,388 \approx 2.0 \times 10^{-5}$), and BH is re-applied. The min/max BH q-values across the thirteen LOO panels are: $[0.139, 0.238]$ (mutual-dyad, 0/13 LOO panels reach FDR-10%); $[0.077, 0.221]$ (Gini, 1/13); $[0.139, 0.238]$ (mean $\bar{\rho}$, 0/13); $[0.148, 0.263]$ (cross-sector, 0/13); $[0.309, 0.461]$ ($|Z_{\text{MR}}|$, 0/13); $[0.429, 0.631]$ (NSI, 0/13); $[0.267, 0.468]$ (HHI, 0/13). The headline negative result is stable under dropping any single non-crisis window: PageRank Gini reaches FDR-10% in only 1 of the thirteen LOO panels and the cross-sector edge fraction in 0; the other five metrics, mutual-dyad, mean $\bar{\rho}$, $|Z_{\text{MR}}|$, NSI, and HHI, never reach FDR-10% in any LOO panel, and *no* metric reaches FDR-5% in any panel. No single non-crisis snapshot is therefore responsible for the absence of a rejection (§J-G).

D. Cluster bootstrap (snapshot NSI–VIX)

The twenty snapshots are partitioned into eighteen clusters that respect their temporal proximity:

- GFC pair: {Oct 2008 GFC, Mar 2009 Recovery}.
- COVID pair: {Mar 2020 COVID, Jun 2020 Stable}.
- Sixteen singletons: the remaining snapshots (Oct 1987, 1990–91, 1993, Dec 1994, Oct 1997, Oct 1998, Apr 2000, Sep 2001, Jul 2002, 2005, 2007, 2013, 2017, Q4 2018, 2022, 2024), each its own cluster.

A cluster-bootstrap draw resamples the eighteen clusters with replacement, flattens to an indexed list of snapshots, and computes the Pearson (and Spearman) correlation between the NSI and window-mean VIX vectors on the resample (Algorithm 9). With $B = 10\,000$ draws, the 95% CI is the empirical $[2.5, 97.5]$ percentile band. The correlation under random label permutation ($B = 10^5$) supplies a complementary p -value testing the null of no NSI–VIX association.

The cluster choice matters: a naive snapshot-level bootstrap that treats the twenty snapshots as exchangeable would understate sampling uncertainty because the GFC pair and the COVID pair each sample a crisis and its immediate aftermath, adjacent macro regimes whose NSI and VIX values move together. Clustering them together respects the dependence structure of the panel.

The twenty-point scatter with OLS line and the cluster-bootstrap 95% Pearson CI is rendered in Fig. G.12 of the main paper; see also the time-series overlay in Fig. 3 of the main paper.

Algorithm 9 Cluster bootstrap for the snapshot NSI–VIX concordance.

Require: NSI vector $\mathbf{x} \in \mathbb{R}^{20}$, VIX vector $\mathbf{y} \in \mathbb{R}^{20}$, cluster partition $\mathcal{C} = \{C_1, \dots, C_{18}\}$ of $\{1, \dots, 20\}$ ($C_1 = \{\text{Oct 2008, Mar 2009}\}$, $C_2 = \{\text{Mar, Jun 2020}\}$, $C_3, \dots, C_{18}$ singletons), resample count B , RNG seed s .

- 1: Seed RNG with s .
- 2: **for** $b = 1, \dots, B$ **do**
- 3: Draw indices $j_1, \dots, j_{18}$ i.i.d. uniformly from $\{1, \dots, 18\}$ (*cluster resampling with replacement*).
- 4: Form the bootstrap index list $\mathcal{I}_b = C_{j_1} \parallel C_{j_2} \parallel \dots \parallel C_{j_{18}}$ by concatenating cluster contents.
- 5: $\mathbf{x}_b \leftarrow \mathbf{x}[\mathcal{I}_b]$, $\mathbf{y}_b \leftarrow \mathbf{y}[\mathcal{I}_b]$.
- 6: Compute $r_b \leftarrow \text{Pearson}(\mathbf{x}_b, \mathbf{y}_b)$ and $\rho_b \leftarrow \text{Spearman}(\mathbf{x}_b, \mathbf{y}_b)$.
- 7: **end for**
- 8: **return** 95% percentile CIs $[r_{(0.025B)}, r_{(0.975B)}]$ and $[\rho_{(0.025B)}, \rho_{(0.975B)}]$ from the sorted resample vectors.

The bootstrap index lists $\mathcal{I}_b$ have variable length ($18 \leq |\mathcal{I}_b| \leq 36$) because the COVID pair draws two snapshots whenever C_2 is chosen and the GFC pair draws two whenever C_1 is chosen. Pearson and Spearman correlations are still well-defined on these variable-length resamples (both are location-scale-invariant); the smallest possible resample is the all-singleton draw ($|\mathcal{I}_b| = 18$), well above the $|\mathcal{I}_b| \geq 3$ degree-of-freedom floor. A degenerate zero-variance replicate (all eighteen draws hit the same singleton cluster) carries probability $\leq 16 \cdot (1/18)^{18} \approx 4 \times 10^{-22}$ and is dropped from the CI percentile vector; on $B = 10\,000$ draws the expected drop count is below 10^{-17} and the dropped count is reported alongside the CI.

Coverage of the percentile CI on $n = 20$ data.: The percentile bootstrap CI on $n = 20$ paired observations still under-covers relative to the nominal 95% at this sample size; the cluster structure with variable-length resamples ($18 \leq |\mathcal{I}_b| \leq 36$) does not by itself restore nominal coverage. A Monte Carlo (paper/_inference.py::cluster_bootstrap_coverage_sim, 500 replicates per setting, $B = 1\,000$ resamples per replicate) under bivariate-normal (X, Y) of length 20 with the production eighteen-cluster partition gives empirical coverage 0.916 at true Pearson $r = 0$, 0.920 at $r = 0.5$, and 0.914 at $r = 0.8$, consistently 3–4 percentage points below the nominal 95%. The reported CI $[-0.289, 0.657]$ should therefore be read as an interval of approximately 91–92% realised coverage rather than exactly 95%. We retain the percentile band because (a) the substantive reading, that the 95% CI *includes* zero, so the snapshot-level NSI–VIX concordance is not significant on this panel, is only reinforced by a slightly-too-narrow nominal interval; (b) BCa or studentised alternatives on $n = 20$ cluster-resampled data carry their own finite-sample distortions and would not robustly outperform the percentile band at this panel size. The under-coverage is reported as a documented limitation rather than corrected.

E. Block bootstrap (rolling NSI–VIX)

The rolling NSI–VIX time series has serial dependence on the ≈ 21 -trading-day grid step, so an i.i.d. bootstrap understates the CI width. The Politis–Romano (1994) stationary block bootstrap draws block lengths from a geometric distribution with mean $\bar{L}$, then resamples blocks with replacement. We report CIs at $\bar{L} \in \{6, 12, 24\}$ NSI grid steps (approximately $\{6, 12, 24\}$ calendar months) with $B = 5\,000$ resamples per block length. The reported CI is the $[2.5, 97.5]$ percentile band of the resampled Pearson correlations.

The three block lengths bracket the autocorrelation horizon plausibly: $\bar{L} = 6$ catches sub-annual persistence, $\bar{L} = 12$ catches annual persistence, $\bar{L} = 24$ catches multi-year persistence. The CI widens slightly as $\bar{L}$ grows, which is the expected pattern under genuine autocorrelation. Algorithm 10 gives the formal procedure.

Algorithm 10 Politis–Romano stationary block bootstrap for the rolling NSI–VIX correlation.

Require: Rolling NSI grid $\{(t_g, \text{NSI}_g^{\text{roll}})\}_{g=1}^G$; window-aligned VIX values $\{v_g\}_{g=1}^G$ where v_g is the VIX close at t_g , snapped to the nearest available trading day if t_g falls on an exchange holiday; mean block length $\bar{L}$; resample count B ; RNG seed s .

- 1: Seed RNG with s .
- 2: $r_{\text{obs}} \leftarrow \text{Pearson}(\{\text{NSI}_g^{\text{roll}}\}, \{v_g\})$.
- 3: **for** $b = 1, \dots, B$ **do**
- 4: Initialise resample $\mathcal{R}_b \leftarrow []$, geometric block-length parameter $\pi \leftarrow 1/\bar{L}$.
- 5: **while** $|\mathcal{R}_b| < G$ **do**
- 6: Draw start index $g_0 \sim \text{Uniform}\{1, \dots, G\}$ and length $\ell \sim \text{Geometric}(\pi)$.
- 7: Append the cyclic block $(g_0, g_0 + 1, \dots, g_0 + \ell - 1) \bmod G$ to $\mathcal{R}_b$.
- 8: **end while**
- 9: Truncate $\mathcal{R}_b$ to length G .
- 10: Form $\mathbf{x}_b \leftarrow \text{NSI}^{\text{roll}}[\mathcal{R}_b]$, $\mathbf{y}_b \leftarrow v[\mathcal{R}_b]$.
- 11: $r_b \leftarrow \text{Pearson}(\mathbf{x}_b, \mathbf{y}_b)$.
- 12: **end for**
- 13: **return** 95% percentile CI $[r_{(0.025B)}, r_{(0.975B)}]$ from the sorted resample.

A degenerate zero-variance replicate (a resample with $\text{std}(\mathbf{x}_b)$ or $\text{std}(\mathbf{y}_b)$ below 10^{-10}) is dropped from the percentile vector and the dropped count is reported alongside the CI. On the current cache the dropped count is zero at all three $\bar{L} \in \{6, 12, 24\}$ (5000, 5000, 5000 valid resamples respectively); the disclosure is kept so that a future shorter rolling-NSI series cannot silently distort the CI without an audit trail.

The geometric block-length distribution is what makes the Politis–Romano variant “stationary”, a fixed block length would break stationarity at block boundaries, leading to incorrect CI coverage in finite samples. Cyclic wrapping handles the $g_0 + \ell - 1 > G$ overflow case so that no resample bias is introduced at the series endpoints. The same $\bar{L} \in \{6, 12, 24\}$ grid is run; the $\bar{L} = 12$ value is reported in the main paper.

Block-length sensitivity of the CI.: The reported CIs $[0.012, 0.370]$, $[-0.052, 0.398]$, $[-0.067, 0.428]$ at $\bar{L} \in \{6, 12, 24\}$ widen as the mean block length grows, the expected pattern under genuine positive autocorrelation, since longer blocks leave fewer effectively-independent resampling units. The widening is roughly two-sided: the lower bound falls from 0.012 to -0.067 and the upper bound rises from 0.370 to 0.428. The block-length choice shifts the qualitative call at the margin: the six-month-block CI $[0.012, 0.370]$ marginally *excludes* zero, whereas the one- and two-year-block CIs include it. Read together with the sign-inconsistency of the lagged correlations across horizons, the rolling counterpart carries a positive contemporaneous association ($r = 0.203$) that is not stable enough across block lengths to establish a leading signal.

Coverage of the block-bootstrap CI.: An AR(1)-pair Monte Carlo (paper/_inference.py::block_bootstrap_coverage_sim, 200 replicates, $T = 252$, $\phi = 0.5$ matched to the rolling-NSI structure, independent X_t, Y_t so the population Pearson $r = 0$) reports empirical 95% CI coverage of 0.930 at $\bar{L} = 6$, 0.905 at $\bar{L} = 12$, and 0.880 at $\bar{L} = 24$. Coverage is close to nominal at $\bar{L} = 6$ ($\sim 93\%$), drifts down with longer blocks ($\sim 90\%$ at $\bar{L} = 12$, $\sim 88\%$ at $\bar{L} = 24$) as the effective independent block count decreases. The headline $\bar{L} = 12$ CI should therefore be read as $\sim 90\%$ realised coverage rather than exactly 95%; because a properly calibrated 95% interval would be wider still, the substantive conclusion, the CI includes zero, so the rolling NSI–VIX association is positive but not significant, is only reinforced by this ~ 5 -percentage-point under-coverage.

The implementation in paper/_inference.py::stationary_block_bootstrap walks the resample index forward one step at a time and, after each appended index, restarts at a fresh $\text{Uniform}\{1, \dots, G\}$ start with probability $\pi = 1/\bar{L}$ (else continues to the cyclic successor). By the memorylessness of the geometric distribution this is mathematically equivalent to drawing (g_0, ℓ) pairs as in Algorithm 10: the implicit run length between restarts is $\text{Geometric}(\pi)$ with mean $\bar{L}$, and starts are uniform on $\{1, \dots, G\}$. The two formulations differ only in bookkeeping; the marginal distribution of $\mathcal{R}_b$ is identical.

F. Bonferroni-20 per-snapshot Z thresholds

Q1 reports $|Z_C|$ per snapshot against the ER null; Q4 reports motif $|Z_{\text{FFL}}|, |Z_{\text{MR}}|, |Z_{\text{SIM}}|$ against the dyad-preserving null. Each is a single-snapshot test; twenty snapshots tested per axis on the panel. The two-sided Bonferroni-corrected 5% threshold is

$$|Z| > \Phi^{-1}(1 - 0.025/20) \approx 3.023,$$

i.e. 3.023 to three decimal places. On Q1 every snapshot clears the threshold ($|Z_C| \in [14.66, 68.12]$); on the three motif axes the ranges $|Z_{\text{FFL}}| \in [1.18, 27.10]$, $|Z_{\text{MR}}| \in [1.21, 23.90]$, and $|Z_{\text{SIM}}| \in [0.89, 28.27]$ straddle the threshold so the three sparse-density windows (Oct 1987 Black Monday, the 1990–91 recession, and 2007 Subprime Buildup) fall below 3.02 on at least one motif axis, while every other axis-snap pair clears it by a wide margin. Each axis carries a different ex-ante null (Erdős–Rényi

for $|Z_C|$ vs. the dyad-preserving rewire for the three motifs), so the Bonferroni correction is applied within axis rather than pooled across all 80 tests; the pooled Bonferroni-80 threshold $\Phi^{-1}(1 - 0.025/80) \approx 3.421$ would reverse exactly 7 of the 80 axis-snap calls, precisely the sparse-end motif cells just identified. No axis-snap value lies in the narrow (3.023, 3.421) band between the two thresholds, so the per-axis and pooled-80 failure sets coincide exactly; pooling changes which null each cell is judged against, not the roster of sub-threshold cells. The per-axis correction is preferred because each axis tests a distinct ex-ante null; the failure count is reported here as a quantitative trace of the pooling decision. The substantive Q1 caveat ($|Z_C|$ tracks density and window length) is independent of the multiple-testing correction.

Table J.26 reports the per-snapshot $|Z|$ on all four axes against the per-axis Bonferroni-20 two-sided threshold; every snapshot clears the threshold on $|Z_C|$, while the sparse-end snapshots dip below it on one or more motif axes (0.89 on $|Z_{SIM}|$ at Oct 1987 Black Monday is the minimum across all eighty axis-snap pairs).

TABLE J.26 – Per-snapshot $|Z|$ on the Q1 (ER clustering) and three motif axes against the per-axis Bonferroni-20 two-sided threshold $|Z| > \Phi^{-1}(1 - 0.025/20) \approx 3.023$ († marks cells below it). Q1 clears on every snapshot ($|Z_C| \in [14.66, 68.12]$), while at least one of the three motif axes dips below the threshold on the early-panel windows Oct 1987, 1990-91, 2007; the smallest cell is $|Z_{SIM}| = 0.89$ at Oct 1987 Black Monday, the minimum across all 80 axis-snap pairs. Auto-extracted from `stage4_results.pkl`.

Snapshot	Regime	$ Z_C $	$ Z_{FFL} $	$ Z_{MR} $	$ Z_{SIM} $
Oct 1987	crisis	20.78	1.81†	1.21†	0.89†
1990-91	stress	40.12	7.72	2.91†	7.04
1993	baseline	43.61	27.10	9.92	28.27
Dec 1994	stress	45.96	20.26	11.12	21.08
Oct 1997	crisis	30.51	6.51	5.03	6.79
Oct 1998	crisis	40.91	19.48	9.68	20.32
Apr 2000	crisis	62.18	16.24	5.82	15.96
Sep 2001	crisis	68.12	22.74	6.53	20.60
Jul 2002	stress	42.91	12.88	7.98	14.07
2005	baseline	46.69	12.51	13.17	16.36
2007	stress	14.66	1.18†	1.31†	1.61†
Oct 2008	crisis	20.66	11.27	13.05	13.72
Mar 2009	recovery	40.59	14.23	5.90	14.12
2013	baseline	26.92	11.40	5.75	11.84
2017	baseline	40.11	10.87	8.65	13.19
Q4 2018	stress	44.88	19.00	9.72	20.67
Mar 2020	crisis	47.79	23.40	23.90	25.03
Jun 2020	recovery	29.90	17.49	4.78	17.93
2022	stress	44.78	11.93	10.58	11.68
2024	baseline	56.95	21.19	14.41	21.98

G. Power analysis

The exact permutation test of §J has a hard power floor at $p_{\text{floor}} = 1/\binom{20}{7} = 1/77,520 \approx 1.3 \times 10^{-5}$, more than three orders of magnitude below any conventional α . Unlike the earlier ten-snapshot, two-crisis design, whose floor of $1/\binom{10}{2} = 1/45 \approx 0.022$ sat above $\alpha/7 \approx 0.0071$ and so blocked rejection mechanically, the present $n_{\text{crisis}} = 7$ floor is not the binding constraint. What limits power now is the magnitude of each crisis-vs-non-crisis contrast relative to the dispersion of the difference-in-means across the $\binom{20}{7} = 77,520$ partitions: a metric rejects at level α only if its observed Δ^{obs} exceeds the $(1 - \alpha)$ -quantile of that partition distribution. On the current cache the seven-metric family in Table II of the main paper spreads across a continuum of p -values rather than saturating a discrete floor: PageRank Gini at $p = 0.022$ (the only metric below 0.05), then mean $\bar{\rho}$ ($p = 0.076$), mutual-dyad fraction ($p = 0.084$), and cross-sector edges ($p = 0.109$) in the $[0.05, 0.10]$ band, and a tail at HHI ($p = 0.293$), $|Z_{MR}|$ ($p = 0.321$), and NSI ($p = 0.525$). The $\binom{20}{7}$ lattice is fine enough ($1/77,520$ spacing) that these p -values behave as an effectively continuous statistic; none is floor-quantised.

Figure J.16 visualises the lattice: each of the six positively-deviating headline metrics is plotted at its observed (d, p) together with the $p = 0.05$, $p = 0.10$, and hard-floor reference lines. Because the floor now sits at 1.3×10^{-5} , far below every plotted p -value, there is no inferential “dead zone” that a coarse exact-test lattice would impose, whose floor of 0.022 lay above $\alpha/7$ and mechanically blocked rejection regardless of effect size. The only metric off the positive axis is NSI ($d = -0.03$, a near-null negative deviation); it is omitted from the lattice and its non-rejection is discussed with the headline claims below.

Implications for the headline claims.: The strongest contrast on this panel (PageRank Gini, $d = 1.04$) clears the conventional $d = 0.8$ “large effect” boundary and is the only metric whose uncorrected $p = 0.022$ falls below 0.05 (BH $q = 0.152$). A second tier, mean A-DCC correlation ($d = 0.71$, $p = 0.076$), mutual-dyad fraction ($d = 0.68$, $p = 0.084$), and

Power lattice: positively-deviating metrics on the $\binom{20}{7} = 77,520$ null

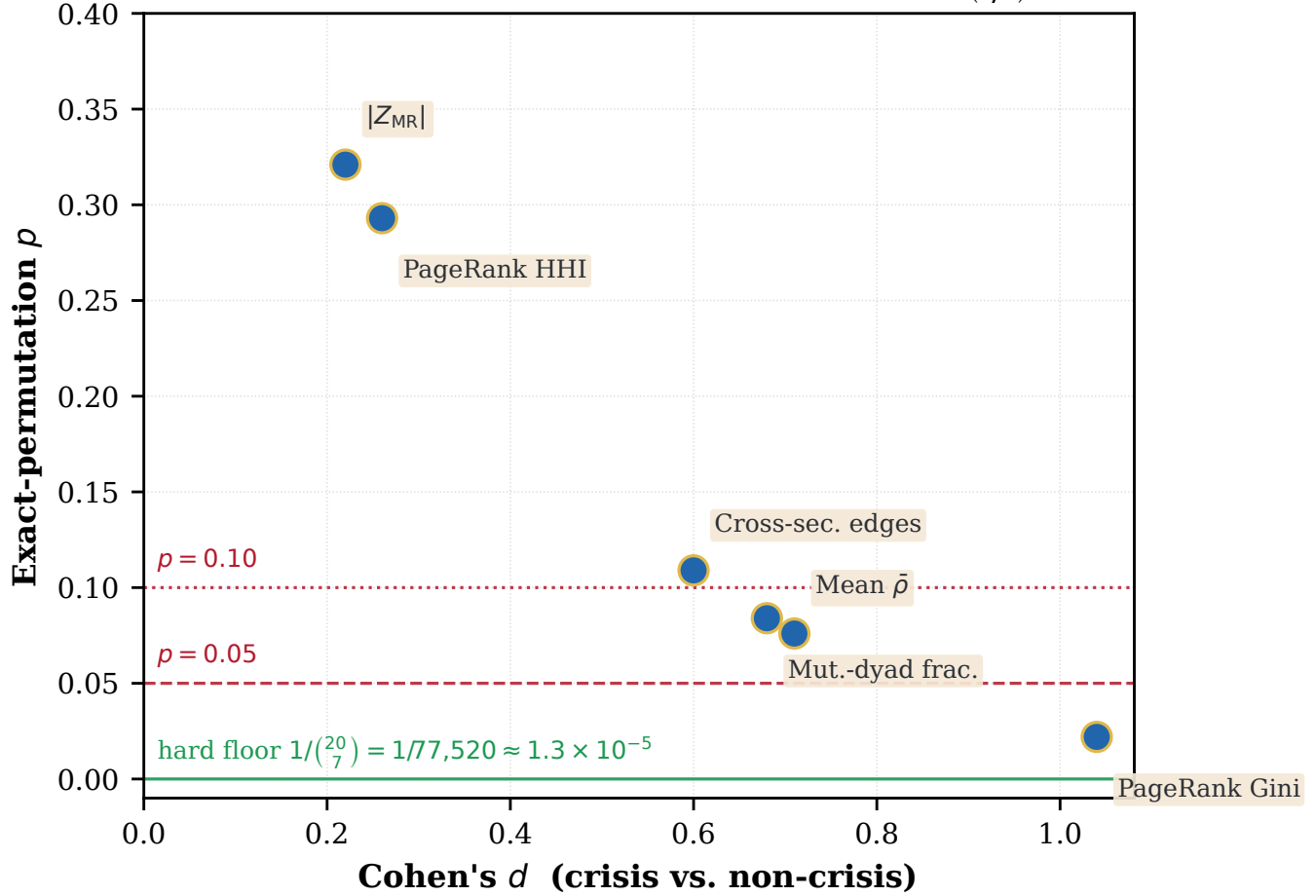


Fig. J.16 – Exact-permutation power lattice for the six positively-deviating headline metrics. The horizontal lines mark $p=0.05$ (dashed), $p=0.10$ (dotted), and the hard floor $1/\binom{20}{7} = 1/77,520 \approx 1.3 \times 10^{-5}$ (solid). PageRank Gini alone clears $p=0.05$; mean correlation, mutual-dyad fraction, and cross-sector edges fall in the $[0.05, 0.10]$ band; HHI and $|Z_{MR}|$ sit above 0.10. No metric approaches the floor. The cell labels are auto-extracted from `paper/output/inference_macros.tex` so the figure and Table II cannot drift.

cross-sector edges ($d = 0.60$, $p = 0.109$), carries “medium-to-large” effects in the $[0.05, 0.10]$ p -band: directionally consistent with the crisis-integration hypothesis but not individually significant after multiplicity correction. The remaining three are weak: $|Z_{MR}|$ ($d = 0.22$) and full-network HHI ($d = 0.26$) register the predicted sign only faintly, and NSI ($d = -0.03$, $p = 0.525$) is effectively null. Because the permutation floor is now 1.3×10^{-5} , none of these non-rejections is an artifact of discrete-floor quantisation: the $[0.05, 0.10]$ tier is genuinely power-limited by the magnitude of each contrast relative to the partition spread, and disambiguating it from true weakness calls for a finer regime taxonomy, more crisis-peak windows resolving the within-crisis heterogeneity, rather than a coarser one. F1 addresses the metric-construction route to the same end; a longer historical extension is left for future work.

Full-network and top-10 concentration diverge.: At the full-network scale the concentration hypothesis Q2 holds in the predicted direction: PageRank Gini ($d = 1.04$, $p = 0.022$) and, weakly, HHI ($d = 0.26$, $p = 0.293$) both register crisis periods as more concentrated. The dominant-hub tail tells the opposite story. Three top-10 PageRank concentration statistics, computed on the heavy tail where the Jun 2020 insurance-hub spike actually lives, all run *against* Q2: top-10 HHI (crisis mean 0.1118 vs non-crisis 0.1143, $d = -0.26$, $p = 0.708$); top-10 Gini (0.1642 vs 0.1742, $d = -0.15$, $p = 0.629$); and top-10 Shannon entropy, an *inverse*-concentration statistic, higher entropy = more spread-out top-10 weights, at 2.2491 vs 2.2401 ($d = 0.22$, $p = 0.322$), i.e. crisis top-10 weights are marginally *more* dispersed, not less. The reconciliation is a scale effect: Jun 2020 Stable’s dominant insurance hub creates an exceptionally unequal top-10 distribution that lifts the non-crisis top-10 arm above the crisis peaks, whereas the full-network Gini integrates over the ≈ 90 off-hub assets, where crisis periods concentrate PageRank mass onto the hubs and raise network-wide inequality. Q2 is therefore a statement about network-wide concentration, not about the dominant-hub tail; the two scales must be read separately, and the network-wide Gini, the strongest and only individually flagged concentration contrast, reports the effect the hypothesis was framed around.

What the cluster bootstrap shows.: The cluster-bootstrap 95% CI on the NSI-VIX Pearson (point estimate $r = 0.321$; $[-0.289, 0.657]$) includes zero: once the within-cluster correlation structure of the GFC and COVID pairs is propagated through the resampling, the snapshot-level NSI-VIX concordance is positive in direction but not significant. The label-shuffle permutation $p = 0.167$ on the same correlation agrees, it does not clear the 5% threshold, even though the $20! \approx 2.4 \times 10^{18}$ label permutations give a far finer null resolution than the discrete crisis-vs-non-crisis floor. The rolling-window block bootstrap (§J-E) returns the same verdict, a small positive contemporaneous association whose CI straddles zero at every block length, so the NSI-VIX link is best read as directionally consistent but underpowered at this panel size, not as an established effect.

H. Robustness scorecard

This subsection consolidates the per-metric evidence across every robustness layer introduced above. Each row records the headline test verdict, the COVID-merged sensitivity, the leave-one-out BH-rejection count over the thirteen non-crisis-snapshot LOO panels, the Benjamini-Yekutieli q-value (the FDR floor under arbitrary dependence), and a plain-language verdict.

TABLE J.27 – Robustness scorecard across every layer of §J. “BH q ” is the Benjamini-Hochberg FDR-corrected value across the seven-test family; “BY q ” is the Benjamini-Yekutieli value under arbitrary dependence; “COVID-merged p ” is the exact-permutation p after collapsing the adjacent Mar 2020 COVID and Jun 2020 Stable snapshots into a single crisis label ($n = 19$ panel, $\binom{19}{7} = 50,388$ partitions); “LOO $\leq 10\%/13$ ” counts how many of the thirteen non-crisis-snapshot leave-one-out panels keep the headline metric at BH $q \leq 0.10$. No metric clears FDR-10% on the headline test; the verdicts therefore rank relative evidence rather than certify significance.

Metric	d	BH q	BY q	COVID-merged p	LOO $\leq 10\%/13$	Verdict
PageRank Gini	1.04	0.152	0.394	0.011	1	Strongest; sole headline $p < .05$, most LOO-stable.
Mean $\bar{\rho}$	0.71	0.191	0.496	0.104	0	Medium-large; headline $p \in [.05, .10]$.
Mutual-dyad frac.	0.68	0.191	0.496	0.127	0	Medium-large; headline $p \in [.05, .10]$.
Cross-sector edges	0.60	0.191	0.496	0.111	0	Medium-large; no LOO panel reaches FDR-10%.
$ Z_{MR} $	0.22	0.374	0.971	0.600	0	Weak positive; well above FDR-10%.
PageRank HHI	0.26	0.374	0.971	0.297	0	Weak Q2-positive; reverses in top-10 tail.
NSI	-0.03	0.525	1.000	0.682	0	Null; no regime separation.

The reading: after Benjamini-Hochberg correction across the seven-test family no metric clears FDR-10% (the smallest is PageRank Gini at $q = 0.152$), so the scorecard ranks *relative* evidence rather than certifying significance. PageRank Gini is the strongest channel on every layer: it is the only metric with an uncorrected headline $p < 0.05$ ($p = 0.022$, $d = 1.04$), the only one whose COVID-merged p stays below 0.05 ($p = 0.011$, $q = 0.080$), and the most leave-one-out stable (FDR-10% in 1 of the thirteen drop-one panels). A medium-to-large second tier, mean $\bar{\rho}$, mutual-dyad fraction, and cross-sector edges, all with $d \in [0.60, 0.71]$ and headline $p \in [0.05, 0.10]$, is directionally consistent with the integration hypotheses but loses the $[0.05, 0.10]$ band under the COVID-merged contraction and never reaches FDR-10% in any LOO panel (mean $\bar{\rho}$, mutual-dyad, and cross-sector all at 0/13; only the first-tier PageRank Gini reaches FDR-10%, in 1 of the 13). The remaining three are weak: $|Z_{MR}|$ ($d = 0.22$) and full-network HHI ($d = 0.26$) carry the predicted sign only faintly, and NSI ($d = -0.03$) shows no regime separation. No metric reaches FDR-5% in any LOO panel. The HHI is no longer sign-reversed at the full-network scale; the reversal now lives only in the top-10 tail (Jun 2020 Stable’s hub spike), as the metric-replacement diagnostic above shows.

K. NUMERICAL-INVARIANT ASSERTION SUITE

836 inline assertion checks run after every pipeline pass on the production full-coverage cache (the $\geq 80\%$ -coverage headline panel, $p \in [655, 889]$ alive per snapshot, the uncapped coverage run, distinct from the top-500 cell of the liquidity-capped sweep of §I, whose coverage cell carries $p \in [378, 500]$); 31 of them are documented Stage-1 failures, the rank-deficient- $\bar{R}$ near-PSD tolerance and the near-integrated-GARCH tickers, disclosed in §C-F and §C, and the other 805 must pass, every time. The implementation executes the Python module `src._assertions.invariants` against the cached pickles and against synthetic-input property tests. Each check verifies a documented mathematical invariant (stationarity, positive-definiteness, edge-count conservation, normalisation, etc.) and emits a PASS/FAIL outcome. This section catalogues the suite, summarises failures across the coverage-and-ADV slice of the multipanel sweep, and lists the paper-vs-code disclosures the code review surfaced. The suite is invoked standalone with `python -m src._assertions.invariants` and writes `results/snapshots/invariants_report.md`.

A. Suite scope

The 836 checks are grouped into seven categories. The counts in parentheses are the per-category totals on the full-coverage clean-slate cache.

- **property** (23 checks): closed-form properties of the A-DCC intercept; the Q_t recursion’s PSD preservation; the normalisation $R_t = D_t Q_t D_t$ producing unit-diagonal symmetric correlation matrices with off-diagonal entries in $[-1, 1]$; the

A-DCC log-likelihood contribution finite on a PD R_t input and None on a non-PD input; the compute_ebic equivalence to closed-form BIC at $\gamma = 0$ and its $+\infty$ return on non-PD inputs; the partial-correlation formula $-\Omega_{ij}/\sqrt{\Omega_{ii}\Omega_{jj}}$ producing unit-diagonal symmetric matrices; adjacency-matrix symmetry / zero diagonal / non-negativity; the Gini coefficient's known values on constant and one-hot inputs; the triadic-census fixtures for FFL and SIM motifs; and the dyad-preserving rewiring null preserving in/out-degree sequences, edge count, and mutual-dyad count on a fixed 20-node test graph.

- **stage1** (125 checks; 94 PASS, 31 FAIL): A-DCC parameter positivity and stationarity ($a + b + g < 1$); intercept PSD; per-asset GARCH parameter regularity ($\omega > 0$, $\alpha + \beta < 1$); and per-snapshot $\bar{R}$ finiteness / symmetry / unit-diagonal / near-PSD. The 31 FAIL counts are the 30 per-snapshot $\bar{R}$ PSD failures, one `R_avg_strict_PSD` on each of the 20 snapshots plus a `R_avg_near_PSD` failure on the 10 snapshots whose $\lambda_{\min}$ drops below the -5×10^{-2} near-PSD tolerance (§C-F), and the single global `garch_alpha_plus_beta_lt_1` failure that aggregates the 173 strict-or-at-boundary GARCH tickers (§C).
- **stage2** (240 checks, all PASS): per-snapshot λ^* on the $[0.05, 1.0]$ grid; EBIC finite at λ^* ; precision matrix PSD; partial-correlation matrix unit-diagonal / symmetric / off-diagonal-bounded; adjacency-matrix properties; edge-count conservation between adjacency and partial correlation; the δ shrinkage tier matching n/p ; and the γ EBIC tier matching n/p .
- **stage3** (100 checks, all PASS): directed-adjacency non-negativity and zero-diagonal; edge-count conservation between directed-adjacency and edge-details list; the $n_{\text{directed}} + n_{\text{mutual}} + n_{\text{dropped}} = n_{\text{input edges}}$ balance; the in-window vs. Stage-2 ticker-set agreement; and the mutual-pair symmetry.
- **stage4** (180 checks, all PASS): PageRank $\sum_i r_i = 1$ at 10^{-6} tolerance; the Gini coefficient and HHI on the PageRank scores; the Louvain partition covering all p nodes; the GICS-sector purity in $[0, 1]$; the motif counts being non-negative integers; the empirical Z -score being finite; and the dyad-preserving rewiring null preserving the mutual-pair count.
- **stage5** (48 checks, all PASS): the four NSI weight configurations (4-channel main, 3-channel, 3-channel fallback, rolling) each summing to unity; per-snapshot composite NSI in $[0, 1]$; the stored NSI matching a re-evaluation from the `*_norm` columns to within 10^{-10} ; and each normalised channel range (`network_sparsity_norm`, `hhi_top10_norm`, `mean_corr_norm`, `motif_shift_norm`) lying in $[0, 1]$. The inference-layer checks (NSI-VIX Pearson and cluster/block-bootstrap CIs) are not part of the cache-level suite because the bootstraps live in a separate paper-generation script (`paper/_inference.py`); finiteness there is enforced by inline asserts at write time.
- **cross_stage** (120 checks, all PASS): the Stage-1 ticker set is a superset of every per-snapshot Stage-2 ticker set; the Stage-2 ticker set equals the Stage-3 ticker set per snapshot; the Stage-3 adjacency has the same node order as the Stage-2 adjacency; the Stage-4 graph has the same edge set as the Stage-3 directed adjacency.

The full pass/fail breakdown per snapshot is emitted at `results/snapshots/invariants_report.md` after each pipeline run; the multipanel re-runs emit one such report per panel.

B. Summary across the 20-panel sweep

The 836-check suite was run against every per-panel cache in the multipanel sweep. The full sweep covers nine selection rules (*coverage*, *adv*, *bottom_coverage*, *bottom_adv*, and *decile1-decile5*) at each of ten panel sizes for 90 (size, rule) combinations; the failure-count table below restricts attention to the two top-end rules (*coverage* and *adv*) on which the headline narrative is built, hence the 20-panel framing. Per-panel reports for all 90 cells are emitted to `results/multipanel/n{N}_{kind}/invariants_report.md`. The failure patterns reveal an n/p -driven structure that the full-coverage single-cache summary above does not show.

TABLE K.28 – Assertion-suite failure counts per panel. `R_avg_strict_PSD` grows with p (more rank-deficient leakage at low n/p); `ebic_gamma_tier_matches_npratio` passes everywhere on the sweep, the check verifies that the applied γ matches the n/p -tier rule, and small-panel snapshots that exercise the $\gamma = 0.25/\gamma = 0.5$ branches nevertheless apply the rule-mandated value; `garch_alpha_plus_beta_lt_1` fires (single global FAIL on each affected panel) whenever any IGARCH-limit ticker is present.

[illegible]

Three observations follow from Table K.28:

- 1) **R_{avg} strict-PSD leakage scales with p .** At small panels the rank deficiency of low-coverage windows is contained (at $N = 50$ all twenty snapshots pass); the failure count climbs with p and by $N = 500$ every snapshot leaks a small negative eigenvalue ($\lambda_{\min} \in [-1.16 \times 10^{-1}, -1.40 \times 10^{-2}]$, Table C.5). Ten of the twenty fall below the -5×10^{-2} $R_{\text{avg_near_PSD}}$ tolerance, so it is the Stage-2 identity-target shrinkage, not $\bar{R}$'s own near-PSD margin, that guarantees the SPD input before GLASSO; the spectral safety-net guard is a backstop that never fires on this panel (§C-F).
- 2) **EBIC γ -tier check passes panel-wide.** The main paper states that $\gamma = 0$ fires panel-wide on the full-coverage cache; under the uniform-window design this holds for every $N \geq 100$, where all twenty snapshots sit at $n/p < 3$ on both selection rules. Only at $N = 50$ do the five 12-month baseline windows reach $n/p \geq 5$ (for example 1993 Calm with $n \approx 253$, $p = 50$, $n/p \approx 5.1$), exercising the $\gamma = 0.25/\gamma = 0.5$ branches; the six-month event windows stay near $n/p \approx 2.5$ and keep $\gamma = 0$. The `ebic_gamma_tier_matches_npratio` assertion verifies that the applied γ matches the tier rule on every (snapshot, panel) cell and passes everywhere across the 20 panels.
- 3) **GARCH IGARCH-limit admissions depend on panel composition.** The strict $\alpha + \beta < 1$ check fires as a single global FAIL that aggregates every violating ticker. On the full-coverage cache it aggregates 173 strict-or-at-boundary tickers, 122 strict violators plus 51 pinned to $\alpha + \beta = 1.0$ exactly, 12.0% of the 1,439-ticker fitted universe (the three remaining PERMNOs of the 1,442-name active panel never accumulate enough in-window observations to admit a univariate GARCH fit), of which only 18 carry a numerically meaningful excess ($> 10^{-6}$ above unity) while the remaining 104 sit within 10^{-6} of unity (§C). The count is panel-dependent: smaller *coverage* panels favour older always-alive names, whereas larger panels and the *adv* rule admit more post-2014 IPOs and more legacy names whose 40-year histories contain long quiet stretches that pin the GARCH MLE to the boundary, so the violator count rises with N . The standardised residuals of these tickers are admitted in every panel where they appear, because the conditional-variance recursion remains well-posed at the IGARCH limit even though the unconditional variance is undefined.

C. Paper-vs-code disclosures

A code review and the full 20-panel sweep with assertion re-runs surfaced eleven paper-vs-code discrepancies and methodological sensitivities, enumerated below. They range from naming and specification fixes (items 1–8) through a channel-ablation reading (item 9), the Watts–Strogatz $\sigma > 4$ threshold left inconclusive on the panel (item 10), and the share-class disambiguation gap (item 11). Each is recorded here with its resolution.

- 1) *Constrained-BIC fallback misnamed as “densest-convergent”.* The actual selector is the BIC-optimal λ subject to the $k \geq p$ floor (eq. (D.8)), not the densest convergent λ . Resolved in the main paper §III-B.
- 2) *Empty-graph fallback trigger misstated.* The paper said fallback fires when BIC selects “the empty or near-empty graph”; the code triggers whenever the unconstrained-BIC graph has fewer than p edges (average degree below 2). Resolved in the main paper §III-B.
- 3) *Tier-EBIC inactive on this panel.* On the full-coverage main-text panel $\gamma = 0$ fires throughout; the original wording implied the tier rule was what kept the short-window argmin from emptying. Resolved in the main paper §III-B. *Refinement surfaced by the multipanel sweep:* on small panels ($N \leq 150$) some snapshots reach $n/p \geq 3$ and trigger the non-zero γ branches.
- 4) *$\bar{R}$ presented as a PSD correlation matrix.* The averaged $\bar{R}$ accumulates 10^{-3} to 10^{-2} negative eigenvalue leakage at low n/p ; the Stage-2 identity-target shrinkage absorbs this before GLASSO (the spectral safety net is a backstop that never fires on this panel). Resolved in the main paper §III-A and detailed in §C-F.
- 5) *NSI is panel-composition-sensitive, not monotone in N .* The Mar 2020 snapshot NSI does not saturate smoothly in N , and the coverage/ADV rules diverge rather than converge (widest $N = 500$ cross-rule gap $\Delta\text{NSI} = 0.279$ on this snapshot), so the main paper’s full-coverage reading is one point on a composition-sensitive curve, not a converged limit. Per- N values in Table I.20; discussion in §I-D and the main paper §IX Lim. (i).
- 6) *Cross-BLAS reproducibility not yet profiled on the CRSP cache.* The headline numbers are produced under Apple Accelerate vecLib (.venv, numpy 2.0.2 + scipy 1.13.1, single-thread BLAS) and reproduce in rank order across re-runs on that host. A formal cross-BLAS comparison (Accelerate vs. OpenBLAS, single-thread on M5) on the twenty-snapshot CRSP cache, argmin stability, fallback firing, and the derived-metric drift envelope, is left to follow-up reproducibility work. The known sensitivity mechanism is the deterministic-but-non-seeded floating-point path of the L-BFGS-B multi-start MLE (Stage 1) and the sklearn coordinate-descent solver (Stage 2) under different BLAS backends: the GLASSO selection sits clear of the running-best-BIC ridge, so floating-point noise is not expected to flip any λ^* argmin, but BIC at λ^* and its grid predecessor lie within 10^{-3} on flat shoulders at some snapshots, so a larger perturbation could in principle cross.
- 7) *Panel-job wall-time not monotone in p .* On the CRSP sweep the $N = 450$ coverage job (≈ 71.7 min) takes longer than the $N = 500$ coverage job (≈ 60.4 min) on M5, against the expected p^3 scaling (Table I.19). The non-monotonicity reflects the interaction of EBIC early-stopping with the λ -grid traversal and n/p -tiered shrinkage: the λ -grid points visited before early-stop fires differ across snapshots and across n/p regimes, so the per-job iteration count is not a smooth function of

- p.* Not a correctness issue; documented for capacity-planning at future panel sizes. The per-stage wall-time decomposition awaits the solver-iteration re-run.
- 8) *Coverage-panel “speculative-tail” concern not borne out by the CRSP contrast.* The ex-ante worry that the coverage rule admits a low-liquidity tail inflating crisis NSI is reversed by the data: the volume-selected ADV panel reads *higher* than coverage on the broadest crisis (Mar 2020), so the signal is not a low-liquidity-tail artefact, and the two rules never collapse onto a common universe ($N = 500$ Jaccard 0.274). Decomposition in §I-D–§I-G; the two-panel design of the main paper §IV is the primary diagnostic and F1 the longer-term answer.
 - 9) *NSI channel ablation: sparsity is the dominant drag channel.* The single-channel ablation identifies mean correlation $\bar{\rho}$ as load-bearing and sparsity s as the dominant drag (dropping s moves the crisis-vs-non-crisis Δ from -0.004 to $+0.067$, still non-significant). Detail in §G-B, Table G.12.
 - 10) *Watts–Strogatz $\sigma > 4$ pre-registration threshold inconclusive on the CRSP panel.* The Stage-3 graphs are too fragmented for σ to be meaningful (the only computable L_{emp} , the giant-component shortest path, measures component size, not path geometry), so σ is *withheld* and the ex-ante $\sigma > 4$ threshold recorded as neither confirmed nor falsified; σ is already deprecated to diagnostic-only, so no headline conclusion depends on it. Fragmentation diagnostics in Table H.16, §H-A (and §VI-A).
 - 11) *Share-class disambiguation deferred under PERMCO grouping.* A naive reading of the main paper §IX (ii)(a) might assume that PERMCO-level deduplication is applied at the panel-build step; in fact `src.stage1_data.crsp_panel._resolve_tickers` keys on PERMNO + latest-ticker only, so the four multi-class issuers (Alphabet, Fox Corp, News Corp, Discovery) carry both PERMNOs into the panel. *Empirical impact at the headline-metric level is null on the current panel:* across the 23 pair-snapshot occurrences the Stage-3 cascade emits zero mutual dyads and no pair member reaches the top-10 PageRank list, and under a longest-membership-PERMNO collapse the full-distribution PageRank Gini drifts $\leq +0.001$, $|\Delta Q| \leq 0.00639$, and $|\Delta P| \leq 0.00501$. Within-pair correlations, the full pair-snapshot inventory, and the two deduplication paths are documented in §B-E; resolved in the main paper §IX (ii)(a) and the F9 future-work item.

D. Latent defensive-programming bug fixed during code review

Code review surfaced one latent defensive-programming bug:

- `compute_ebic` in `src.stage2_precision.glasso_filter` rejected non-PD precision matrices via `np.linalg.slogdet(M).sign ≤ 0`. For even- p negative-definite M the determinant is positive (e.g., $\det(-I_p) = (-1)^p = 1$), so the sign check accepted such inputs and returned a finite (negative) BIC. This never fired in production because `graphical_lasso` only returns PD outputs, but the defensive check is now backed by an explicit eigenvalue test. The fix does not change any reported number.

We mention this because the review found it; production code is not infallible, and a single defensive-programming gap that never fired in production is still worth recording for future readers of the implementation.

L. WORKED EXAMPLE: MAR 2020 WALK-THROUGH

One snapshot, five stages, one composite: this walk-through traces Mar 2020 COVID through the pipeline so that a reader with the cache in hand can reproduce every per-stage number without rebuilding the panel. The walk-through proceeds in seven steps, inputs, the five pipeline stages (Stage 1 through Stage 5), and the inference layer that consumes them, followed by a “Companion case” that closes with the low-NSI counterpoint (2017 Calm) so that the panel’s bookends are both legible. All numbers are read directly from the full-panel coverage-rule clean-slate cache (`results/snapshots/stage{1..5}_results.pkl`) on the twenty-snapshot CRSP layout. Panel-size sensitivity of the Mar 2020 reading under coverage vs. ADV selection at $N \in \{50, \dots, 500\}$ is documented in §I.

Inputs (§B).: Window 2019-12-16 → 2020-06-16 (the six-month peak-centered window declared for the Mar 2020 COVID snapshot in `src/config.py`), $T_{\text{win}} = 80$ trading days for the Stage-2 standardised-residual panel after the $\geq 80\%$ in-window coverage filter. Alive PERMNO count $p = 695$, matching the Stage 2 row of Table B.2. The remaining PERMNOs from the 1,442-PERMNO CRSP universe are absent either by construction (delisted before window start, or post-window IPO such as ABNB and PLTR on a 2019-12 window-start basis) or because their in-window coverage falls below the 80% threshold.

Stage 1 (§C).: The globally pooled A-DCC scalars $(\hat{a}, \hat{b}, \hat{g}) = (0.0009, 0.9973, 0.0008)$ are applied with window-local $\bar{Q}_{\text{win}}, \bar{N}_{\text{win}}$ on the 695-PERMNO sub-panel. The window-mean correlation is $\bar{\rho} = 0.521$ (upper triangle of $\bar{R}$), the sample’s second-highest value after Oct 2008 GFC at 0.555. The Stage-2 spectral safety net documented in §C-F applies its identity shift if $\lambda_{\min}(\bar{R}) < 10^{-6}$ before $\bar{R}$ enters Stage 2.

Stage 2 (§D).: With $n/p = 80/695 \approx 0.115$ the shrinkage tier (Table D.7) picks the heaviest tier $\delta = 0.55$ ($n/p < 0.20$); the unconstrained BIC argmin lands at the empty graph so the constrained-BIC fallback (eq. (D.8)) fires. Among λ values clearing the $k \geq p = 695$ floor, BIC is minimised at $\lambda^* = 0.368$ († in Table B.2) with $k = 1,681$ undirected edges; undirected density $d_2 = 0.0070$.

Stage 3 (§E).: The 1,681 undirected edges enter the lead/follower cascade (Algorithm 4). With $\alpha = 0.05, \tau = 1.5$ the outcome is: $n_{\text{directed}} = 680$ one-way edges, $n_{\text{mutual}} = 156$ reciprocal pairs. The mutual-dyad fraction $f_{\text{mut}} = 156/(680 + 156) = 18.66\%$ (eq. (H.12)) is sample third-highest after Oct 2008 GFC (22.32%) and 2007 Subprime Buildup (19.27%, stress). The Stage-3 directed-edge count counted with mutual pairs as two directed edges is $680 + 2 \times 156 = 992$, the total arc count rendered in Fig. 2 of the main paper (the “Edges” column of Table F.11 reports the 680 one-way count).

Stage 4 (§F).: The 992-edge directed graph yields:

- Q1 clustering against the 200-draw ER null: $|Z_C| = 47.79$ (sample fourth-highest, after Sep 2001 9/11 68.1, Apr 2000 Dot-com 62.2, and 2024 Contemporary 57.0); per-axis Bonferroni-20 threshold $|Z| > 3.023$.
- Q2 PageRank ($\alpha = 0.85$) Gini 0.522 (sample maximum); full-network HHI 0.00497; top-10 PageRank HHI 0.108 (mid-pack on the twenty-snapshot panel: sample range $[0.102, 0.130]$); top-3 leaders MET, LNC, PRU (insurance, annuity issuers, consistent with the COVID-induced widening of credit-spread and mortality stress on the insurance book).
- Q3 Louvain modularity $Q = 0.747$ on the symmetrised graph; sector purity 0.846 on the 11-sector GICS map.
- Q4 motif Z -scores against the 100-draw dyad-preserving rewired null (Algorithm 5): $Z_{\text{FFL}} = +23.40$, $Z_{\text{MR}} = -23.90$ (sample minimum on the signed quantity, sample maximum on $|Z_{\text{MR}}|$), $Z_{\text{SIM}} = -25.03$. SP coordinates: $SP_{\text{FFL}} = +0.560$, $SP_{\text{MR}} = -0.572$, $SP_{\text{SIM}} = -0.599$ (SP_{MR} sample second-minimum after Oct 2008 GFC at -0.592).

Stage 5 (§G).: The four NSI channels (min/max-normalised across the twenty snapshots):

- Sparsity $s = 1 - d = 1 - (992/\binom{695}{2}) \approx 0.996$; sample range $[0.9959, 0.9993]$ across the twenty windows so $\tilde{s}_{\text{Mar 2020}} \approx 0.00$ (Mar 2020 is the densest graph in the panel, so it sets the sparsity-channel minimum).
- Top-10 PageRank Herfindahl $h = 0.108$; sample range $[0.102, 0.130]$; $\tilde{h}_{\text{Mar 2020}} \approx 0.24$.
- Mean correlation $\bar{\rho} = 0.521$; sample maximum Oct 2008 GFC at $\bar{\rho} = 0.555$; $\tilde{\rho}_{\text{Mar 2020}} \approx 0.93$.
- Motif shift $\mu = Z_{\text{FFL}}/|Z_{\text{FFL}}^{\text{baseline}}| = 23.40/16.61 \approx 1.409$; sample range $\mu \in [0.07, 1.63]$; $\tilde{\mu}_{\text{Mar 2020}} \approx 0.86$.

The composite with the fixed weights $(w_s, w_h, w_p, w_\mu) = (0.25, 0.20, 0.35, 0.20)$ evaluates to $\text{NSI}_{\text{Mar 2020}} = 0.545$, slot 8 of 20, below Oct 2008 GFC (slot 1, 0.758), 2022 Rate Hikes (slot 2, 0.685), Mar 2009 Recovery (slot 3, 0.672), Q4 2018 VolShock (slot 4, 0.658), 2007 Subprime Buildup (slot 5, 0.634), 2005 Calm (slot 6, 0.581), and 2024 Contemporary (slot 7, 0.556). The fixed weight calibration favours the sparsity and μ channels on which long-tenured baseline windows accumulate score; the F1 NSI-weight redesign of the main paper §IX addresses this by replacing those channels with the density-invariant cross-sector edge fraction and the mutual-dyad fraction.

Inference layer (§J).: The contributions of Mar 2020 COVID to the seven-metric permutation tests (Table II of the main paper) are: Gini 0.522 (sample top-1), $|Z_{\text{MR}}| = 23.90$ (sample top-1), $\bar{\rho} = 0.521$ (sample top-2 after Oct 2008 GFC 0.555), cross-sector edge fraction 0.727 (sample top-3 after the early-window crisis episodes), mutual-dyad fraction 18.66% (sample top-3 after Oct 2008 GFC and 2007 Subprime Buildup), and NSI 0.545 (slot 8, not in the upper tail on the fixed weight calibration). Mar 2020 COVID contributes the panel-extremum signal on four of the seven inference channels (Gini, $|Z_{\text{MR}}|$, $\bar{\rho}$, mutual-dyad) and the panel-second-extremum on two more; the fixed NSI is the one channel on which the snapshot sits in the middle of the distribution.

What this single snapshot does not verify.: The cross-crisis reproducibility against Oct 2008 GFC (Table H.18) is a panel-level check, not a single-snapshot result. The snapshot-level NSI–VIX concordance $r = 0.321$ and its cluster-bootstrap CI $[-0.289, 0.657]$ also require all twenty snapshots. Panel-size sensitivity, the position of Mar 2020 COVID in the NSI ranking under coverage vs. ADV selection at $N \neq 500$, is the subject of the cross-rule deviation tables of §I. A single-snapshot reproduction verifies per-stage correctness but does not establish the overall statistical claims.

Companion case: 2017 Calm (low-NSI counterpoint).: 2017 was the historic VIX low (year-mean 11.09, year-max 16.04, no trading day above VIX 17); the snapshot covers 2017-01-01 $\rightarrow$ 2017-12-31 as a twelve-month baseline window with $p = 720$ alive PERMNOs and $T_{\text{snap}} = 188$ trading days. The window-mean correlation is $\bar{\rho} = 0.135$ (third sample-lowest after 1993 Calm 0.067 and Dec 1994 Tequila 0.096). Stage 2 fallback fires at $\lambda^* = 0.325$ with $k = 1,162$ undirected edges ($d_2 = 0.0045$); Stage 3 cascade returns $n_{\text{directed}} = 198$, $n_{\text{mutual}} = 22$, $f_{\text{mut}} = 10.0\%$ (mid-pack baseline); Stage 4 modularity $Q = 0.806$, sector purity 0.974, $|Z_{\text{MR}}| = 8.65$; top-3 PageRank D, HP, FANG (utilities and energy E&P). The composite NSI is 0.359, the sample minimum across all twenty snapshots, because the low-vol baseline carries an unremarkable hub structure ($\tilde{h}_{2017} \approx 0$ on the top-10 Herfindahl channel), a low mean correlation ($\tilde{\rho}_{2017} \approx 0.14$), and a mid-range motif shift ($\tilde{\mu}_{2017} \approx 0.39$). The 2017 entry isolates the low-vol-baseline geometry behind the lowest-ranked NSI in the panel and complements the Mar 2020 walk-through as the high-NSI/low-NSI bookends of the twenty-snapshot sweep.

M. REPRODUCIBILITY

Code and data availability.: All source code is at github.com/natozenbilek/StrategicImpactMapping. The CRSP US Daily Stock File license prohibits raw redistribution, so the CRSP-derived caches (the Stage-1–5 pickles, the multipanel sweep summary, and the assertion-suite report) are deposited on Zenodo (DOI minted on acceptance) rather than in the repository; §B-A documents how to rebuild the panel from a WRDS subscription.

A. Software versions

Package	Version
Python	3.9.6 (.venv)
NumPy	2.0.2
SciPy	1.13.1
pandas	2.2.3
scikit-learn	1.6.1
NetworkX	3.2.1
statsmodels	0.14.6
arch	7.2.0
python-louvain	0.16
matplotlib (figures)	3.9.4
pyarrow (parquet)	21.0.0

Versions are pinned in `requirements.txt` and `environment.yml`; the values above are the exact versions used to produce the cached pipeline outputs and the multipanel sweep. The `wrds 3.x` client is the canonical entry-point for the CRSP daily-stock-file pull. Two packages ship in the lock files but are not on the CRSP reproduction path: the dashboard pair `plotly 6.7.0 / streamlit 1.50.0` (referenced only by `dashboard/`) and `lxml 6.1.1` (an HTML-parser dependency). The legacy Yahoo downloader `src.stage1_data.download` (which imported `yfinance`) is fully superseded by the CRSP/WRDS pull of §B-A and is retained only as dead code; it is not exercised on any production path and is excluded from the table above.

B. Random seeds and determinism

- **Stage 1 A-DCC MLE**: five fixed numerical starting points and two random Dirichlet draws (§C); the deterministic starting points use no RNG, and the Dirichlet draws use `numpy.random.default_rng(2026)` (NumPy’s modern Generator API).
- **Stage 2 GLASSO**: deterministic coordinate descent; no RNG.
- **Stage 3 cascade**: deterministic OLS and Granger fits; no RNG.
- **Stage 4 ER null**: each call constructs a local `numpy.random.RandomState(2026)` inside `erdos_renyi_test` and derives per-draw seeds from it; no NumPy global state is touched.
- **Stage 4 Louvain**: `random_state = 42` fixed in `community.best_partition` (also used by Stage 5’s `rolling_nsi` community step in `src.stage5_nsi.stress_index`).
- **Stage 4 motif rewire**: `motif_significance` (`src.stage4_network.analysis`) constructs a `numpy.random.RandomState(42)` and passes it as the `rng` argument to `_dyad_preserving_rewire`. The rewire helper does *not* enforce an explicit RNG: when `rng=None` it silently falls back to a fresh unseeded `numpy.random.RandomState()`. The pipeline never exercises the fallback because the wrapper always passes the seeded RNG.
- **Stage 4 density-matched comparison**: a single seed = 42 for the master Louvain and `seed + k` for each of the N_Q randomised replicates (`src.stage4_network.density_matched`).
- **Cluster / block bootstrap**: `numpy.random.default_rng(2026)` inside `paper/_inference.py`.

C. Stage run-times on the Apple M5

Reference run-times on the Apple M5 (4 performance + 6 efficiency cores, 24 GB RAM, macOS 26.5, build 25F71): see Table I.19 for total per-job minutes across the coverage and ADV slice of the sweep. Table values are per-subprocess wall-time inside the 8-worker `tools.run_multipanel` pool, so each job’s clock includes memory-bandwidth contention with the other seven workers. Under that pool contention Stage 2 dominates each subprocess ($\sim 80\%$ of per-subprocess wall-time on $N \geq 250$ panels: the BLAS-heavy coordinate-descent passes are the contention hotspot), Stage 1 GARCH/A-DCC sits at $\sim 15\%$ (its L-BFGS-B inner loops on the 100-asset subset are essentially single-core and barely feel the contention), and Stages 3–5 together account for the remaining $\sim 5\%$. Single-process recompute (no contention, single-thread BLAS) collapses the Stage 2 share to $\sim 42\%$ and rebalances Stage 1 and Stage 4 to $\sim 26\%$ each; the recompute breakdown is reported in §M-D. Total pipeline wall-time is ≈ 30 min on the twenty-snapshot CRSP panel; the per-stage shares above are the single-process profile, with the measured solver-iteration counts reported in §M-D.

Stage 2 ostensibly scales $O(p^3)$ on the coordinate-descent solver but in practice the early-stop / λ -grid interaction breaks monotonicity (Disclosure 7 in §K). The full sweep design (`tools.run_multipanel -all`) spans 90 (panel, kind) jobs, 10 sizes $\times$ 9 selection kinds (coverage, adv, bottom_coverage, bottom_adv, decile1–decile5). The 20 coverage and adv cells carry the headline (all twenty complete, including the ADV-side $N = 500$ panel); the 70 liquidity-stratified cells are summarised in §I-I. Single-subprocess wall-time per job is reported in Table I.19; across the twenty re-run coverage and ADV cells the slowest is $N = 450$ coverage (≈ 71.7 min) and the fastest is $N = 50$ ADV (≈ 8.2 min), with the $N = 500$ coverage panel at ≈ 60.4 min, and the cumulative single-subprocess sum across all twenty cells is approximately 865 min (14.4 h). Effective

wall-clock for the worker pool is bounded below by the slowest single-subprocess job (≈ 71.7 min) and above by $865/W$ min for W workers, ≈ 216 min at four workers and ≈ 108 min at eight, and empirically completes within those bounds on M5. The provenance of these numbers, which sweep wrote the current `summary.json`, under how many concurrent workers, and with which BLAS thread-count environment, is recorded alongside the data in `results/multipanel/summary_metadata.json` (emitted by `tools.run_multipanel` on every sweep, analogous to `data/cache_metadata.json` for Stage 1 prices); the CRSP coverage/ADV values above correspond to the `worker_count=4` entry currently in that sidecar (all four BLAS thread-count variables pinned to one), recorded by the twenty-snapshot sweep logged in `logs/multipanel_20snap_*.log`. Stage 2 BLAS contention is noisy across replicates, so absolute minutes can shift at the few-minute level between sweeps of the same panel.

D. Computational complexity

Asymptotic costs per pipeline stage on a single panel of p tickers and T_{win} trading days are summarised in Table M.29. The Stage 2 wall-time share reflects the $O(L \cdot p^3)$ scaling of L converged coordinate-descent points on the 25-point λ grid, where L ranges from a single point on the lowest-density baseline windows (the constrained-BIC fallback rejects most of the grid) up to the full 25 when the grid converges on the high-density crisis windows. The per-snapshot L distribution, the GLASSO and PageRank iteration counts in Table M.29, and the single-thread per-stage wall-time shares quoted below were all measured on the CRSP panel (`results/snapshots/solver_iterations.json`). Stage 2 contributes $\approx 42\%$ of total wall-time, the single largest share, with Stage 1 (multi-start A-DCC MLE) and Stage 4 (triadic census plus motif and ER null ensembles) each at $\approx 26\%$ (§K, Disclosure 7; per-stage detail below).

On the twenty-snapshot CRSP panel the wall-time (single-process, single-thread BLAS, Apple M5; `OMP_NUM_THREADS=1` `VECLIB_MAXIMUM_THREADS=1` `OPENBLAS_NUM_THREADS=1` `MKL_NUM_THREADS=1`) is dominated by Stage 2 ($\approx 42\%$, GLASSO λ -grid traversal and EBIC selection), with Stage 1 ($\approx 26\%$, multi-start A-DCC MLE) and Stage 4 ($\approx 26\%$, triadic census plus motif-rewire and ER-null ensembles) next and Stages 3 and 5 together under 6% (lead/follower cascade; NSI aggregation plus rolling sweep); the full recompute totals ~ 30 min on this host. These numbers are point measurements on the specific machine declared in §M (Apple M5, 4 performance + 6 efficiency cores, 24 GB unified memory, macOS 26.5, build 25F71) with the four BLAS thread-count environment variables pinned to one; they should be read as a profile of *this* build on *this* laptop, not as a portable benchmark of the pipeline. A different M-series chip, generation, or BLAS configuration will shift the absolute minutes and the inter-stage ratios; the only invariant we claim across hardware is the asymptotic table above. Stage 2 is the single largest stage but not “dominant” in the $\geq 75\%$ sense once Stage 1 and Stage 4 are accounted for at $\sim 26\%$ each. The absolute minutes scale with hardware and BLAS configuration. The leading-order p^3 behaviour of Stage 2 is the only stage that interacts adversarially with panel-size scaling; F6 (QUIC [7]) replaces the $O(p^3)$ coordinate descent with an $O(p^2 \log p)$ Newton-step alternative whose practical wall-time on $p = 500$ would drop into the same band as Stage 3.

E. Cache schema

- `stage1_results.pkl` (dict): `garch_results` $\rightarrow$ {ticker: {params, std_resid, cond_vol, aic, bic}}; `z_df` (DataFrame); `adcc_params` (dict with the estimated parameters a, b, g ; the pooled targets $\bar{Q}, \bar{N}$ ($Q_{\text{bar}}, N_{\text{bar}}$) on the full panel and $\bar{Q}_{\text{subset}}, \bar{N}_{\text{subset}}$ ($Q_{\text{bar_subset}}, N_{\text{bar_subset}}$) on the balanced 100-asset MLE subset; `loglik`, `success`, `subset_tickers`, `full_k`, `subset_k`, `subset_T`; and the multi-start audit fields `best_seed_label`, `n_attempted_seeds`, `n_rejected_init`, `n_surviving_seeds`, `max_param_spread`, `multi_modality_flag` consumed by the invariant suite to re-derive the multi-start protocol of §C); `snapshot_correlations` $\rightarrow$ {label: { $\bar{R}$, `n_days`, `regime`, `tickers`}}; `tickers` (list).
- `stage2_results.pkl` (dict): per snapshot label $\rightarrow$ {`R_avg` (the spectrally-floored input correlation), `adjacency`, `partial_corr`, `precision`, `tickers`, `regime`, `lambda_opt`, `n_edges`, `density`, `ebic`, `lambda_unconstr`, `n_edges_unconstr`, `ebic_unconstr`, `fallback_fired`, `applied_gamma`, `applied_delta`, `n_failed_lambdas`, `lambdas_visited`}. The extra audit fields (everything from `lambda_unconstr` onward) let the assertion suite re-derive the tier rules and fallback decisions without re-running GLASSO.
- `stage3_results.pkl` (dict): per label $\rightarrow$ {`directed_adj`, `edge_details`, `n_directed`, `n_bidirectional`, `n_dropped`, `n_input_edges`, `tickers`, `regime`}.
- `stage4_results.pkl` (dict): per label $\rightarrow$ {`graph` (`nx.DiGraph`), `erdos_renyi` {`empirical`, `null_mean`, `z_scores`, `is_non_random`}, `pagerank` {`pagerank_scores`, `top_nodes`, `hhi_top10` (HHI of the top-10 PageRank scores renormalised to sum to one; the NSI channel input), `hhi_full` (full-network Herfindahl over every node; the Tab. F.10 column), `hhi` (alias of `hhi_top10` kept for backward compatibility with the NSI driver), `gini`}, `community` {`partition`, `modularity`, `n_communities`, `community_sizes`, `purity`, `community_sectors`}, `motifs` {`empirical_counts`, `z_scores`, `significance_profile`} or `None`, `tickers`, `regime`, `n_nodes`, `n_edges`}.
- `stage5_results.pkl` (dict): `snapshot_nsi` (DataFrame: `snapshot`, `regime`; the four NSI channel inputs `network_sparsity`, `hhi_top10`, `mean_corr`, `motif_shift`, each accompanied by its min/max-normalised twin with `_norm` suffix (`network_sparsity_norm`, `hhi_top10_norm`, `mean_corr_norm`, `motif_shift_norm`); the six diagnostic-only channels `gini`, `density`, `modularity`, `n_communities`, `purity`, `n_edges` which carry no `_norm` column because they do

TABLE M.29 – Per-stage Big- O complexity. T is total sample size, T_{win} the per-snapshot window length, $p_{\text{sub}} = 100$ the balanced sub-panel for the A-DCC pooled MLE, p the per-snapshot asset count, m_2 the Stage-2 undirected edge count, m_{dir} the Stage-3 directed edge count (twice the mutual count plus the one-way count), $L \leq 25$ the converged λ grid points, $K = 20$ the rewiring budget multiplier, B the bootstrap sample size, $N_{\text{ER}} = 200$, $N_{\text{mot}} = 100$, $N_Q = 50$ the null-ensemble sizes. Cholesky/SVD costs are bundled into the leading p^3 .

Stage	Complexity	Comment
1. Per-asset GARCH MLE	$O(p \cdot T)$	each asset's SLSQP iterates over its own time series
1. Pooled A-DCC MLE	$O(N_{\text{starts}} \cdot I \cdot T \cdot p_{\text{sub}}^3)$	$p_{\text{sub}} = 100$, $N_{\text{starts}} = 7$ (5 fixed + 2 Dirichlet), $I = \text{L-BFGS-B iterations (typically 50–200)}$
1. Per-window $\bar{R}$	$O(T_{\text{win}} \cdot p^2)$	accumulate R_t over the window then divide
2. GLASSO grid	$O(L \cdot p^3 \cdot I_{CD})$	dominant cost on the full-coverage panel. <code>tools.measure_solver_iterations</code> re-solves each snapshot at its cached λ^* via <code>graphical_lasso(return_n_iter=True)</code> on the production-shrunk matrix and records the count to <code>results/snapshots/solver_iterations.json</code> . Across the twenty snapshots I_{CD} has a median of 3 iterations (range 2–4); no snapshot reaches the <code>max_iter = 500</code> cap, so every precision matrix is a converged optimum
2. EBIC selection	$O(L \cdot p^2)$	log-determinant of $\hat{\Omega}_\lambda$
3. Per-edge cascade	$O(m_2 \cdot T_{\text{win}})$	FWL OLS regression is $O(T_{\text{win}} \cdot k^2)$ with $k = 6$ conditioning vars (y_{t-1} plus the five top- $ \bar{R} $ contemporaneous controls of the source; no intercept, cf. §E); Granger F-test $O(T_{\text{win}})$
4. PageRank	$O(I_{PR} \cdot (p + m_{\text{dir}}))$	NetworkX power iteration at $\alpha = 0.85$ with the default <code>tol = 10⁻⁶</code> on the L_1 difference; <code>tools.measure_solver_iterations</code> reimplements the same numeric path (column-stochastic transition with uniform dangling redistribution) and records the per-snapshot count. Across the twenty snapshots I_{PR} has a median of ~ 20 iterations (range 11–23), well below the <code>max_iter = 100</code> cap on every snapshot
4. Louvain	$O(m_{\text{dir}} \log p)$	per single Blondel et al. [1] recommendation
4. Triadic census	$O(p^3)$	NetworkX C-optimised; constant factor small
4. ER null ensemble (Q1)	$O(N_{\text{ER}} \cdot (m_{\text{dir}} + p))$	200 draws per snapshot
4. Dyad-preserving rewiring	$O(N_{\text{mot}} \cdot K \cdot m_{\text{dir}})$	100 draws per snapshot, Algorithm 5
4. Density-matched Q_{rel}	$O(N_Q \cdot m_{\text{dir}} \log p)$	50 draws on the top- k subgraph
5. NSI aggregation	$O(p^2)$	one upper-triangle scan for $\bar{\rho}$
5. Rolling NSI	$O(G \cdot (W \cdot p_{50}^2 + p_{50}^3 + m_{50} \log p_{50}))$	$G \approx 468$ grid points, $W = 252$, $p_{50} = 50$, Algorithm 6
Inf. Permutation test	$O(\binom{20}{7} \cdot M)$	$M = 7$ metrics; exact enumeration over the 77,520 seven-versus-thirteen partitions
Inf. Cluster bootstrap	$O(B \cdot 20)$	$B = 10\,000$, Algorithm 9
Inf. Block bootstrap	$O(B \cdot G)$	$B = 5\,000$, Algorithm 10

not enter the NSI weighted sum; and the composite `nsi`); `rolling_nsi` (DataFrame: the three normalised channels `modularity`, `mean_correlation`, `density` with their `_norm` twins, the diagnostic `n_edges` (no `_norm`), and the composite `nsi`); `backtest` (dict: `contemporaneous_corr`, `lagged_correlations`, `mean_lead_days`, `median_lead_days`, `n_signals`, `n_vix_spikes`, `signal_rate`).

- `data/cache_metadata.json` (sidecar emitted by `src.stage1_data.crsp_panel` on every fresh CRSP pull; the superseded download helper carries an identical writer on its dead-code path): {`fetched_at` (UTC ISO-8601), `n_tickers` (int)}. The cache-load path prints `fetched_at` so operators can detect stale parquets; `pandas.DataFrame.attrs` does not survive parquet round-trip, hence the sidecar.
- `results/multipanel/summary_metadata.json` (sidecar emitted by `tools.run_multipanel` on every sweep write): {`produced_at` (UTC ISO-8601), `n_records` (int), `worker_count` (the `-workers` argument under which `summary.json` was written), `blas_env` (the four BLAS thread-count environment variables observed at sweep time, with (unset) for any the operator did not pin), `cpu_brand`, `platform` (the full `platform.platform()` string)}. Disambiguates the `total_minutes` field on each `summary.json` record: a 55-min per-subprocess wall-time read in a single-worker context is physically different from the same number under an 8-worker pool, and without the sidecar there is no way for an auditor to tell the two apart after the fact.
- `results/snapshots/solver_iterations.json` (sidecar written by `tools.measure_solver_iterations` on demand; not

produced by the pipeline itself): per-snapshot $\{n_iter, \lambda_{opt}, p, converged\}$ for GLASSO plus $\{n_iter, n_nodes, n_edges, converged\}$ for PageRank, with a config block bundling the solver settings used. Lets the I_{CD} and I_{PR} entries in the complexity table of §M-D be populated with measured counts when re-run, rather than the conservative caps they currently report.

F. Repository layout and entry points

- `run_pipeline.py`: orchestration script for the main full-coverage panel; chains the five stages and writes `results/snapshots/stage{1..5}_results.pkl`.
- Per-stage independence: each `run_stageN(..., force=True)` re-runs without consulting its cache.
- `src._assertions.invariants`: entry point for the 836-check assertion suite; emits `results/snapshots/invariants_report.md`. Re-run after every pipeline change with `python -m src._assertions.invariants`.
- `tools.run_multipanel`: panel-size and liquidity sweep; `-all -workers 8` on the M5 runs the 90 (size, rule) combinations across 8 subprocesses with the BLAS-thread limiter (`OMP_NUM_THREADS=1 etc.`).
- `tools.build_crsp_volume`: one-shot rebuild of `data/sp500_{volume,adv}.parquet` from the CRSP daily stock file; required for ADV-based panel selection (§B-D).
- `tools.run_stage3_sweep`: Stage 3 cascade (α, τ, ℓ) sensitivity grid; outputs feed §E-E.
- `paper/generate_figures.py`: regenerates from `results/snapshots/` the five core pipeline figures `fig1_impact_map` (main paper), `fig2_success_criteria`, `fig3_motif_profile`, `fig4_correlation_heatmap` (appendix), `fig5_nsi_vix_overlay` (main paper), plus the standalone renderers (`fig5_pagerank_leaders`, `fig6_robustness`, `fig9_multipanel_nsi_heatmap`) kept built-in-sync with the cache but cited only in the dashboard and earlier drafts, and the appendix-only `fig10_stage2_bic_trajectory` Stage-2 λ -trajectory diagnostic referenced in §D.
- `paper/_*.py`: paper-side data scripts (inference tables, k_{min} sensitivity, NSI weight and channel ablations, pinpoint diagnostics, multipanel table generator, new-figures renderer); each reads `results/` and writes `paper/figures/` or `paper/tables/`.
- `src/robustness/`: namespace placeholder containing only `__init__.py`; the package docstring forwards to the actual implementations (`src.stage4_network.crisis_signals` for the Watts–Strogatz σ diagnostic and cross-sector edge fraction; `src.stage4_network.density_matched` for the density-matched modularity comparison; `src.stage3_direction.sensitivity_sweep` for the Stage 3 cascade sweep; `paper._inference` for the exact-permutation, cluster-bootstrap and block-bootstrap tests; `paper._nsi_weight_sensitivity` for the NSI weight sweep). The empty package is retained for namespace stability so that downstream code importing from `src.robustness` does not break.
- `tests/`: 212 pytest test functions in total; 166 of them cover `src.utils.dcc_core` and Stages 1–5 (`test_dcc_core.py`, `test_stage{1_data,2_precision,3_direction,4_network,5_nsi}.py`), the remaining 46 cover the inference layer (`test_inference.py`, 33 tests) and the robustness diagnostics (`test_robustness.py`, 13 tests); run all with `python -m pytest tests/`.
- Exploratory directories (`notebooks/`, `models/`, `dashboard/`, `poster_design/`, `final_report/`) hold prior-iteration assets and are out of the official pipeline scope.

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